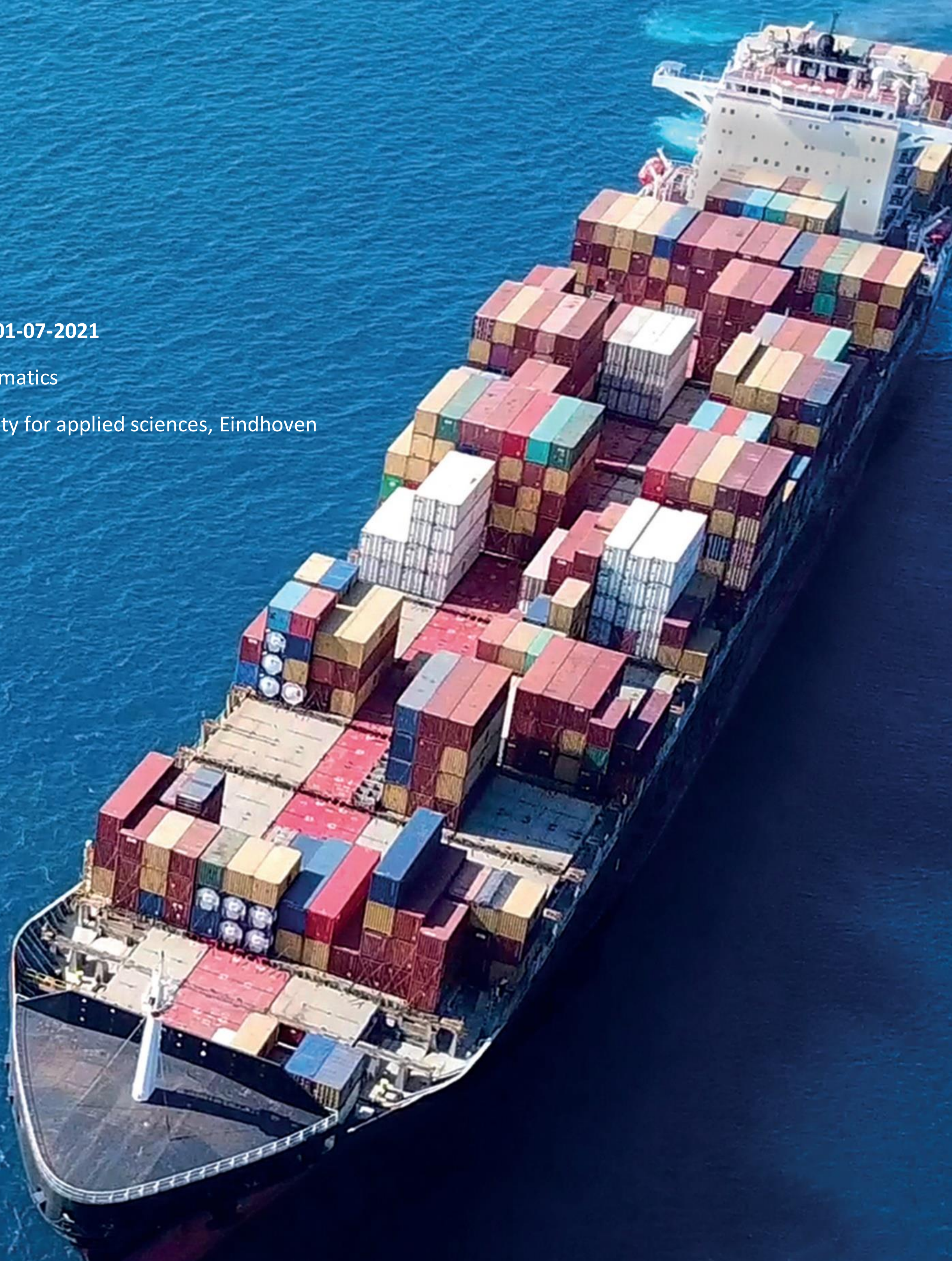
A Quantitative Model to Increase Control Over Individual Containers in Barge Transportation



Thesis

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Preface

To finalize my education of applied mathematics, specializing in operations research supported by data analysis, at the Fontys university of applied sciences in Eindhoven, I have written this thesis about my internship at Van Berkel Logistics in Veghel. For this research, a problem was submitted which I solved by using a model I designed during this internship that lasted one hundred days. An application was designed, built and implemented for this model. This report introduces the problem at hand, how I believe the problem solved, and why certain decisions were made.

Thanks to the help I received from my mentor at Fontys Charlotte Vergouwen-de Rooij, and Michel van Dijk, director logistics at Van Berkel Logistics, I have had a pleasant and instructive internship where to my believe I contributed solving the problem at hand, for which I would like to thank them sincerely. I also want to thank my father Frank, who helped out checking this thesis, and my girlfriend Kimberley for helping me with my competence portfolio which was also mandatory for me to finalize my education.

Furthermore, I would like to thank Janarthanan Karunakaran for helping with the definition and implementation of the mathematical problem. Floyd Schwiebbe and the other colleagues at customer service and bargeplanning who helped me out especially in the beginning of my internship.

Abstract

Van Berkel Logistics transports containers from **deep-sea terminals** to its **inland terminals** and vice versa. It is a growing company in the complex world and tight supply chain of container logistics. Events such as the recent blocking of the Suez canal (march 2021) are a perfect example of how disturbance can have a drastic and long term effect on the whole chain. Everyday factors like delaying **deep-sea vessels** and conditions of the weather can impact agreements made between customers, Van Berkel Logistics, **deep-sea terminals** and others.

The problem at hand is to remain in control of the containers, i.e. if something does go wrong somewhere in the chain, which occurs every day, what are the effects of these global influences on individual containers, how can these be monitored and dealt with in such a way that Van Berkel logistics gets better control of its containers. Due to the growth of the company a requirement that becomes essential since the number of customers is increasing.

After a root cause analysis of the problem, a model has been designed that serves both the customer service and the bargeplanning department. This model and its related application are based on data which is already present in the database of Van Berkel Logistics. The required data is retrieved from the database, and transformed in such a way that customer service can quickly identify which containers cause errors due to missing values or changed data fields, i.e. containers that threaten to go out of control.

The same data serves as basis for a **mathematical program**, which allocates containers to voyages. This is a process which is currently done manually by barge planners based on skills and knowledge from memory. This will be partly automatized with an online model so that schedules are less sensitive to human errors and always produce a feasible solution based on mathematics. Next to less errors the solution also saves time.

Since the model is real-time connected with the database, any changes in this database will always be considered when solving the scheduling program, and any flagged errors in the data can be changed or linked to corresponding significant information for customer service to act on. By supporting both divisions, schedules and connections are being made based on a mathematical quantitative approach, therefore providing an increased control over the containers to Van Berkel Logistics. The model provides a base solution for scheduling containers on barges which is designed to be ex, and displaying relevant data to support Van Berkel Logistics

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Chapter 1: Introduction

1.1 Motive

Van Berkel Logistics, further referred to as ‘the company’, is a Dutch based container transport company serving a variety of customers. They are specialized in inland shipping, but also transport by truck. They handle import and export from deep-sea ports like Rotterdam, to their customers and vice versa. The company operates 3 **inland terminal**s in the region (Veghel, Oss, and Cuijck) and sails with a fleet of barges between these **inland terminal**s and deep-sea ports to ensure containers are delivered on time to the customer (import) or to the terminal at the deep-sea port (export). As stated, the company uses ***multimodal*** *transport* (barges and trucks) to make this happen.

The 2004 founded company grows fast. At this point, the company transports 75.000 **TEU** (*twenty-foot equivalent unit)* per year. It all started in Veghel, later the terminals in Oss and Cuijck where also added. Most of the barges are independent, but the company also owns two barges (*Van Berkel Shipping)*. Barges are boats that sail the hinterland, from a **deep-sea terminal** to a so-called **inland terminal** and the other way around. On top of that, the company also owns 37 trucks which transport containers, and they work with a lot of trucking companies to transport even more in the region. At the port (Rotterdam for instance), there are multiple **deep-sea terminals**, at which the **deep-sea vessels** dock. At these terminals, the barges deliver containers for export on the **deep-sea vessels** and collect containers for import from the **deep-sea vessels** to return to customers.

Multiple uncertainties can occur in a given scenario. With third parties such as the barges, trucking companies, and terminal operators, a constant flow of communication and deal-making is acquired to make sure the containers are delivered on time and on the right place. In addition, factors such as weather circumstances (for barges), or traffic (for trucks) also contribute to uncertainty. This, combined with the fact that the company is still growing in demand, and not likely to cease, there are many factors that impact the voyage of any given container. Top priority at the company is to deliver goods at the right place and the right time, this is referred to as **inland date**. At all times the **inland date** must be met. The term **inland date** refers to the date a container has to be at the customer, the inland. In this research the term will be used to refer to both the date a container has to be at the customer (import), and the date a container has to be at the deep seaport (export).

1.2 Problem

Because of all the involved parties, and the constant growth of the company, control of the containers and **inland date** by the company becomes more and more difficult. This means, that due to all these impacted factors, it gets more difficult to determine whether an individual container will arrive on time at its destination, and if not, when will it?

All customers rely on receiving goods on time to avoid production delays; hence the **inland date** is key for them. If a container is delayed, for instance because the **deep-sea vessel** is delayed at the port of Rotterdam, they need to know this as soon as possible. All other processes in the company serve the goal of meeting the **inland date**.

Much data is available to support decision making, whether it’s provided by the customer, the port, or the institute of meteorology (weather). All of this must be considered. A main root cause of the problem (be in control of the containers) is the manual work related to processes like planning the barges. This is not only time consuming but also error prone. A skilled team of professionals secure that all incoming bookings are processed correctly, however this is time consuming and requires knowledge and experience to handle. For example, barge planning alone is done daily and takes all day. The department bargeplanning plans voyages and assign containers to these voyages. They shift containers on and off every day to make sure every booking is handled correctly. All decision making is mainly done based on gut feeling. Planning is done visually, relying on drafts and experience.

On the other hand, customer service must be in contact with the barge planners to secure whether they can handle a booking or not. Continuous checks occur to determine for example, if there are sufficient empty containers at the **inland terminal** that can be assigned to a booking. They also have frequent contact with the deep-sea port terminals to remain updated on the latest **deep-sea vessel**s **ETA**s (*estimated time of arrival)*. If these **ETA**s change, they must know which bookings/containers are affected and check whether the containers can still make the **inland date**. If not, they must contact customers and make new arrangements.

There many factors that can influence all the processes, some of which can not be foreseen like a malfunction of a lock. But there also is much data that can be used to automate some of the day-to-day operations. How can this data be used to contribute to being more in control of the individual containers? That is the problem at hand.

1.3 Research scope

The problem as stated in chapter 1.2 can be interpretated in many ways. There are many departments and processes at the company, and since they all serve the main goal, getting the container at the destination on the **inland date**, any improvement will contribute to solving the issue. However, the scope must be determined for this research. The focus will be on the two departments that have the most impact and are in contact with the customer directly, *bargeplanning* and *customer service*. *Bargeplanning* because this department is the company core business, transporting containers by barge. And *customer service* because they have to rearrange things when certain plans cannot be met for any reason.

Truck planning is also a big department handling transport from the **inland terminal** to the customer and vice versa. However, they also travel to Rotterdam when for instance a container cannot be collected by barge on time to still meet the **inland date**. Since truck operations runs well this department is not part of the scope. Even if not one of the company own trucks can transport the goods, there are allot of other trucking companies who can drive day and night to handle last-minute solutions.

Also, due to the research time limit, it will focus on the export of goods from the **inland terminal** at Veghel to the port of Rotterdam. Taking into consideration all **inland terminal**s, ports, and combine the import with the export, will simply take too much time. This research will take in consideration though the possibilities of extending the scope of the research and it outcomes. Although most containers are shipped to or from Rotterdam, focusing on export side only will already significantly contribute to the desired solution.

1.4 Practical relevance

What would be the practical relevance of solving the problem? The company is not in direct danger, the company is still growing and has been since it's foundation. But if part of the problem as described in chapter 1.2 can be solved, the company can focus on other things. Taken the future into consideration, improving now will contribute to further growth. Nowadays data is everywhere, it has been there for a long time, but really using this data in such a way to significantly improve day to day operations, is not as easy as it seems as indicated.

The barge planners are for sure capable of planning, and the *customer service* department knows where and how to look for information, but if day-to-day processes could be automated, or even optimized, then why not do so? This will not only contribute to the company's growth ambition, but it will also allow its employees to spend more time on other tasks. Automating processes will not replace the employees (directly), a model will always be a model and must be supervised. No model is correct, because it can never completely represent the real world, only some are useful. But if base solutions for everyday operations and data requirement are given, it will also be free of human errors.

If all the data, from different resources, can be combined into a model where both *customer service* and *bargeplanning* could profit, it will contribute to being in control of the containers. *Customer service* needs to have all data centralized, so they can react quickly to any deviations. *Bargeplanning* needs to assign containers to voyages. If they could have a base solution where all containers go to the right destination, on time, and fit on the barge, this would be a base solution that can be derived from the data already at hand. These two concepts combined will contribute to being in control of the containers and therefore the **inland date***s*

1.5 Research questions

Based on the problem at hand, and the scope of the research, questions have been formulated. This has been done via a main question, divided into sub-questions. By answering each sub-question individually, the main question will be answered as well. In this way a contribution to the problem can be delivered.

<u>Main question:</u>

How can the process of planning barges be supported, and customer service be helped with insights, given (real time) data, in such a way that the company becomes more in control of its containers and the **inland date***s*?

<u>Sub-questions:</u>

1. How can the current situation be made transparent?
2. Which factors/parameters are of significant for the problem at hand?
3. What conditions must be considered when making a barge planning?
4. How is the quality of a planning determined?
5. How can a quantitative model be made to support/improve barge planning?
6. How can the results and consequences be presented transparent for *customer service*?

Chapter 2: Theoretical background

This chapter addresses the background of the problem. First the organization of the company will be explained and relevant objects to the problem and their relations will be addressed. By analyzing the current situation at the company, the practical relevance becomes more clear. Existing literature will be used to find relevant information about the problem, the possible root causes and how this can be solved.

2.1 Background Van Berkel

Van Berkel Group consists of a holding company, which includes various operating companies. The organization chart below *(figure 2.1)* gives an overview of the different companies. This research and internship were held at Van Berkel Logistics;

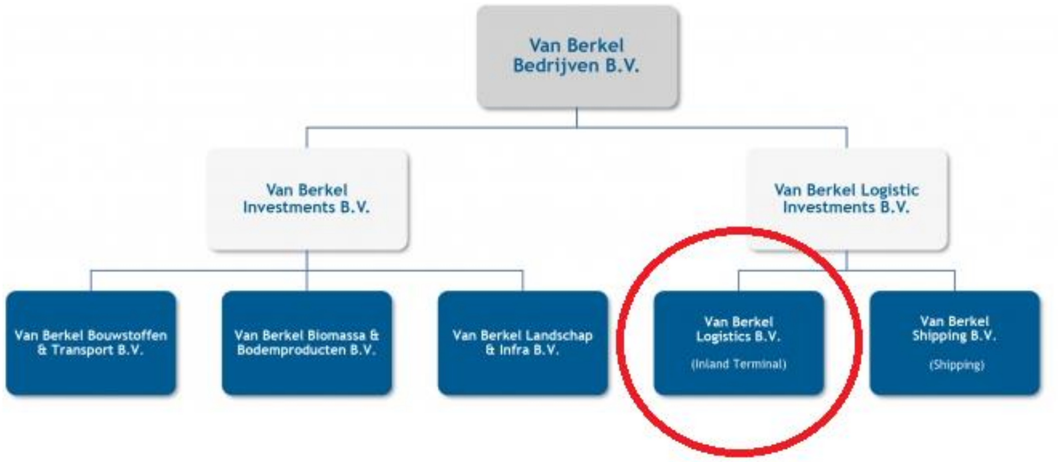


Figure 2.1: organization chart of Van Berkel Group

To understand the relevance of this study better, it is important to get a better insight in how Logistical operations are managed at the company (Van Berkel Logistics). Particularly how they process (incoming) bookings, and how *bargeplanning* combines these into voyages. *Figure 2.2* illustrates a schematic view of the process of booking at the company.

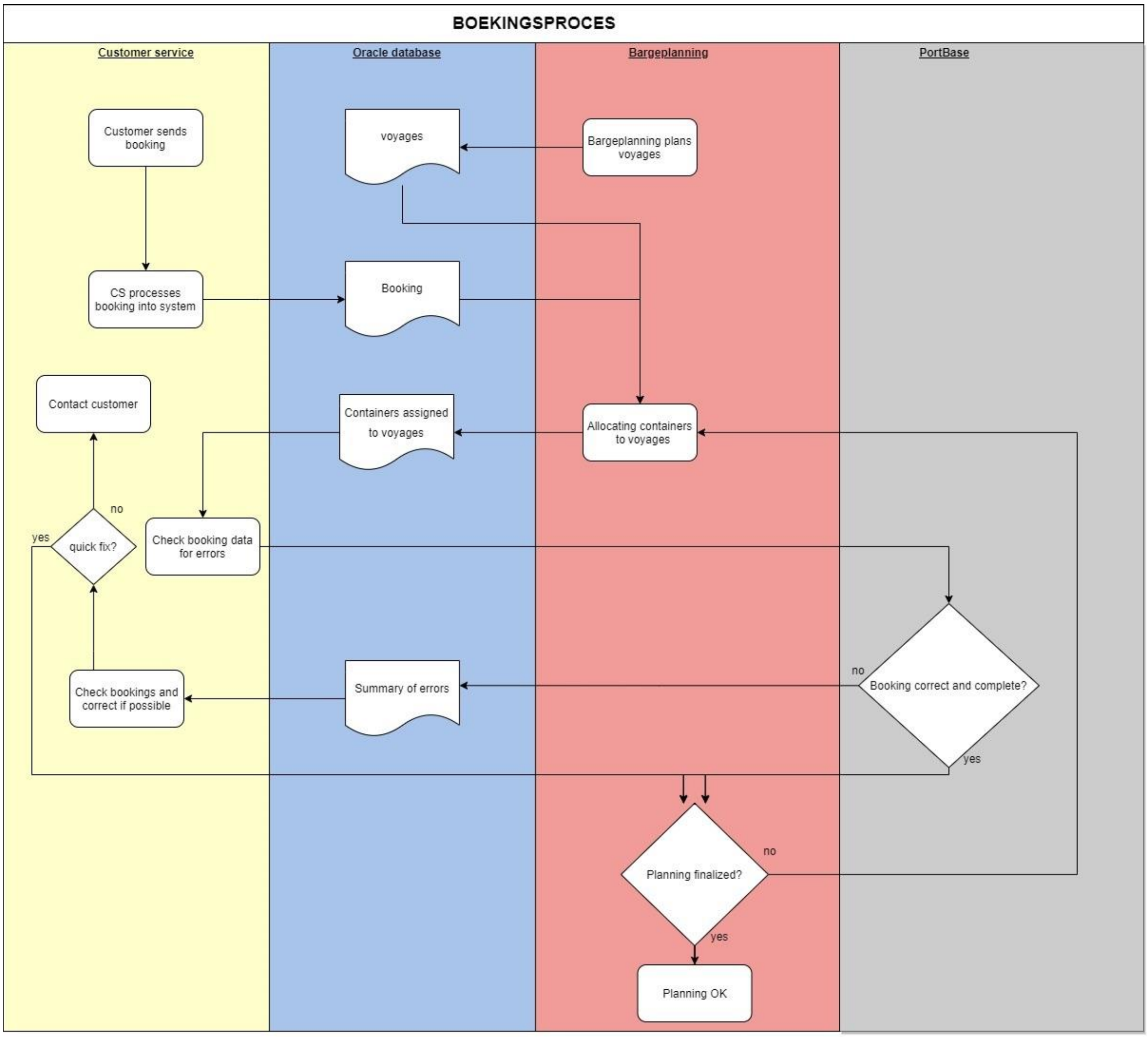


Figure 2.2: Schematic view of booking process

2.1.1 Incoming booking

As shown in *figure 2.2*, the booking process starts with an incoming booking from the customer(s). Everyday tens or even hundred of bookings arrive at *customer service* either by phone or email. Customer Service, further referred to as CS, processes these bookings into the system using ***Modailty***. This is the software used by the company to work with the database. It is used to process and retrieve information from the underlying oracle database

All relevant data of the booking is processed and stored in the oracle database of the company. Information such as number of containers, net weight, destination, import or export etc. are all processed and saved to the database. CS also assigns containers to the booking if this has not yet been done. For example, a customer wants to export goods, CS assigns an empty container manually to the booking, which can be delivered to the customer to fill, be picked up again, and shipped to its destination at the deep seaport.

2.1.2 Open and closing datetimes

An important parameter of the bookings is the open/close datetime of the container at the terminal, that is relevant for export bookings. And as stated in chapter 1.3, this research will focus on these export bookings. That's why its important to understand what an open/close datetime is. When the customer wants to export goods to a destination anywhere in the world, they have to book a place at a **deep-sea vessel**. The company facilitates the transport from and to the deep seaport terminals, but from there on it's in the hands of the shipping company responsible for the transport across the oceans.

These shipping companies sail with huge ships with a capacity up to 24.000 **TEU** of containers. They are in constant rotation over the world. When a customer initially books goods to be shipped with the shipping company, they receive a time window in which the goods can be delivered at the terminal where the **deep-sea vessel** will dock. In most cases this time window is a couple of days up to a week or even more occasionally. These open/close dates however, are based on **ETA**s of the **deep-sea vessel**s, and can alter as these **ETA**s change, which often happens. So, for a container to be on time, the open/close date for the corresponding terminal and **deep-sea vessel** must be monitored and frequently updated.

One of customer services' tasks is to monitor these **ETA**s by checking the terminal websites or contacting the terminals by phone if no online data is available. One can understand, that shifts in these time windows have a direct and significant impact on the allocation of containers on barges.

2.1.3 Bargeplanning

When customer service processes and updates all the bookings, a collection of bookings is created and stored in the database. Simultaneously bargeplanning is planning voyages. Based on experience they have with planning and the bookings; they assign available barges to voyages.

To each voyage a destination is assigned, and corresponding **ETA**s of the barges at the terminal, based on departure date. Each destination could result in a different terminal at the port of Rotterdam for example. Based on the assigned barge, that voyage has a certain capacity. In combination with the open and close dates for the containers, a schedule is made by the planners. It's obvious that all containers must go to the right destination, on the right time, and they have to fit on the barge in terms of weight and **TEU** capacity.

Bargeplanning continuously works on creating and updating these schedules. They define voyages and assign containers to these voyages. In case the number of bookings increase, or as open and close dates change over time, these schedules are manually manipulated so that they fit all the requirements. Until the last possible moment, the barge planners 'shift' containers from one voyage to the other. This shifting and allocating of containers is done manually, with Excel sheets and models from memory based on experience. It's their job to make everything fit daily.

It is important to understand this manual process because if this will be automated (partially) and optimized, this will decrease the number of errors (due to manual entry) and saves allot of time for the planners. At the end contributing to become in control of the containers that was indicated as the initial.

2.1.4 Data check

When containers have been assigned to voyages, *customer service* (and sometimes *bargeplanning*) checks whether all the necessary information is available. Since the booking at the **deep-sea vessel** is made by the customer, and the booking is processed manually by *customer service*, human errors can easily occur. This can be anything from a typo to a missing customs document, or a missing closing date. When this *export/import check* is done, the information is verified by *Portbase*.

Portbase essentially is a computer network to which all terminals, inland shipping companies, and deep see shipping companies in the port of Rotterdam are connected. If for instance customs papers have been released, but not yet verified, *Portbase* will detect this and returns a summary of errors.

In case this happens, *customer service* must analyze these errors and correct them if possible. If for example the error of a booking is a missing closing date, this can easily be tracked and quickly updated by customer service. Not every issue can be solved by customer service easily, for example when a customs document is missing. In this case they must reach out to the customer and seek for the reason of the error.

2.1.5 Fluidity of a planning

When all bookings are correct and complete, and assigned to a voyage, the planning is complete. However, as mentioned earlier, parameters can shift. Closing and opening times change over time, **ETA**s of barges can change, the weather can delay certain processes etc.

So, a planning is only complete when it is confirmed everything remains unchanged. For this reason, *bargeplanning* is planning continuously, they must reevaluate their schedules and change them if necessary. This is time consuming and forces the planners to keep changing initial planned voyages continuously.

By automating this, a schedule can be generated at the last possible moment which will be accurate and up to date, because any last-minute changes are processed and hence be taken into account.

2.1.6 Truck planning

When a container cannot be shipped in time, the company has a lot of trucks available to transport containers from and to Rotterdam. Although this is an expensive alternative, it will be executed because the **inland date** must be met no matter what. These trucks mainly drive containers from the **inland terminal**s to the customer and back but are also used to ship containers to and from the ports if no other options are available. So, a better planning of barges, results in less use of trucks and thus less operational costs for the company.

2.1.7 Journey of the container

When bookings are confirmed, complete and the schedule is made, the containers start their journey. *Figure 2.3* illustrates a schematic overview of an export journey. Import journeys are similar but reverse. Since the scope of this research focuses on export, this process has been illustrated.

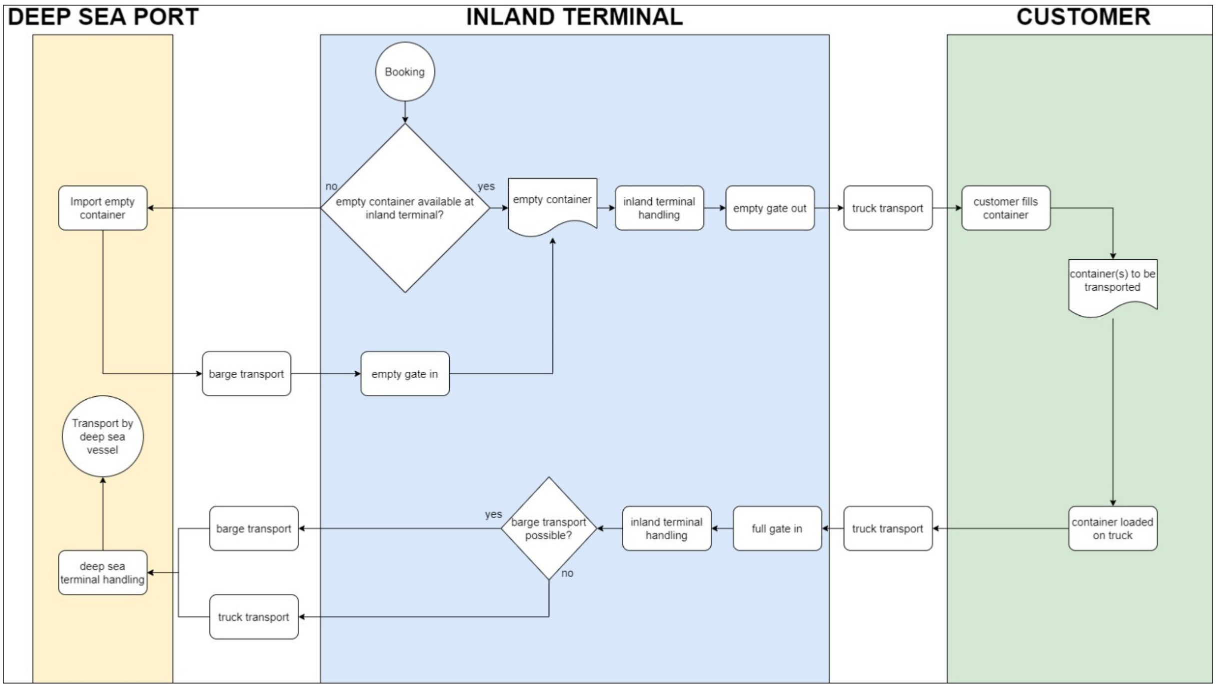


Figure 2.3: Schematic view of an export journey

When the booking is confirmed, an empty container is needed for the customer to load goods. Empty containers may be available at the **inland terminal**, but sometimes they must be imported at the **deep-sea terminal**. If it must be imported, it will be picked up by a barge that is importing goods from that terminal. The empty gate in move means the empty container is received by the **inland terminal** and thus available for the customer. After "**inland terminal** handling", the container is ready to move through the empty gate out. From there it's been transported by truck to the customer where container(s) are filled with goods by this customer. When ready, a truck picks up the container to returns it to the **inland terminal**.

A full gate in move is made, meaning the container is now registered as a full container at the **inland terminal** ready to be exported. If the container has been assigned to a voyage, the container is being moved to the corresponding barge where it will start its journey from the **inland terminal** to the **deep-sea terminal**.

If barge transport is not possible, the container is moved by truck to the **deep-sea terminal**. Once the containers arrive at the **deep-sea terminal**, they will be unloaded. Sometimes, a barge can both unload and load containers at a **deep-sea terminal**. They have a certain amount of moves they can make at any given terminal. These moves can be loading or unloading of a container. How many moves a barge can make, is pre-determined by agreements made between *bargeplanning* and the **deep-sea terminal**.

After the containers have been unloaded, they are being handled at the **deep-sea terminal** where they will be loaded on the **deep-sea vessel**. From here on, the container reaching its destiny is the responsibility of the shipping company. This is the final stage of the container's journey from a the company point-of-view.

2.1.8 Interview customer service

To better understand the current situation and the main problem, an interview has been conducted with the *customer service* department. This to determine how the problem effects the department, and how *customer service* handles the effects of it. The full interview can be found at the *attachments* section of this report.

When requested how often *customer service* is not in control of a container, they respond it occurs hundreds of times a week. This is because agreed time windows with **deep-sea terminals** cannot be met. The actual **inland date** is met more often, so the loading/unloading at the customer. When the inland date is not met, it is because customers for example wait until the last moment to plan this date. They check the portal of the company, and only when they see a rather stable **ETA** of their container(s), they start planning to have it delivered. Ideally this is planned when confirming the booking. This situation typically occurs with small customers. Big customers with many orders during a given period have a very strict planning without many problems.

How does *customer service* know when they are not able to meet agreements? The daily import/export check reveals which containers will not be able to be shipped in time. When this happens, in most cases, it is because open/close dates at the **deep-sea terminals** have changed. For example, when the **deep-sea vessel** is delayed. This often happens a few days before the initial planned pick-up date, therefore *customer service* is always too late when happens. The big deep sea shipping companies rule the entire value chain. When they face delays, or early arrivals, **deep-sea terminals** at the port have to adapt to this as well as the company and their.

Therefore, it is crucial for *customer service* to check the open and closing dates for all the terminals frequently. The big terminals have websites for this, smaller terminals must be contacted by mail or by telephone to check their status.

2.2 Literature study

Existing literature on the topic has been studied to better understand the concept of container shipping in general, and to examine how similar planning challenges are addressed. How the problem of not being in control of a container can arise, it is important to understand the global concept of container transportation.

2.2.1 History of container transport

When the first steam ships and trains were introduced during the industrial revolution, transportation of goods became easier and more efficient than ever before. However, loading and unloading them was a very time consuming and labor intensive. Everything was transported in crates, barrels, and sacks, resulting in the fact that ships spend most of their times in ports (Bernhoffen, El-Sahil, & Kneller, 2016). All of this changed in 1955 when Malcolm McLean, an American entrepreneur, thought that shipping had to become easier. He believed that each individual piece of cargo needed to be handled only twice – at their origin when stored in a standardized container box, and at their destination when unloading. So, he purchased a small tanker company, adapted its ships, and started transporting truck trailers; the birth of the container as we know it (Beazley, 2000). The size of containers is defined by *Twenty-foot Equivalent Units* (**TEU**s), which is a measurement of the length of the container. Most common are the 20 foot and 40 foot containers, so 1 and 2 **TEU** respectively, although some containers are 45 foot (Zweers B. G., 2021).

2.2.2 Global container shipping

Ever since container shipping became popular in the 1960's, it has grown nonstop. Worldwide container port throughput increased from 36 million **TEU** in the 1980, to 266 million **TEU** in 2002 (Nottenboom, 2004). Growing even more in the latest years.

It is impossible to imagine global trade without the use of containers. Nowadays, 75% of the global trade volume is carried by sea, and half of that is shipped by container (Lee & Song, 2017). That being said, a container travels through different stages on its journey from its origin to its final destination. A rough estimate states, that between 40 and 80% of the total costs for the transport of a container is made in the hinterland (Nottenboom, 2004) although (most of the times) its only a small fraction of the distance of the total trip. The use of barges reduces these costs and is an eco-friendly solution to container transport, especially compared to transport by truck (Zweers B. , 2019). Also, bundling containers at an **inland terminal** is more cost efficient and it releases some pressure of the operations in **deep-sea terminal**s (Zweers B. , 2019). So, planning and executing barge transport correctly and efficient, will result in a more cost efficient and stable transport of goods in general.

2.2.3 Transportation modes

Three types of transportation are closely connected to container hinterland transportation: **intermodal**, **multimodal** and **synchromodal** .

Intermodal transportation is a form of transportation in which the goods are shipped in only one transportation unit during the entire shipment. So transport by container is **intermodal**. **Multimodal** transport means that the goods are shipped by at least two modes of transportation, and in **synchromodal** transportation there is synchronization between multiple modes of transport were based on conditions the best way of transport is chosen, thus a form of **multimodal** transportation (SteadieSeifi, Dallaert, Nuijten, van Woensel, & Raou, 2014).

These terms are relevant to the problem since the company operates this way. **Intermodal** because they ship containers, **multimodal** because barges and trucks are used, and **synchromodal** because, for instance, the container is picked up by truck if transport by barge is not possible. One more term completes the description of the situation at hand, which is interterminal transport. Which means that transportation deals with the transportation of containers from one terminal at a deep seaport, to an **inland terminal**. Knowing the way transportation is done in general, how does existing literature contribute to the problem of being in control of a container. Planning and executing the transport is at the core of the business, automating and/or optimizing this process could contribute to the problem, how is this done in existing literature?

2.2.4 Categories of problems

Based on the decision made, problems in existing literature are divided into three categories:

A. Problems in which containers are assigned to existing barge services

B. Problems in which the routes of barges are determined for a given demand

C. Problems in which both the assignment of containers to barges and the route of barges are decided upon (Zweers B. G., 2021).

Given the background of the company, the scope of this research, and the problem at hand, category A fits most to the problem. Since *bargeplanning* defines voyages, and only then assign containers to them. This also must be done because making appointments at **deep-sea terminals** must be done a couple of days in advance. For further study, or expansion of this research, a problem of category C could be defined as well, the scope of this research however doesn't allow it to be seen this way yet because voyages will be predetermined.

An example of an **multi-objective function** of the third category is designed by Zweers B. in his study and is formulated as follows.

$$
\begin{aligned}
\min \sum_{v=1}^{b} \left(c_v^B \left(u_v - \sum_{i=1}^{n} t_i X_{iv} \right) \right) \\
+ \sum_{i=1}^{n} c^T X_{i0} + \sum_{v=0}^{b} \sum_{i=1}^{n} (c_{iv}^D + c_{iv}^S) X_{iv} \\
+ \sum_{v=1}^{b} \sum_{r=1}^{R} \pi_{rv} Y_{rv} + \sum_{v=1}^{b} \rho_v Z_v,
\end{aligned}
$$

This objective function consists of five sums:

1. The first sum makes that for each unit of **TEU** that is not used on barge, a penalty of c_{Bv} has to be paid. As a result, this sum ensures that the capacities of the barges are utilized as much as possible. The first sum is equivalent to a situation in which at first $u_v c_{Bv}$ is paid to the barge operator and the barge operator refunds c_{Bv} for each **TEU** that is actually transported on barge *v*.
2. The second sum contains all shipping costs that are made by truck shipment.
3. The third sum is the total of all demurrage and storage costs.
4. Furthermore, the fourth sum contains the penalties that are paid for visiting a terminal by barge.
5. Finally, the fifth sum is all the fixed costs that need to be paid to use the barges (Zweers B. , 2019).

The objective function shows that the **ILP** can be made complex and take into account multiple aspects of a planning. This objective function serves as an inspiration for possible extensions of the model. It considers different types of costs and alternative ways of transport (synchro modal transportation).

The **single objective function** has been applied for this research. Although multiple objectives (costs and utilization) can be considered, being in control of the containers is the single and far most important for now. For that reason, the objective function chosen will minimize the number of containers delivered both too early **and** too late. In practice this will result in containers that cannot be transported.

When a container cannot be moved, it is important to become notified. Therefore all containers that cannot be moved will be assigned to the same fictional voyage, from there the company can decide how to move them (truck or alter the voyages). It is important though to know that the possibility exists to also let a mathematical model take these factors into account.

2.2.5 Best fit for the company

When looking at the first category, assignment of containers can be done both offline and online. In an offline planning, the assignment of containers to barges is done once all the information is complete and available. An example of such a study is that of Baykasoglu and Sublan (2016) where a multi objective planning problem is described.

Other ways of solving such problems are by using a Markov chain (Perez & Mes, 2016), or using a decision tree (Van RIessen, Negenboom, & Dekker, 2016). A Markov chain is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event. A countably infinite sequence, in which the chain moves state at discrete time steps, gives a discrete-time Markov chain (DTMC).

A decision tree is a decision support tool that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to display an algorithm that only contains conditional control statements.

Because of the dynamics in demand, and the need of being more in control at any given time during the process, an online allocation model would be preferred. Markov chains and other heuristics have proven to work as well as described above, but due to the relatively simple nature of the problem in this scope allocating containers to voyages so they arrive in time, the most direct and simple way would be to formulate an **Integer Linear Program (ILP)** which later could be extended. The single objective function can be transformed into a multi objective function and constraints can be added (like import, number of **inland terminal**s).

No week looks the same, and not being able to fit everything on existing voyages is not the core of the problem, as long as it is clear which containers can not be moved and what their status is, the company will be (more) in control.

2.2.6 Software & solvers

When choosing to use an **ILP**, this **mathematical program** must be implemented and solved using software. Different software packages such as AIMMS are made to design, build, and solve **linear programs**. But with an eye on useability for the company, Python has been chosen as the language to write the **linear program**. Python can import many libraries which consist of premade functions to ensure implementation of certain problems. Linear programing can be done directly in Python with self defined functions, but a library especially made to write and solve **linear programs** already exists and is, like Python, free to use; PuLP. Because it is free to use it can be implemented directly at the company.

PuLP is a library for the Python scripting language that enables users to describe **mathematical program**s. Python is a well-established and supported high level programming language with an emphasis on rapid development, clarity of code and syntax, and a simple object model. PuLP works entirely within the syntax and natural idioms of the Python language by providing Python objects that represent optimization problems and decision variables and allowing constraints to be expressed in a way that is very similar to the original mathematical expression. To keep the syntax as simple and intuitive as possible, PuLP has focused on supporting linear and mixed-integer models (Mitchell, O' Sullivan, & Dunning, 2011).

Once defined, the program is solved using solvers. These are pieces of software used to find (optimal) solutions within a given area defined by the constraints and driven by the objective function of the **linear program**. Within PuLP many solvers for **linear programs** are available, both commercial (CPLEX, Gurobi) and open source (CBC, GLPK). For the model introduced in this research, the default (open source) solvers will be used and will be powerful enough to optimize the **linear program**. However, it is good to know that, if the model would be extended, different solvers can be used to solve the program. Because both the definition as the solution of the **linear program** is all combined within one environment, it is easy to extend. Therefore, this approach has been chosen in this research.

Chapter 3: Research method

This chapter will review available methods of research and will explain why a certain research method has been chosen. The problem will be further defined in terms of what is quantifiable and what is not. A hypothesis will be introduced and a plan of action will be discussed.

3.1 Solution strategy

As stated before, the two departments which are closest involved of being in container control are *bargeplanning* and *customer service.* This research will focus on finding a solution that will serve these departments, and therefore the company, of being in control of a container. The company has much data available, essentially all the data regarding bookings and barges are stored in the database. Data that is not direct available, can partly be quantified in such a way that it can be considered in a numeric model. For these reasons this research will use a quantitative approach of the problem.

For *bargeplanning* all available data is in the system. Whenever a booking is added, its parameters are stored in the database. This is also the case for capacities of barges and defined voyages. Therefore, there must be a quantitative approach to (partially) automating and/or optimizing the planning of barges. One of the factors that causes not being in control, is that schedules are made throughout the day, and change all the time. When generating a schedule can be postponed until the last possible moment, all actual bookings/data up until then can be taken into account.

As explained in chapter 2, this research will use an **ILP** for the allocation of containers to barges. This is because it can easily be extended in the future, and because feasibility can be assured using certain methods. The main method to guarantee feasibility, is to introduce a GHOST voyage. This will be explained in more detail in chapter 5 but in essence is needed to avoid infeasibility. When using real data, it is almost certain that, given the predefined voyages, some containers can not be moved (on time). By allocating all these containers to the GHOST voyage, they will be bundled to review by *bargeplanning*. They can inspect the containers assigned to the GHOST voyage and verify why they cannot be moved. This can be due the fact that they have missing data, or that data is wrong. It can also mean that no defined voyage will sail to the container's destination (on time). By adding/altering voyages, and rerunning the **ILP**, the containers then will be assigned to a suitable voyage.

C*ustomer service* indicates that a dashboard with quantitative data could be helpful to gain more control. The most important parameters for them are the **ETA**s of (relevant) **deep-sea vessels**, and their corresponding open and close dates at their terminals. Information can as stated in *chapter 2.17*, partially be found online for the big terminals. When this data can be made visible for *customer service*, it can also serve the planning model for *bargeplanning*. When times shift, they should be considered when making a schedule.

One of the aims of this research is to provide a solution based on a quantitative model that will serve *bargeplanning* and *customer service* simultaneously. The objective is to make an online model (as referred to in chapter 2.2.5) because it must be able to process real time data.

The main goal is to increase control of containers. So, when certain containers cannot be moved, it will be made clear which containers this concerns and why they cannot be transported. This can happen because no (defined) voyage will arrive at the destination terminal (in time). The model provides insight into the actual situation and returns a basic solution.

Because it concerns an online model, all changes in the database can instantly be considered. For example, if a certain container cannot be transported because it doesn't arrive in time, changing the **ETA**/**ETD** of the voyage assigned to the corresponding destination terminal (and still has capacity for it) will result in the container being allocated to the voyage. As mentioned, due to the scope of the project, this part of the model where containers are assigned to voyages, will focus on export. Eventually the model could be extended. More on this in chapter 7.4.

3.2 Problem quantification

Before a model can be defined, it is important to understand what can be quantified, and what not. There will always be factors that impact the **inland date** and the level of control of a container, which can not be foreseen. One of the most common factors is weather. If (unpredicted) weather delays barges for example, its effects cannot be quantified directly. The barge will have a delay, but how much delay can be unclear.

Other examples are lock malfunctions. Between the **inland terminals**, and the **deep-sea terminals**, there is a network of channels with locks and bridges. If a lock malfunctions, barges cannot pass them. Lock operators must repair them, however since it's unclear how much delay this will cause, it cannot be quantified or considered.

More recent examples are events like the Suez Canal incident (March 2021) where a **deep-sea vessel** blocks the canal because it got stuck between its banks causing huge delays for up- and downstream traffic. This delay impacted the whole chain of deep-sea transportation. Deep-sea terminals in Rotterdam got overloaded, and they must run extra shifts to get all the delayed containers in and out of the port.

Because the nature of this research is quantitative, it will focus on the stable and predictable data only. Or the factors mentioned above will not be included in the model because they are unpredictable and when they occur, must be dealt with either way.

So what data is quantifiable? The following global parameters and variables will be taken into account when constructing the model.

- <u>Capacity of barges</u>
  How many containers can be loaded on a barge, in terms of **TEU**, weight etc. These are fixed parameters which can be obtained from the database.

- <u>Planned voyages</u>
  Voyages are created by *bargeplanning*. A certain (available) barge is assigned to a voyage, and the voyage will have one or more destinations depending on the agreements between *bargeplanning* and the **deep-sea terminals**. Although this can still change over time, the data (which barge on which voyage, which destination and **ETA**/**ETD**) is available in the database.

- <u>Bookings</u>
  Which bookings are currently in the system, what containers are attached to it, and what are their dimensions in terms of weight, container class, **TEU** etc. as well as departure and arrival time? All these variables are populated with data when a booking is created. These are classified as variables because they might change over time. But either way, they are quantifiable.

- <u>Real time open/close dates</u>
  Although not available for all **deep-sea terminals**, the big (most frequently visited) terminals update their most recent open and closing dates online. Today this is checked manually, but during this research, the company acquired a tool (realized via another internship), that scrapes the websites of these big terminals. So next to the open and close dates that are provided during booking creation (and which often shift), this data can also be included when creating the model.

The company transports containers for its customers. *Figure 3.1* introduces relevant objects and explains its relations.

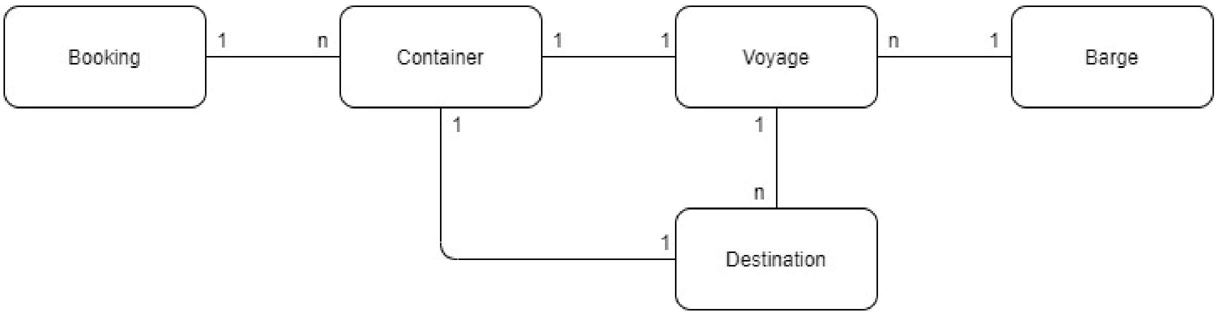


Figure 3.1: Significant objects for container transport and its relations

The company receives bookings from its customers. One booking can contain multiple containers, although it can also contain only one container. Each container has exactly one location to where it should be shipped. When importing, this destination is an **inland terminal** of the compan**y** (Veghel, Oss or Cuijck) from where on it is further transported to the customer. For export, the destination is a terminal at a deep seaport from where on it is loaded on a **deep-sea vessel** to reach its final destination. Containers must be assigned to predefined voyages, each container goes on one voyage. These voyages can have multiple destinations in one trip. A barge is assigned to the voyage, every voyage has one barge, when the voyage is over the barge is then used for other voyages.

3.3 Approach

The final model which must contribute to solving the problem, will be created from different modules data sources. This will be described in detail in chapter 4 while the approach defining such a model will be explained here.

Essentially two main parts of the model will serve both *bargeplanning* and *customer service*. For the *bargeplanning* a **mathematical program** will be defined and solved to come up with a basic solution for a (export) planning. For *customer service*, priority is to provide insight in the data, both from the database and outside sources like the internet.

The input and output of the **mathematical program** used for *bargeplanning*, is also of relevance for *customer service.* When they have insight which containers cannot be moved, they can either communicate with *bargeplanning* to change the voyages allowing the containers can be moved, or when this is not possible, contact the customer and make new arrangements. If this data, together with the data from the **deep-sea terminal** websites, can be displayed real time to customer service, it will provide the monitor function they required to become in control of the containers.

First step starts with data acquisition. There is allot of data from various sources that needs to be combined and linked in order to come up with an online real-time model, acquiring and exploring all the different data sources (this will be addressed in detail in chapter 4.2). Centralizing data will be done via spreadsheets because exporting output to spreadsheets is rather simple, and most people/employees have basic skills to work with spreadsheets.

Besides this, it’s also perfect to use as an input for the **mathematical program** and to create *Power BI* dashboards. The Oracle database contains hundreds of tables. The “bookings” table dates back to 2005 and each booking contains more than 200 parameters. For this reason, it is essential to first fully understand the data. Also, data processing and interaction had to be investigated.

Once the data is analyzed, centralized and understood, the coding of the **mathematical program** is the next step. Constraints, decision variables and object functions must be defined. Implementation and deployment of the application must be considered as well in an early stage.

Similar to spreadsheets and *Power BI*, the aim in this research is to use software the company/employees are familiar with or at least are able to use. Licensed software cannot be used (in this stage of the research). Solvers must be used to come to a solution of the **mathematical program**, some are free, some are not. A solver is a piece of mathematical software, possibly in the form of a stand-alone computer program or as a software library, that 'solves' a mathematical problem. A solver takes problem descriptions in some sort of generic form and calculates their solution. In a solver, the emphasis is on creating a program or library that can easily be applied to other problems of similar type. Once again, the aim is to establish a model within the given scope and time frame of this research which can be used straight away. Extending the model may result in the choice or purchase of other software, but this is not part of the scope.

After design but before implementation, the application must be tested. First by using test data, and later on with real data. The results of both cases must be examined and compared to come up with conclusions. When this has proven to work, the total model must be tested. Before going live, it should be tested via ‘shadow runs’. Which means, it should run with real data and be monitored, while *bargeplanning* still makes schedules the way they are used to do, the output of the model must be compared to these ‘real’ schedules and discussed.

The main goal is to provide a model which is mathematically correct. Displaying all the data and output of the mathematical model on a dashboard, will be done afterwards. Once again, it is not yet sure whether the time frame of this research will allow this, but it should be a small task when all the data is at hand and available. Either way a sketch will be made to demonstrate its purpose.

Chapter 4: Model

This chapter contains and explains the model that has been created to counter the problem of this research. It will address the model as a whole and will define the **mathematical program** used. Also, all the used data and its acquisition will be examined.

4.1 Setup global model

Before going into detail about the different modules of the designed model, a global schematic view is presented and explained. *Figure 4.1* contains a schematic view of the designed model. Programs used for implementation are included in this view, but will be explained in the next chapter. Data requirements and acquisition will be addressed further in this chapter combined with the details of the used **mathematical program**.

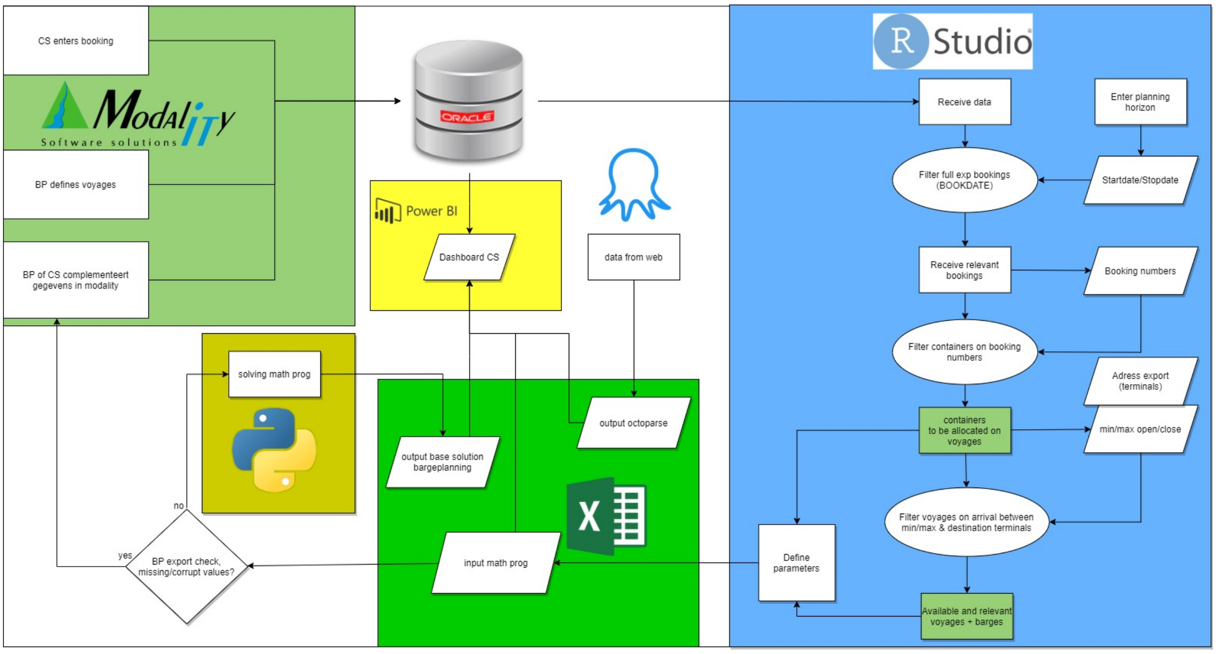


Figure 4.1: Schematic view of global model

The model contains many individual modules and sources that are connected to eventually contribute to the problem at hand. The model 'starts' at the top left corner of *figure 4.1*. CS (*customer service)* enters bookings into the system (chapter 2.1.1) using **Modailty**. **Modailty** is the software used by the company. It is used to enter bookings, create voyages and everything else in the company. It is essentially the software that the company uses to write and read the underlying oracle database. Voyages are entered by BP (*bargeplanning)* in a similar way. All bookings and planned voyages are stored in the oracle database. The database contains more information such as parameters of barges, available trucks, customer information and much more.

Data is received from the oracle database with the R script, based on a given horizon. The bookings are then filtered based on this horizon. For example, if the horizon is set on tomorrow until the day after, all bookings which have closing date between that horizon are being received from the oracle database. Reason for this is that the closing date is the ultimate deadline for delivering a container at the **deep-sea terminal**. A filter is used to retrieve export bookings only.

When all relevant bookings are received, the booking numbers are used to determine and retrieve the corresponding containers. One booking can contain multiple containers, each of them having a certain (net) weight and dimension. The destination and open/close dates are in most cases the same for all containers on one booking, but they do not have to be. As stated in chapter 3.3, containers have more than 200 parameters, only the relevant ones are filtered out. When finished, all containers retrieved are the ones which have to be assigned to a voyage.

Based on the destinations of the containers, and the open and close datetimes, voyages are being retrieved from the database. All voyages with corresponding destinations and **ETA**s between the smallest open datetime and largest closing datetime are being received. Two sets are created; one set with all containers to be moved in a certain period, and one set with all voyages which go to the corresponding destinations in the same period. Last step is to retrieve the underlying barges of the voyages from the database. The capacity of a voyage is based on to which barge this voyage is assigned. This information is essential and has to be retrieved also finalizing the third and final set of data.

After these three tables are received (containers, voyages and barges) parameters for the math program are created. The math program requires data in given format allowing it to be executed. Columns of tables are isolated, 2D binary matrices are created representing the information and barge capacity is translated to voyage capacity. Also, any missing or corrupt value is automatically detected and solved in such a way the math program will not run into errors later on. This could be, for example, an opening time that is missing. When not entered, the NA value is replaced with zero and the container is marked to show later on it is missing this data field.

When the parameters are final, they will exported into a Excel spreadsheet as input for the math program. They will be represented as tables, and as 2D binary input matrices. A final check can be done on this data by *bargeplanning*. Missing values or other deviations can be tracked and addressed if necessary. Since all the information is derived from the database, any changes should be made there as well. When validated and confirmed ok, the data will serve as input for the **mathematical program**. The program will be solved using a solver, and its output will be written back in a spreadsheet file. The solution of the **mathematical program** is two dimensional binary decision matrix indicating which container should go on which voyage.

Together with data from the internet, there are now essentially three 'new' data sources; one containing all containers and voyages in the planning horizon, one containing the allocation of containers on those voyages according to the math program, and one file containing the most recent open and close dates with corresponding **ETA**s for the deep sea vessels. The latter is obtained through the websites of the **deep-sea terminals**. All this information can be combined and displayed on a dashboard for *customer service*, and the output from the math program can be used as a basic solution for allocating export containers on voyages for *Bargeplanning*.

4.2 Data requirement

As explained in the previous paragraph, data is required input for the model. This paragraph will describe how the data is stored in the database, and which parameters are being retrieved in order to build the parameters required for the **mathematical program**.

4.2.1 Oracle

The main data source is the oracle database where all the data of the company is stored. Oracle is a database based on SQL. A query is a command for the database to receive data from it, hence these can be used to retrieve data. In the case of the company, the database contains many tables from which three data frames will be constructed: containers, voyages and barges. To construct these data frames, four tables are needed, BOOKING, VOYAGE, VOYTERM and UNIT. The complete structure of the database is pictured in *figure 4.2*.

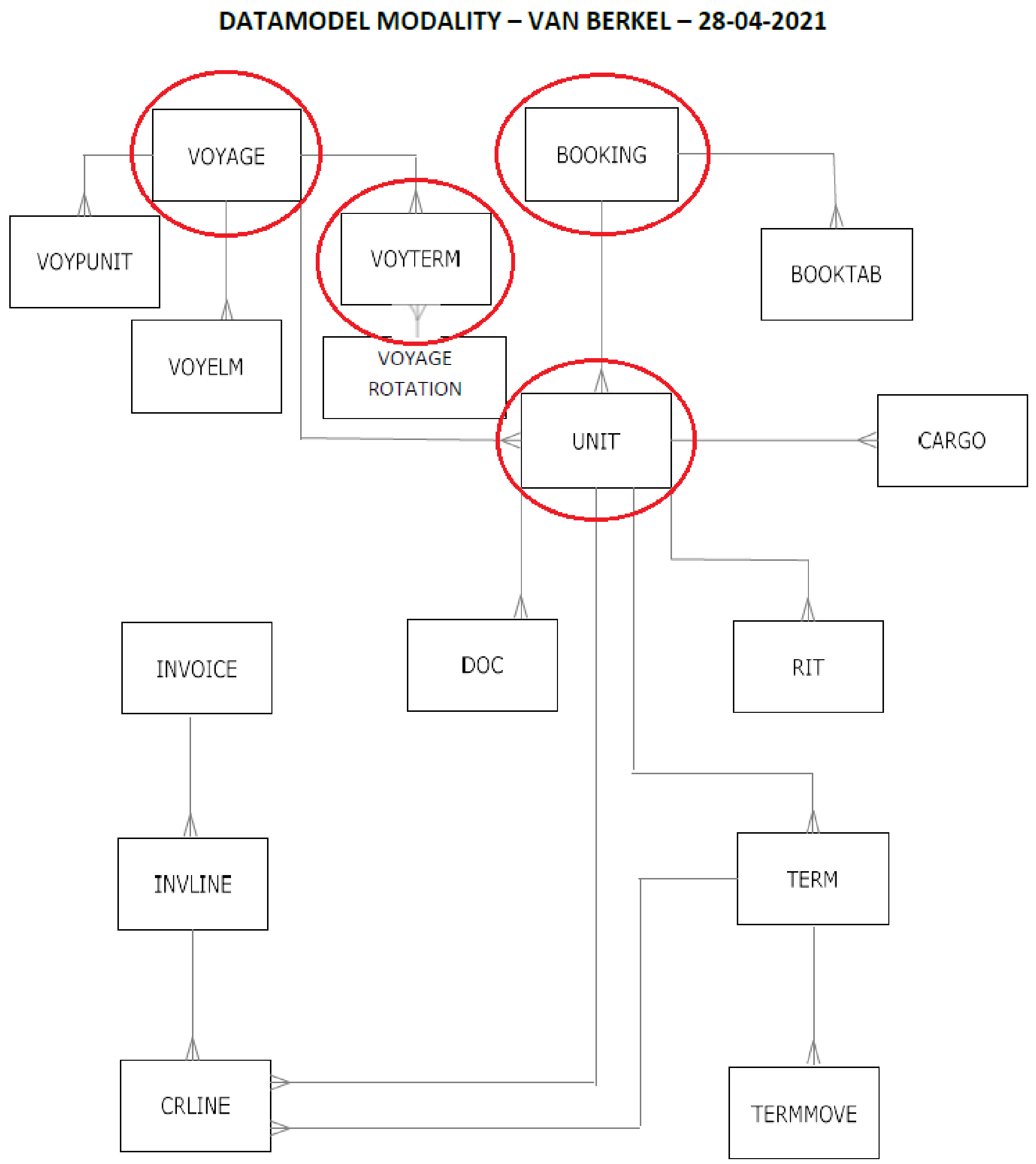


Figure 4.2: Complete database structure of the Oracle database

4.2.2 Bookings

First, bookings within the given time horizon are being retrieved from the table BOOKING. This table contains all the bookings as processed by *customer service*. The original table contains 87 columns with a variety of booking information. Only a few of these are needed, *table 4.1* contains all the columns which are received with a brief description of the value.

Parameter	*Description*	*Data type*
SEQ	Booking number	Integer
ADRESS_EXP	Destination for containers in the booking (deep sea terminal)	String
DELDATE	Delivery date, the closing date (last possible date to deliver)	Date
DELTIME	Delivery time, the closing time on the closing date	Integer
FIRSTDELDATE	First delivery date, the opening date (first possible date to deliver)	Date
FIRSTDELTIME	First delivery time, the opening time on the opening date	Integer

Table 4.1: Parameters received from BOOKING table

All rows where the closing date (DELDATE) is between the given start and stop date are received. Also, only the rows where 'FULL_EXP' is set to true, are being received. This is because this indicates that the bookings are for export, this way the import bookings are filtered out. This column is not needed after the filtration, and therefore not shown in *table 4.1*. This is because only the export bookings are required. The resulting data frame (*dfBOOKING*) is a table with the six columns shown in *table 4.1*. For practical reasons later in the model, DELDATE and DELTIME are being merged into a datetime value, and then converted to a numeric value so calculations can be made with them. The same is done with FIRSTDELDATE and FIRSTDELTIME. They are respectively converted to DELNUM and FIRSTDELNUM. An example of a final table *dfBOOKING* with real data is depicted in *table 4.2*;

	SEQ	ADDRESS_EXP	DELDATE	DELTIME	FIRSTDELDATE	FIRSTDELTIME	FIRSTDELNUM	DELNUM
1	199263	ECTDDE	2021-06-09	0200	2021-05-26	0001	550.02	888.00
2	201077	ECTDDE	2021-06-09	0200	2021-05-30	0001	646.02	888.00
3	201585	RWG	2021-06-09	1200	2021-06-01	0001	694.02	898.00
4	202116	APMR2	2021-06-09	1400	2021-06-04	0001	766.02	900.00
5	202225	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
6	202229	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
7	202239	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
8	202271	RWG	2021-06-09	1500	2021-05-25	0001	526.02	901.00
9	202323	ECTDDE	2021-06-09	0700	2021-05-27	0001	574.02	893.00
10	202324	ECTDDE	2021-06-09	0700	2021-05-27	0001	574.02	893.00

Table 4.2: Example of dfBOOKING

4.2.3 Units

When booking data is completed, the corresponding containers have to be received in order to create a table of containers to move. This data is stored in the table UNIT, the booking numbers (SEQ) from the booking data frame *dfBOOKING* are being used to retrieve the corresponding containers from the table. This is one of the largest tables of the database, with even more parameters (359). *Table 4.3* shows which columns are received and what they stand for.

Parameter	*Description*	*Data type*
BOOKING	Booking number	Integer
CNTR	Container number	String
NETT	Net weight (Kg)	Double
ADRESS_EXP	Destination for container in the booking (deep sea terminal)	String
UNITTYPE	Container type	String
FIRSTDELDATE	First delivery date, the opening date (first possible date to deliver)	Date
FIRSTDELTIME	First delivery time, the opening time on the opening date	Integer
DELDATE	Delivery date, the closing date (last possible date to deliver)	Date
DELTIME	Delivery time, the closing time on the closing date	Integer

Table 4.3: Parameters received from UNIT table

Some parameters are the same as in *table 4.1*, some are new. The booking number is BOOKING instead of SEQ. CNTR stands for the container assigned to the cargo. Sometimes, this container number is not known yet. The booking already states the net weight, and the type of container needed, but if none has been assigned yet this field remains empty. In this case a fictional container number is being created. The format is the booking number merged with a alphanumeric character. Example: booking number 123 has 3 containers, but no containers have been assigned yet. The fictional container numbers will be 123A, 123B and 123C. This way, the program can assign the container (without a name, there is nothing to assign) and the end user of the model (*customer service/bargeplanning)* sees no container is yet assigned. Because the dimensions and type of the container have been specified before, this can be done.

The net weight of a container filled with goods explains itself. UNITTYPE stands for the type of container being used. As mentioned in chapter 2.1, there are mainly 3 types of containers; 20 foot, 40 foot and 45 foot containers. Although more types of 40 ft containers can exist, for this model only the dimensions of the container (20 ft, 40 ft or 45 ft) are relevant.

This column (UNITTYPE) is being used to construct 2 more additional columns, **TEU** and CAT. They, respectively, represent the numeric value of **TEU** (20, 40 or 45) and a text representation of the category, i.e. the category represented by a text string. This is needed because, the capacity of a barge is not only limited by total weight and total **TEU**, but also the capacity for each category of containers. For instance, when a barge can ship 200 **TEU**, it does not automatically mean it can ship 10 containers of 20 **TEU**. Because of the ship's construction, and where containers can be stored, it could mean that only 8 containers of 20 **TEU** and 1 container of 40 **TEU** can be shipped with a total of 200 **TEU**.

Furthermore, missing values are being filled in. Sometimes data is incomplete caused by various reasons, as explained in chapter 2.1.4. Values must be filled in to avoid the **mathematical program** not to crash. When this is done, containers will be flagged later on in the model so they can be reviewed. For now, it is important they do not cause errors in the **mathematical program**.

Also, similar as in *dfBOOKING,* the date and time values for the open/close dates and times are being merged and converted to a numeric value the same way. An example of how *dfUNIT* is shown in *table 4.4*.

	BOOKING	CNTR	NETT	ADDRESS_EXP	UNITTYPE	FIRSTDELDATE	FIRSTDELTIME	DELDATE	DELTIME	FIRSTDELNUM	DELNUM	TEU	CAT
1	202324	BEAU 546386 0B	20100.0	ECTDDE	40HC	2021-05-27	0001	2021-06-09	0700	574.02	8.9300e+02	40	40ft
2	201585	BMOU 521601 5A	19145.0	RWG	40HC	NA	0001	NA	0001	1.00	2.0000e+07	40	40ft
3	203983	CAAU 563723 8H	1338.0	APMR	40HC	NA	0001	2021-06-09	0300	1.00	8.8900e+02	40	40ft
4	202229	CAIU 790251 8A	6072.0	ECTDDE	40HC	2021-06-02	0001	2021-06-09	1400	718.02	9.0000e+02	40	40ft
5	202271	CAIU 924367 8A	20809.0	RWG	40HC	2021-05-25	0001	2021-06-10	1100	526.02	9.2100e+02	40	40ft
6	202602	CMAU 772248 4A	23305.0	ECTDDE	40HC	2021-06-05	0001	2021-06-09	2000	790.02	9.0600e+02	40	40ft
7	202354	FFAU 317749 8A	23488.0	RWG	40HC	2021-06-04	0001	2021-06-09	1500	766.02	9.0100e+02	40	40ft
8	203618	FSCU 827620 5A	20300.0	APMR2	40HC	2021-06-03	0001	2021-06-09	1400	742.02	9.0000e+02	40	40ft
9	199263	GESU 623815 6G	25500.0	NA	40HC	NA	0001	2021-06-09	0200	1.00	8.8800e+02	40	40ft
10	201077	HAMU 122222 3A	0.1	NA	40HC	NA	0001	NA	0001	1.00	2.0000e+07	40	40ft

table 4.4: Example of dfUNIT

4.2.4 Voyages

Next up are the candidate voyages. The table VOYTERM contains all defined voyages processed by *bargeplanning*. Again, a huge table with 81 columns. A query is made to extract only the voyages, which have the same addresses as the containers to be moved, and an **ETA** which lies between the earliest opening date, and the latest closing date of all containers. This way, only eligible voyages will be retrieved. The columns retrieved are shown in *table 4.5*.

Parameter	*Description*	*Data type*
VOYAGE	Voyage number	String
EDATE	**ETA** (date)	Date
ETIME	**ETA** (time)	Integer
ADRESS	Destination of voyage	String

Table 4.5: Parameters received from VOYTERM table

In most cases this data is self-explaining. It is important to understand a voyage can have multiple destinations, which is often the case. The remaining info missing is the barge assigned to the voyage. This information can be found in another table, called VOYAGE. This table is filtered based on the voyage numbers. The barges are retrieved and linked to the existing data frame *dfVOYAGE* resulting in the final table (see example in *table 4.6)*. The dates and times are once again merged and converted to EDATENUM.

	VOYAGE	EDATE	ETIME	ADDRESS	BARGE	EDATENUM
1	ALF0635	2021-05-25	1250	RWG	ALF	538.83
2	ALF0639	2021-05-29	0425	RWG	ALF	626.42
3	ALF0641	2021-06-02	1700	APMR2	ALF	735.00
4	ALF0641	2021-06-02	1600	RWG	ALF	734.00
5	ALF0643	2021-06-05	0730	APMR	ALF	797.50
6	ALF0645	2021-06-06	1900	APMR2	ALF	833.00
7	ALF0647	2021-06-09	1330	ECTDDE	ALF	899.50
8	ALF0649	2021-06-11	1805	ECTDDE	ALF	952.08
9	ALF0651	2021-06-13	1430	APMR	ALF	996.50
10	ALF0653	2021-06-15	1445	EMX	ALF	1044.75

Table 4.6: Example of dfVOYAGE

4.2.5 Barges

Finally, the parameters for the barges assigned to the voyages have to be retrieved from the database. This data can be found in the table BARGE. Below a description of the retrieved parameters (*table 4.7)*.

Parameter	*Description*	*Data type*
BARGE	Code of the barge	String
DESCR	Full name of the barge	String
MAXTEU	Maximum **TEU** to be loaded on barge	Integer
MAXWEIGHT	Maximum weight to be loaded on barge	Integer
MAX20FT	Maximum amount of 20ft containers	Integer
MAX40FT	Maximum amount of 40ft containers	Integer
MAX45FT	Maximum amount of 45ft containers	Integer

Table 4.7: Parameters received from BARGE

All barges used on the retrieved voyages will be used to execute a query on the BARGE table. All capacities are received from the table so they can be linked to the voyages alter on. Note, *table 4.8* shows an example of *dfBARGE*, where it appears that mostly MAXTEU is the same as the corresponding maximum of a category times its dimensions (row 1, MAXTEU is 1080, MAX20FT is 54, 1080 divided by 20), but as stated in chapter 4.2.3, it does note mean it is just a simple division. Note that some barges are not able to carry 45 ft containers at all.

	BARGE	DESCR	MAXTEU	MAXWEIGHT	MAX20FT	MAX40FT	MAX45FT
1	MOR	MORDOMIA	1080	1702000	54	24	0
2	VIR	VIRAGE	3120	3000000	156	72	52
3	VIC	VICTOR	2080	1200000	104	52	9
4	FAR	FAR YSKAR	1800	1507000	90	45	36
5	ALF	ALFA NERO	1800	1500000	90	45	0
6	MRB	MER-BLUE	1800	1500000	90	45	0
7	SIN	SINCERO	2160	1200000	108	54	0
8	ERC	ERCULANO	1980	1750000	99	45	0
9	ESP	ESPERO	2160	2000000	108	48	0
10	SPE	SPECTER	3120	3000000	156	72	54

Table 4.8: Example of dfBARGE

Note that, MAXTEU is now the max of total foot of containers. As stated in chapter 2.2.1, 1 **TEU** is a twenty-foot equivalent unit. The numbers shown in *figure 4.4* under MAXTEU are actually the maximum **TEU**s times 20, this has been done for convenience.

4.3 (Key) Performance Indicators for the model

To determine the quality of the model and the planning it generates, (key) performance indicators have been assigned. This paragraph will introduce and describe them, later on in chapter 6 they will be used to measure the quality of the results. The key performance indicators (KPI) mentioned below are determined because these enable both *bargeplanning* and *customer service* to control shipments. The performance indicators (PI) provide more planning detail.

Key Performance Indicators (KPI)

- <u># of containers to be moved</u>
  The total amount of containers to be allocated on voyages
- <u># of voyages defined</u>
  The total amount of defined voyages that have **ETA**s in the given horizon
- <u>Percentage of defined voyages used</u>
  The percentage of the total amount of defined voyages which is actually used to move containers according to the outcome of the **ILP**
- <u>Percentage of containers moved on time</u>
  The percentage of the total amount of containers to be allocated on voyages that arrive on time (after opening, before close) according to the outcome of the **ILP**
- <u>Percentage of containers not moved (on time)</u>
  The percentage of the total amount of containers to be allocated on voyages that do not arrive on time (before opening, after close) according to the outcome of the **ILP**

Performance Indicators (PI)

- Booking close dates horizon
  Dates used to filter the bookings used in the model
- Opening dates of containers between
  Horizon of opening dates for containers containing to filtered bookings
- Closing dates of container between
  Horizon of opening dates for containers containing to filtered bookings
- Percentage of close dates of containers between close dates of bookings
  Percentage of the total number of containers to be moved that have a closing date on the same horizon as the one used to filter the bookings
- # of barges used
  Total amount of barges used
- # of containers moved on time
  Total amount of containers moved on time
- # of containers on GHOST voyage
  Total amount of container assigned to the GHOST voyage i.e. not moveable (on time) by defined voyage
- # of containers delivered before opening time
  Number of containers delivered before opening time
- # of containers delivered after closing time
  Number of containers delivered after closing time
- # of defined voyages used
  Number of defined voyages actually used to ship containers according to the outcome of the **ILP**
- Mean used capacity in weight for used voyages
  Mean percentage of used capacity in weight for all voyages that also actually move containers
- Mean used capacity in **TEU** for used voyages
  Mean percentage of used capacity in **TEU** for all voyages that also actually move containers
- Used capacity in weight for most loaded voyage
  Percentage of used capacity in weight for the most loaded used voyage
- Used capacity in **TEU** for most loaded voyage
  Percentage of used capacity in **TEU** for the most loaded used voyage
- Used capacity in weight for least loaded voyage
  Percentage of used capacity in weight for the least loaded (non zero) used voyage
- Used capacity in **TEU** for least loaded voyage
  Percentage of used capacity in **TEU** for the least loaded (non zero) used voyage

4.4 Mathematical model

The **mathematical program** (**ILP**) used in the model will be addressed here. It will be defined and explained in this section. The following sets will be used in the **mathematical program**;

$i \triangleq containers\ to\ be\ moved$

$b \triangleq used\ barges$

$v \triangleq planned\ voyages$

$t \triangleq destination\ terminals$

$c \triangleq category\ of\ containers = \{20ft, 40ft, 45ft\}$

When creating a planning following simple question needs to be answered: which container will be assigned to which voyage? The answer is a little bit more complex though. When defining a **mathematical program**, the first thing to determine is, what are the decision variables and what is the objective function?

The decision variable can be directly related to the main question at hand, allocating containers to voyages. Therefore, the decision variable for the math program has been defined as follows;

$$x_{i,v} \triangleq \begin{cases} 1\ if\ container\ i\ goes\ on\ voyage\ v \\ 0\ if\ otherwise \end{cases}$$

Two more variables, yo_i & yc_i ,are added to the program, they are introduced to indicate whether a container is tardy or not. One for indicating whether the container arrives before the opening date, and one to indicate whether it arrives after closing. Containers should never arrive early or late, but if this is a hard constraint, the program may be infeasible because it could be no voyage is defined yet that can bring the container to its destination on time.

$$yo_i \triangleq \begin{cases} 1\ if\ container\ i\ arives\ before\ opening \\ 0\ if\ otherwise \end{cases}$$

$$yc_i \triangleq \begin{cases} 1\ if\ container\ i\ arives\ after\ closing \\ 0\ if\ otherwise \end{cases}$$

Using the data described in chapter 4.2, the following parameters are derived.

$$W_i \triangleq Weight\ of\ container\ i$$

$$D_{i,t}^{con} \triangleq \begin{cases} 1\ if\ container\ i\ has\ to\ go\ to\ terminal\ t \\ 0\ if\ otherwise \end{cases}$$

$D_{t,v}^{voy} \triangleq \begin{cases} 1 \text{ if terminal } t \text{ is visited by voyage } v \\ 0 \text{ if otherwise} \end{cases}$

$T_i \triangleq Length\ in\ feet\ of\ container\ i\ (TEU * 20)$

$Cat_{i,c} \triangleq \begin{cases} 1 \text{ if container } i \text{ is of category } c \\ 0 \text{ if otherwise} \end{cases}$

$Cw_b \triangleq Capacity\ in\ weight\ of\ tons\ (kg)\ of\ barge\ b$

$Cap_{c,b} \triangleq Capacity\ of\ category\ c\ containers\ on\ barge\ b$

$V_{b,v} \triangleq \begin{cases} 1 \text{ if barge } b \text{ is on voyage } v \\ 0 \text{ if otherwise} \end{cases}$

$A_{t,v} \triangleq Arrival\ datetime\ of\ voyage\ v\ at\ terminal\ t$

$O_{i,t} \triangleq Open\ datetime\ of\ terminal\ t\ for\ container\ i$

$C_{i,t} \triangleq Closing\ datetime\ of\ terminal\ t\ for\ container\ i$

$B_{b,v} \triangleq \begin{cases} 1 \text{ if barge } b \text{ is assigned to voyage } v \\ 0 \text{ if otherwise} \end{cases}$

$Tcap_b \triangleq total\ capacity\ in\ feet\ of\ containers\ of\ barge\ b\ (capacity\ in\ TEU * 20)$

Finally, some parameters must be calculated using the parameters above. This is done to aggregate information about barges and voyages, and to construct other parameters which help defining constraints of the program.

$P_{i,v} \triangleq \begin{cases} 1 \text{ if container } i \text{ has same destination as voyage } v \\ 0 \text{ if otherwise} \end{cases} = D_{i,t}^{con} * D_{t,v}^{voy}$

$Cw_v^{voy} \triangleq Capacity\ in\ weight\ of\ tons\ (kg)\ of\ voyage\ v = Cw_b * B_{b,v}$

$Tcap_v^{voy} \triangleq Total\ capacity\ in\ feet\ of\ containers\ of\ voyage\ v = Tcap_b * B_{b,v}$

$O_{i,v}^{voy} \triangleq Open\ datetime\ for\ container\ i\ on\ voyage\ v = O_{i,t} * D_{t,v}^{voy}$

$C_{i,v}^{voy} \triangleq Close\ datetime\ for\ container\ i\ on\ voyage\ v = C_{i,t} * D_{t,v}^{voy}$

$A_{i,v}^{voy} \triangleq Arival\ time\ for\ container\ i\ if\ on\ voyage\ v = D_{i,t}^{con} * A_{t,v}$

$Cap_{c,v}^{voy} \triangleq Capacity\ of\ category\ c\ containers\ on\ voyage\ v = Cap_{c,b} * B_{b,v}$

With all these parameters, the following **integer linear program** (**ILP**) can be constructed.

$$\min \sum_i (yo_i * M) + (yc_i * M) \quad (1)$$

subject to:

$$\sum_i w_i * x_{i,v} \leq Cw_v^{voy} \; \forall \, v \quad (2)$$

$$\sum_v x_{i,v} * P_{i,v} = 1 \; \forall \, i \quad (3)$$

$$\sum_i Cat_{c,i} * x_{i,v} \leq Cap_{c,v}^{voy} \; \forall v, c \quad (4)$$

$$\sum_i T_i * x_{i,v} \leq Tcap_v^{voy} \; \forall \, v \quad (5)$$

$$\sum_v x_{i,v} = 1 \; \forall \, i \quad (6)$$

$$yo_i \geq \frac{(O_{i,v}^{voy} * x_{i,v}) - (A_{i,v}^{voy} * x_{i,v})}{M} \; \forall i, v \quad (7)$$

$$yc_i \geq \frac{\left(A_{i,v}^{voy} * x_{i,v}\right) - (C_{i,v}^{voy} * x_{i,v})}{M} \; \forall i, v \quad (8)$$

The objective function in (1) is rather simple, because the main objective of the overall model is just to allocate the containers in a way they arrive on time. ‘M’ represents a big number, so the program will try to keep both yo_i and yc_i at 0 if possible.

There are alternative objective functions that could be applied, like one that maximizes the use of capacity of each voyage, this will be addressed in chapter 7.3. Eventually, this objective function was chosen because it is the foremost goal of the model to allocate containers in such a way that they are shipped on time

Constraint (2) secures that each voyage carries no more weight than possible.

Constraint (3) secures that each container is assigned to the right destination.

Constraint (4) secures that the maximum amount of a certain category of container is moved by each voyage is not exceeded.

Constraint (5) is also a capacity constraint, securing that the total amount of **TEU** will not exceed the **TEU** capacity of the voyage.

Constraint (6) secures that each container is assigned to exactly one voyage.

Finally, constraint (7) and (8) make yo_i and yc_i take a value of 1 if the container does not arrive in the open/close window, and 0 if they are on time.

The latter constraint may need some explanation. Since both yo_i and yc_i are forced to be binary values, they are either 1 or 0. When a container has an opening datetime on day 3 of a certain horizon, meaning that $O_{i,t}$ = 4, and it arrives on day 2, meaning $A_{i,v}^{voy}$ = 2, then the upper term of the division will be 4-2 = 2. Clearly, the container is too early (2 days). When this term ($O_{i,t}$- $A_{i,v}^{voy}$ = 2) is divided by a large number, it will take a positive value between 0 and 1, and since the constraint forces it to be larger than the right-hand side, it will be set to 1. When the numbers are the other way around, and thus the subtraction gives a negative outcome, the variable will take on the value of 0 because a value of 1 will give a penalty to the object function. The same principle works on constraint (8).

4.5 Dashboard

The results of the model will be displayed on the dashboard as described before. The dashboard will be rather simple at first but can be extended easily due the fact it relates to the outcomes of the model, with the database of the company, and the output of the tool which extracts information from the web. The setup of the dashboard will initially contains following two tabs:

Allocation tab

This tab will show a table of all containers, and the voyage to which the **ILP** has assigned them. It will also display a table with all the defined voyages, their used capacity and the maximum capacity. The tab will have 7 slicers, which filter the displayed data. All the slicers can be set to 1 or 0 to display all the corresponding containers and voyages that satisfy the criteria, the slicers can also be combined if containers satisfy the conditions.

1. **Yc**
   When set to 1, all containers in the model that have value of 1 for yc_i will be displayed, i.e. all containers that arrive after the closing date according to the outcome of the **linear program**. When set to 0, containers in the model that arrive before the closing datetime will be displayed
2. **Yo**
   When set to 1, all containers in the model that have value of 1 for yo_i will be displayed, i.e. all containers that arrive before the opening date according to the outcome of the **linear program**. When set to 0, containers in the model that arrive after the opening datetime will be displayed
3. **NOCONTAINER**
   When set to 1, all orders that have not yet been assigned to a container in the model will be displayed, when set to 0, all other containers in the model with an defined container (number) will be displayed
4. **NODEST**
   When set to 1, all containers with no defined destination in the model will be displayed, when set to 0, all other containers in the model with a defined destination will be displayed
5. **NOFIRSTDELDATE**
   When set to 1, all containers in the model with no defined opening datetime will be displayed, when set to 0, all containers in the model with a defined opening datetime will be displayed
6. **NODELDATE**
   When set to 1, all containers in the model with no defined closing datetime will be displayed, when set to 0, all containers in the model with a defined closing datetime will be displayed
7. **NOWEIGHT**
   When set to 1, all containers in the model that have no defined weight will be displayed, when set to 0, all containers in the model with a defined weight will be displayed

Occupation tab

The second tab displays the occupancy of the voyages. A slicer has been added where voyages can be selected. The selected voyage will then be displayed using gauges. Five gauges exist displaying the maximum capacity of the chosen voyage, and its occupancy for that type of capacity. The five categories are: **TEU** capacity, weight capacity, 20ft container capacity, 40ft container capacity and 45ft container capacity.

Chapter 5: Implementation & testing

This chapter explains the implementation of the model. It will briefly substantiate the selected software that is used to support this and how parameters are set up for the **mathematical program**. A fictional test set used to test the model will be introduced and explain.

Software is used for the implementation of the model. Software had to be chosen in a way that the company can use the implementation. This way the model can be extended and no additional purchases have to be made by the company (software, solvers etc.) As shown in *figure 4.1* (chapter4.1), different environments are used for different modules of the program. Why will be explained here.

5.1 Parameter building with test set

For retrieving data and build the parameter files **R studio** was selected. This program is very suitable for data acquisition and no license is required (free to use).

The company stores data in an **Oracle** database. This database is linked to **Modailty**, a tool to enter and view data. With R studio, a connection to this database can be established to retrieve data directly. Once imported, parameters can be built based on the retrieved data and will serve as an input for the **mathematical program**.

The different data frames described in chapter 4.2 contain all the information needed for the **mathematical program** in chapter 4.4. However, to translate the information from the data frames to parameters for the **mathematical program**, some necessary steps are required in order to achieve this.

This paragraph will describe how parameters are created after data has been received. A small fictional data set will be used to provide insight in how the parameters look, and how the R script builds these using the data frames.

A set of 3 bookings and 10 containers is introduced, the data frame has the same format as described in chapter 4.2 and is depictured in *table 5.1*;

BOOKING	CNTR	NETT	ADRESS_EXP	UNITTYPE	FIRSTDELDATE	FIRSTDELTIME	FIRSTDELNUM	DELDATE	DELTIME	DELNUM	TEU	CAT
1	i1	100	t1	20hc	1-5-2021	0	44317,00	15-5-2021	0	44331,00	20	20ft
1	i2	200	t2	20hc	3-5-2021	0	44319,00	15-5-2021	0	44331,00	20	20ft
1	i3	300	t3	40hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	40	40ft
2	i4	400	t4	40hc	7-5-2021	0	44323,00	16-5-2021	0	44332,00	40	40ft
2	i5	500	t5	40hc	9-5-2021	0	44325,00	17-5-2021	0	44333,00	40	40ft
2	i6	600	t6	20hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	20	20ft
3	i7	700	t1	20hc	1-5-2021	0	44317,00	15-5-2021	0	44331,00	20	20ft
3	i8	800	t2	40hc	3-5-2021	0	44319,00	15-5-2021	0	44331,00	40	40ft
3	i9	900	t3	40hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	40	40ft
4	i10	1000	t4	45hc	7-5-2021	0	44323,00	16-5-2021	0	44332,00	45	45ft

Table 5.1: dfUNIT test set

The time window equals 3 days (15-5-2021 to 17-5-2021). From this data frame, the smallest opening datetime (1-5-2021 00:00) and the largest closing datetime (17-5-2021 00:00) are retrieved and assigned to variables *tmin* and *tmax*. Based on these values and the addresses (t1 through t6), the voyages in the database with an **ETA** between these dates are retrieved. In the case of the test data set, this are the following 3 voyages (*table 5.2)*;

VOYAGE	EDATE	ETIME	ADRESS	BARGE	EDATENUM
v1	10-5-2021	7	t1	b1	44326,29
v1	10-5-2021	9	t2	b1	44326,38
v2	11-5-2021	7	t3	b2	44327,29
v2	11-5-2021	9	t4	b2	44327,38
v3	12-5-2021	7	t5	b3	44328,29
v3	12-5-2021	9	t6	b3	44328,38

Table 5.2: dfVOYAGE test set

Each voyage visits two terminals, together they visit all the terminals to where the containers must be transported. For example, v1 arrives at the deep seaport on 10-5-2021, first it visits t1 at 07:00, and then it visits t2 at 09:00. Please note the small difference in EDATENUM indicating the time difference. Three voyages are defined, and three barges are assigned to them (b1, b2, b3). So the barges are retrieved to construct *dfBARGE* as depicted in *table 5.3*;

BARGE	DESCR	MAXTEU	MAXWEIGH	MAX20FT	MAX40FT	MAX45FT
b1	Barge1	2200	31000	110	55	48
b2	Barge2	1400	30000	70	35	0
b3	Barge3	1800	32000	90	45	40

Table 5.3: dfBARGE test set

Now that the data frames are retrieved, the parameters can be constructed using this data. Before constructing the parameters the so called GHOST voyage will be explained. This is important.

To counter infeasibility, a fictional voyage is introduced which is not included in the data frames but is included in the parameters and thus the model. When a container can not be moved due to lack of capacity, or because no voyage travels to its destination, the container is assigned to this GHOST voyage. In practice, this means that the container can not be assigned to any voyage. Just like containers that arrive too late or too early, this is a fictional allocation to ensure feasibility of the model, and to flag these containers for the user of the model. The GHOST voyage visits all terminals and will never arrive on time. This way, the container can be assigned if its destination is not visited or if the visiting voyage does not have the capacity for transporting the container. Also, the GHOST voyage has unlimited capacity.

The first parameter build in the script is $B_{b,v}$. This is one of the 2 dimensional binary matrices which links voyages to barges. Using the test data, $B_{b,v}$ will look as follows;

	v1	v2	v3	GHOST
b1	1			
b2		1		
b3			1	
OTHER				1

Table 5.4: Assignment of barges to voyages in matrix form of test set

The R script constructs the matrix, and adds the GHOST voyage to it, note that a fictional barge (OTHER) is created as well.

Next, the destination of the voyages is created. $D_{t,v}^{voy}$ is once again a 2 dimensional binary matrix;

	v1	v2	v3	GHOST
t1	1			1
t2	1			1
t3		1		1
t4		1		1
t5			1	1
t6			1	1

Table 5.5: Destination of the voyages in matrix form of test set

Each voyage visits two terminals, as stated in *table 5.2*. The GHOST voyage visits each terminal to ensure feasibility.

The destination of the containers $D_{i,t}^{con}$ is made up in the same way;

	t1	t2	t3	t4	t5	t6
i1	1					
i2		1				
i3			1			
i4				1		
i5					1	
i6						1
i7	1					
i8		1				
i9			1			
i10				1		

Table 5.6: Destination of the containers i at terminal t in matrix form of test set

Based on $D_{i,t}^{con}$ the open and close datetime parameters can be constructed ($O_{i,t}$ & $C_{i,t}$). This is done by multiplying the corresponding columns in *dfUNIT* with $D_{i,t}^{con}$, resulting in the following matrices;

	t1	t2	t3	t4	t5	t6
i1	44317					
i2		44319				
i3			44321			
i4				44323		
i5					44325	
i6						44321
i7	44317					
i8		44319				
i9			44321			
i10				44323		

Table 5.7: Open time for container I at terminal t of test set

	t1	t2	t3	t4	t5	t6
i1	44331					
i2		44331				
i3			44331			
i4				44332		
i5					44333	
i6						44331
i7	44331					
i8		44331				
i9			44331			
i10				44332		

Table 5.8: Close time for container I at terminal t of test set

With $D_{t,v}^{voy}$ now $A_{t,v}$ can be constructed, the same way as $O_{i,t}$ & $C_{i,t}$ by multiplying $D_{t,v}^{voy}$ with the EDATENUM column of *dfVOYAGE*;

	v1	v2	v3	GHOST
t1	44326,29			-10
t2	44326,38			-10
t3		44327,29		-10
t4		44327,38		-10
t5			44328,29	-10
t6			44328,38	-10

Table 5.9: Arrival time at terminal t of voyage v of test set

Note the ghost voyage is added to the matrix. The numeric arrival value at each terminal is set to -10. This means, that it will arrive everywhere and too early. This is secured by converting datetimes numeric values in such a way that the value always is bigger than 1.

Next up, the capacity per category of container of each barge $Cap_{c,b}$ is written as a matrix;

cap	b1	b2	b3	OTHER
20hc	110	70	90	1E+99
40hc	55	35	45	1E+99
45hc	48	0	40	1E+99

Table 5.10: Capacity of category c of barge b of test set

Note barge OTHER which is assigned to the GHOST voyage has a verry large capacity, virtually infinite. This is because containers that can not be moved are assigned to this voyage. If the capacity is too low, the problem will still be infeasible when too many containers would be assigned to this GHOST voyage.

The last parameters that will be made in the R environment are two of the so-called calculated parameters described in chapter 4.4. The capacity of both weight Cw_v^{voy} and total **TEU** $Tcap_v^{voy}$ of the voyages, based on the corresponding barges;

VOYAGE	MAXWEIGHT	MAXTEU
v1	31000	2200
v2	30000	1400
v3	32000	1800
GHOST	1E+99	1E+99

Table 5.11: Capacities of voyages in weight and total **TEU** *of test set*

Once again, the GHOST voyage has large capacity because it must be able to take all containers which can not be assigned to a voyage.

Once all parameters have been constructed, they are being exported to an .xlsx file from where they will be ready to serve as input for the implementation of the **ILP**. All other non-calculated parameters, such as the weight of each container (W_i) for example, that are not further mentioned in this paragraph, are derived from the original three data frames. The rest of the calculated parameters such as all the possible voyages for the containers ($P_{i,v}$) are calculated in the environment where the **ILP** is solved.

5.2 Solving the ILP

Once the input for the **ILP** is ready, it is imported in the environment where the **ILP** is defined. The implementation of the **ILP** has been written in Python, because it is a well known and easy to use language. The implementation in Python also contributes to the maintainability of the **ILP** and therefore the model. Other programs where considered (see chapter 7.2) but Python got selected.

Coding **linear programs** in Python can be done in several ways. The current implementation uses the PuLP package. This package supports several solvers, some are included and some can be imported. Not all solvers come with a free license. The current implementation uses default solvers which are included in the PuLP package, or were free to import. This paragraph will describe briefly the steps made in the Python script.

1. To initiate the program, some packages are imported including PuLP. These contain functions needed for the script to run and are pre downloaded and installed. After that, the **ILP** is initiated, this includes nothing more than defining a name and stating it is a minimalization problem.
2. The next step is importing the data written to the .xlsx input file from R studio. The initial data frames and the matrices are all imported into Python. The classes are being derived and defined from them, and some mutations are applied to make sure all the parameters have the same form.
3. When all data and parameters have the right form, type and class for Python to work with, the remaining calculated parameters are being defined. This are the possible destinations ($P_{i,v}$), Open and close datetimes for the voyages ($O_{i,v}^{voy}$ & $C_{i,v}^{voy}$), arrival datetimes for the voyages ($A_{i,v}^{voy}$) and the categorial capacity of the voyages ($Cap_{c,v}^{voy}$).
4. With all the data complete in the Python environment, now the decision variables $x_{i,v}$, yo_i and yc_i are defined. All the constraints and the object function are defined and added to the prior defined **ILP**.

Now the problem is complete and ready to solve. As it is, two solvers can be chosen to solve the problem. These are two default solvers for the PuLP package. Which one is used does not matter for the results of this research, the outcomes where exactly the same so was the computing time. When this is done and the program is executed, it will display all nonzero variable values in the python console and writes them to a .xlsx file.

5.3 Display of output

Displaying all the data is preferably done on a dashboard as discussed in chapter 3.3. Prior to this research, the common way to look at data, besides using **Modailty**, was via **PowerBI**. The company has adapted to it's environment, and a link between the oracle database and **PowerBI** was already established. It is being used for several other dashboards already, so displaying the output (and input) of the **mathematical program** in this environment seems logical. Another advantage is that it can then be linked to the Oracle database for comparison.

As far as implementation goes, a dashboard has not yet been designed. The choice of saving everything to .xlsx files enables easy centralizing and linking to the **PowerBI** environment. The setup of the dashboard and what to display has been worked out.

Before solving the problem, the build parameters should be displayed on the dashboard so potentially missing values or incorrect data can be spotted immediately. This could be anything from a container with no weight, or a closing date that lies in the past, i.e. errors that have been made while processing the booking. The idea is to flag these errors so they can easily be filtered on the dashboard and addressed by anyone checking the export.

The result of the **mathematical program** is available via another tab. Which container is assigned to which voyage, how much capacity of each voyage is used, and which containers can not be shipped or are threatened to be tardy?

Because the dashboard will be linked to the Oracle database, underlying references can be made with the database. For instance, if a container has missing or incorrect values, which are not known to the company, the containers that cause errors can be linked to their bookings. Via these bookings the customer's name, contact person and telephone number can be tracked. This enables *customer service* to quickly decide what to do. No time is lost anymore by look up information such as the contact person's telephone number.

One last important aspect of the dashboard is the web input. Simultaneously with this research, another internship was occupied at the company, delivering a tool which scrapes the websites of the big, most frequently visited terminals in Rotterdam port. The tool used is **Octoparse** which essentially scrapes all the last know open and close datetimes for deep sea vessels at terminals from their corresponding websites. These can be displayed on the dashboard as well if they differ with the original processed and used open and closing date. When confirmed, these than can easily be modified in the original booking, which means they are changed in the Oracle database and therefore taken into account with the model.

Chapter 6: Results

This chapter will address the results of the model. First the priorly addressed test set will be used to show test results and how to interpretate these. When completed, three sets with real data will be retrieved from the oracle database and used to the test model. The differences between the test and real data set will be discussed and interpretated.

6.1 Test data

The results of the test data set introduced in chapter 5 will be shown and disused here. First the raw output from the **mathematical program** will be shown, then a brief explanation of the data transformation for the dashboard and finally a sketch of how the results should be shown on the dashboard.

When running the Python file, it is programmed to only return non zero values, i.e. only the variables that are significant. Using the test data, the python script returns the following code;

```
x_var_('i1',_'v1') = 1
x_var_('i10',_'GHOST') = 1
x_var_('i2',_'v1') = 1
x_var_('i3',_'v2') = 1
x_var_('i4',_'v2') = 1
x_var_('i5',_'v3') = 1
x_var_('i6',_'v3') = 1
x_var_('i7',_'v1') = 1
x_var_('i8',_'v1') = 1
x_var_('i9',_'v2') = 1
yo_var_i10 = 1
1
```

Figure 6.1: Raw output from python console for **mathematical program** *of test set*

The lines shown are all non zero values, the top line (*x_var_('i1',_'v1') = 1*) should be interpretated as $x_{i1,v1} = 1$. Which means, that container *i1* is assigned to voyage *v1*. On the second line, it states that container *i10* is assigned to the GHOST voyage, stating it cannot be shipped by any planned voyage. This conclusion can also be derived from the fact that $yo_{i10} = 1$, stating that the container arrives before its opening date. This is because the GHOST voyage always arrives too early. On the bottom line, the problem status is printed. When 1, the optimal solution has been reached, when -1 the problem is infeasible.

Container *i10* is the only 45ft container in *dfUNIT*, and its destination is terminal *t4* as can be seen in *table 5.1*. Furthermore, *table 5.2* states that voyage *v2* is the only voyage going to this destination and is using barge *b2*, and *table 5.3* states that barge *b2* has no capacity for 45ft containers. Therefore container *i10* cannot be moved by voyage *v1*, *v2* or *v3*. Because of the (practical) infinite capacity of the GHOST voyage, and because it sails to every destination terminal in *table 5.1*, container *i10* is assigned to the GHOST voyage. And Because the GHOST voyage arrives never on time, yo_{i10} takes on a value of 1.

The results are exported to the .csv file (*MathProgOut.csv*), which includes all variables, zero and one. This is a raw output file similar *figure 6.1*. To prepare the data for interpretation, a R script transforms the data and creates two tables; one stating the allocation of each container to the voyages, and one which takes all the capacities for the voyages from the original data frames and adds, according to the allocation and the parameters from the original data frames, the total weight, **TEU**, and number of containers per category per container for each voyage;

CNTR	VOYAGE	yo	yc
i1	v1	0	0
i10	GHOST	1	0
i2	v1	0	0
i3	v2	0	0
i4	v2	0	0
i5	v3	0	0
i6	v3	0	0
i7	v1	0	0
i8	v1	0	0
i9	v2	0	0

Table 6.1: Allocation and variable values for each container of test set

VOYAGE	MAXWEIGHT	MAXTEU	totWeight	totTEU	BARGE	max20ft	max40ft	max45ft	tot20ft	tot40ft	tot45ft
v1	31000	2200	1800	100	b1	110	55	48	3	1	0
v2	30000	1400	1600	120	b2	70	35	0	0	3	0
v3	32000	1800	1100	60	b3	90	45	40	1	1	0
GHOST	1E+15	1E+15	1000	45	OTHER	0	0	0	0	0	1

Table 6.2: Capacity and occupation of the voyages of test set

This output serves as input for the **PowerBI** dashboard, where constraint checks can be done visually and *bargeplanning* and *customer service* have a more elegant view of the data. A screenshot of the dashboard (sketch) for the occupancy tab can be seen in *figure 6.2*.

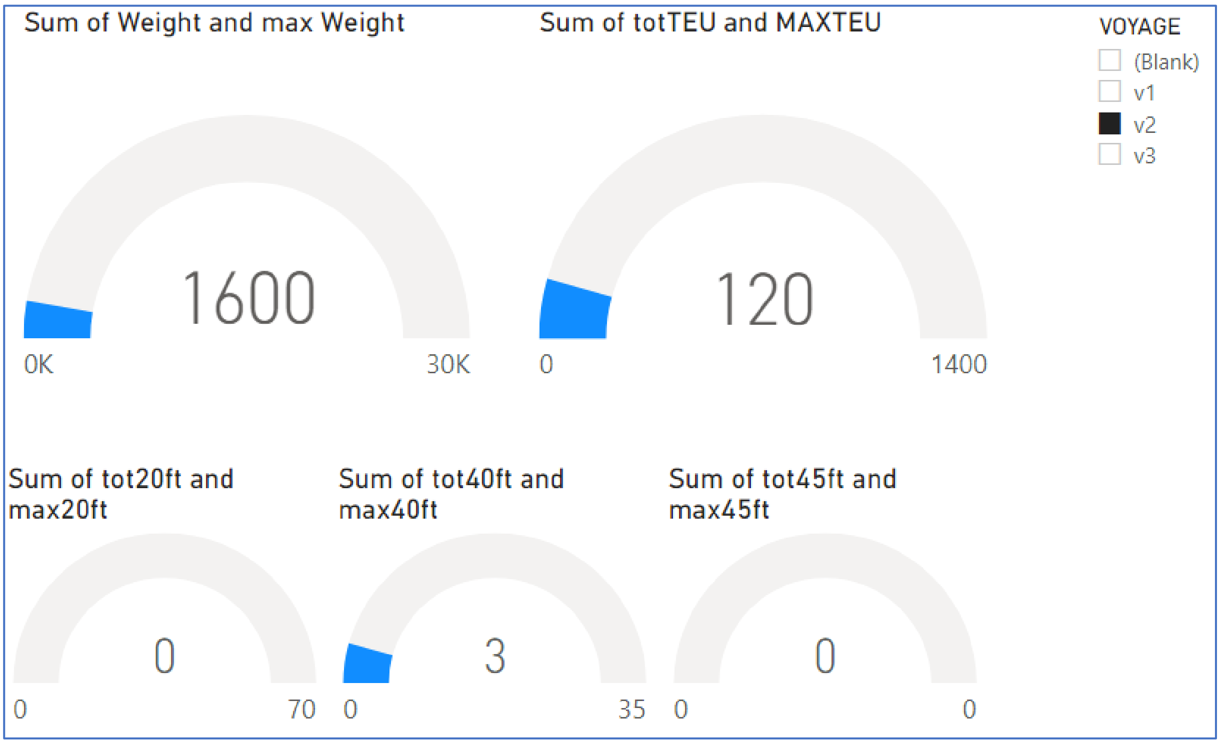


*Figure 6.2: Sketch of interactive dashboard in **PowerBI**, occupancy tab*

In the right top corner, voyages to be reviewed can be selected. In *figure 6.2* voyage *v2* is selected. The dashboard then shows the maximum capacity of all five KPIs (Weight, total **TEU** * 20, 20ft, 40ft and 45ft containers) and how much is currently loaded, according to the outcome of the **mathematical program**, on each voyage. When no selection is made, the grand total over all voyages is shown.

A screenshot of the allocation tab is depicted in *figure 6.3:*

CNTR	VOYAGE
i1	v1
i10	GHOST
i2	v1
i3	v2
i4	v2
i5	v3
i6	v3
i7	v1
i8	v1
i9	v2

VOYAGE	MAXTEU	totTEU	MAXWEIGHT	totWeight
v1	2200	100	31000	1800
v2	1400	120	30000	1600
v3	1800	60	32000	1100

VOYAGE	max20ft	tot20ft	max40ft	tot40ft	max45ft	tot45ft
v1	110	3	55	1	48	0
v2	70	0	35	3	0	0
v3	90	1	45	1	40	0

*Figure 6.3: Sketch of interactive dashboard for test data set in **PowerBI**, allocation tab of test set*

Note that *figure 6.3* not yet contains the filters described in 5.3. The dashboard for the real data set do contain this option.

6.2 Real data

In this paragraph real data will be used to test the model. The first set is the smallest and contains data from the past. The second set is a bigger containing future plan. The third set is created to match reality, a model that can plan a week ahead, and the fourth data set is an alternative of the third data set, using different filters.

The first real dataset will be analyzed in detail, the two bigger datasets will only show the results. All four datasets, and their corresponding input and output can be found in the attachments of this study (*RealDataSet1, RealDataSet2, RealDataSet3, RealDataSet4*). KPIs of the output will be presented as well.

6.2.1 Real data set 1

For the first real data set a horizon of two days was set (2021-06-23 & 2021-06-24). The bookings were filtered using these dates. When the real data set was used, it was discovered that the open and close datetime of a booking is not always the same as its individual containers. The containers to be moved need to be shipped between 2021-06-21 and 2021-07-01.

The voyages are still filtered on these dates, so the problem is feasible. More on this topic can be found in the discussion chapter of this research (chapter 7.2). When extracting the data, 153 containers, 80 voyages and 9 barges were retrieved from the database. During executing the query, these dates lied in the past. This way the program is a bit easier to solve, because voyages are certainly defined already. Screenshots of the dashboard can be found in the attachments (*figure 6.4 to figure 6.11 and table 6.4/6.5)*.

Figure 6.5 shows the container that arrive after closing time, this means no voyage will arrive on time at the terminal. When checking the container in *dfUNIT*, it's clear the container has to go to terminal

APMR, and its opening and closing datetimes are respectively 1198 and 1233 (numeric values). When checking *dfVOYAGE*, several voyages travel to this destination (*table 6.4*);

As shown in the far-right column of *table 6.4*, none of these voyages will arrive in the given open and close datetime, therefore, the container can not be delivered on time. It is assigned to voyage SIN432, but in practice this means nothing. It is flagged and therefore it cannot be shipped, the voyages must be changed, or the container should be shipped by truck according to the data.

Figure 6.6 shows all containers that arrive too early, in this case, all the containers are assigned to the GHOST voyage meaning they cannot be shipped by any defined voyage. Container 200285A has no opening and closing times and must go to terminal ECTDDN, where no voyage sails. Container 202362B and TRIU 865109 0A also have this destination and therefore are assigned to the GHOST voyage. Container DFSU 126544 0C also has no open and close datetime, same as container TRLU 90585 8I and container EUTU 272626 8E. Because of this, they are all assigned to the GHOST voyage, and because this voyage always arrives too early, yo_i is always 1.

Figure 6.7 shows the containers that have not yet been assigned to a container. This sounds weird, but as described in chapter 4.2.3, once a booking is made, it can occur everything is known about the container that is required, but no real container has been assigned yet. The net weight, open and close datetime, type of container (20ft, 40ft, 45ft) and the destination may be known, but no empty container has been assigned yet.

This however is not necessary to still plan the container to a voyage since its parameters are known. It is useful though to flag these containers, so *customer service* can have an easy overview of incomplete bookings. In this case, the first container of booking 200285 and the second container of booking 202362 are unknown, but they are already assigned to a voyage.

The containers in *figure 6.8* have no destination terminal. When R detects a NA value in the ADDRESS_EXP column of *dfUNIT*, the script replaces this with UNKNOWN. The GHOST voyage is the only voyage sailing to UNKOWN, therefore they are always assigned to the GHOST voyage and always flagged. These are already flagged by yo_i as can be seen in *figure 6.6*.

The containers shown in *figure 6.9* have no closing date, therefore they are assigned to the GHOST voyage and yo_i is therefore also set to 1.

The containers shown in *figure 6.10* have no opening date. The same containers pop up, however, one container (not assigned to the GHOST voyage) is also shown. Yet it has been assigned to a voyage. This is because the closing time of container CSNU 77886 7A is known, which is 1435 (2021-07-01 21:00). Voyage VIC1415 travels to the destination for container CSNU 77886 7A (EMX) and arrives at 1101,25 (2021-06-17 at 23:00). So, it will be delivered before the closing, however, it is important to flag this. If the opening lies after 2021-06-17 at 23:00, the container will be delivered before opening and therefore can not be shipped.

By checking the input from the web (**Octoparse**), or by reviewing the original booking, the opening date can be determined, processed, and the model can be run again to allocate the container to the right voyage.

Notice in *figure 6.4*, that there is no option to select the NOWEIGHT containers, this simply means all containers have a defined weight and therefore no new checking is needed. Also notice that the filters can be combined, so for instance selecting al the containers arriving before opening (*figure 6.6*) also allows an option to select all the containers that have no DELDATE. When the option is not shown (like with NOWEIGHT), none of the filtered containers share the condition.

The second tab of the dashboard allows the user to check the occupancy of all voyages according to the allocation given by the math program. *Figure 6.11* shows the occupation for voyage MRB0467

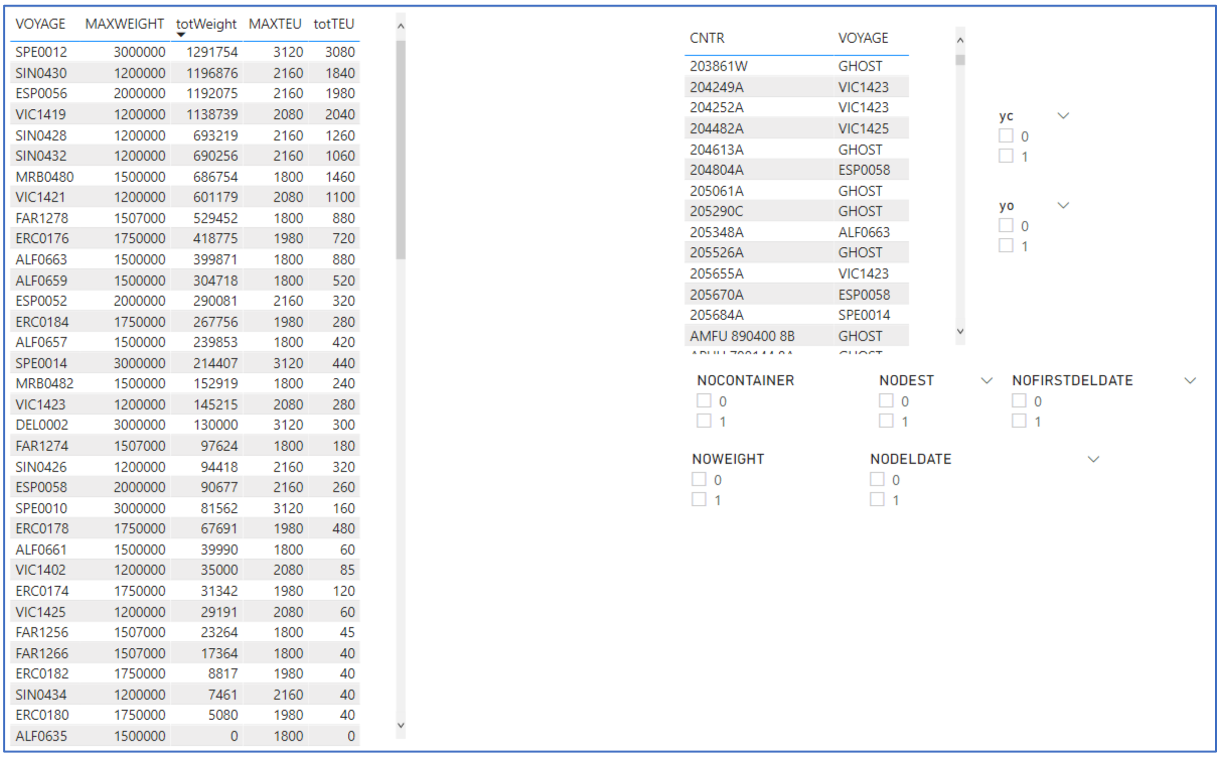
Using this tab, each voyage can be reviewed quickly. This voyage ships 57 **TEU** (1140/20), 650.000 KG of goods, 1 twenty ft container (1**TEU**), and 28 forty ft containers (2**TEU**). There is no capacity for 45 ft containers. A summary of (K)PIs of the results for this data set are shown in *table 6.5*;

KPI Real data set 1	**Value**	
Booking close dates horizon	2021-06-23	2021-06-24
Opening dates of containers between	2021-06-09	2021-07-01
Closing dates of container between	2021-06-21	2021-07-07
Percentage of close dates of containers between close date horizon of bookings	89,86%	
# of containers to be moved	152	
# of voyages defined	79	
# of barges used	8	
# of containers moved on time	145	
# of containers on GHOST voyage	6	
# of containers delivered before opening time	6	
# of containers delivered after closing time	1	
# of defined voyages used	23	
Mean used capacity in weight for used voyages	3%	
Mean used capacity in **TEU** for used voyages	6%	
Used capacity in weight for most loaded voyage	43%	
Used capacity in **TEU** for most loaded voyage	63%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	2%	
Percentage of defined voyages used	29%	
Percentage of containers moved on time	95%	
Percentage of containers not moved (on time)	5%	

Table 6.5: KPIs for allocation of containers in real data set 1

6.2.2 Real data set 2

The second data set used contains 667 containers, 144 voyages and 11 barges. All containers must be shipped between 2021-06-22 and 2021-07-08. Initially, the horizon for the closing dates of the bookings was set from 2021-06-24 to 2021-06-28. The **ILP** was solved and all containers where assigned. The tabs of the dashboard are depicted in the *figure 6.12* and *figure 6.13*;



VOYAGE	MAXWEIGHT	totWeight	MAXTEU	totTEU
SPE0012	3000000	1291754	3120	3080
SIN0430	1200000	1196876	2160	1840
ESP0056	2000000	1192075	2160	1980
VIC1419	1200000	1138739	2080	2040
SIN0428	1200000	693219	2160	1260
SIN0432	1200000	690256	2160	1060
MRB0480	1500000	686754	1800	1460
VIC1421	1200000	601179	2080	1100
FAR1278	1507000	529452	1800	880
ERC0176	1750000	418775	1980	720
ALF0663	1500000	399871	1800	880
ALF0659	1500000	304718	1800	520
ESP0052	2000000	290081	2160	320
ERC0184	1750000	267756	1980	280
ALF0657	1500000	239853	1800	420
SPE0014	3000000	214407	3120	440
MRB0482	1500000	152919	1800	240
VIC1423	1200000	145215	2080	280
DEL0002	3000000	130000	3120	300
FAR1274	1507000	97624	1800	180
SIN0426	1200000	94418	2160	320
ESP0058	2000000	90677	2160	260
SPE0010	3000000	81562	3120	160
ERC0178	1750000	67691	1980	480
ALF0661	1500000	39990	1800	60
VIC1402	1200000	35000	2080	85
ERC0174	1750000	31342	1980	120
VIC1425	1200000	29191	2080	60
FAR1256	1507000	23264	1800	45
FAR1266	1507000	17364	1800	40
ERC0182	1750000	8817	1980	40
SIN0434	1200000	7461	2160	40
ERC0180	1750000	5080	1980	40
ALF0635	1500000	0	1800	0

CNTR	VOYAGE
203861W	GHOST
204249A	VIC1423
204252A	VIC1423
204482A	VIC1425
204613A	GHOST
204804A	ESP0058
205061A	GHOST
205290C	GHOST
205348A	ALF0663
205526A	GHOST
205655A	VIC1423
205670A	ESP0058
205684A	SPE0014
AMFU 890400 8B	GHOST

Figure 6.12: Allocation tab of the interactive dashboard for real data set 2

Figure 6.13: Occupation tab of the interactive dashboard for real data set 2

As can be seen in *figure 6.13*, the selected voyage is almost at full capacity. Although it only ships half the weight it can carry, it is almost at its maximum **TEU** capacity. It can ship 72 40ft containers, which happens. The KPIs for real data set 2 are shown in *table 6.6*;

KPI Real data set 2	**Value**	
Booking close dates horizon	2021-06-24	2021-06-28
Opening dates of containers between	2021-05-24	2021-07-02
Closing dates of container between	2021-06-22	2021-07-08
Percentage of close dates of containers between close date horizon of bookings	67,33%	
# of containers to be moved	668	
# of voyages defined	144	
# of barges used	11	
# of containers moved on time	579	
# of containers on GHOST voyage	82	
# of containers delivered before opening time	85	
# of containers delivered after closing time	3	
# of defined voyages used	33	
Mean used capacity in weight for used voyages	9%	
Mean used capacity in **TEU** for used voyages	15%	
Used capacity in weight for most loaded voyage	100%	
Used capacity in **TEU** for most loaded voyage	85%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	2%	
Percentage of defined voyages used	23%	
Percentage of containers moved on time	86,68%	
Percentage of containers not moved (on time)	13,32%	

Table 6.6: KPIs for allocation of containers in real data set 2

6.2.3 Real data set 3

The third real data set is made to match the real-life scenario best; a schedule for one week ahead in the future. On the day the query was made on the database, the day was 2021-06-24. The horizon on which the closing datetimes of the bookings are filtered starts 2021-06-25 and ends 2021-07-02. The number of containers retrieved from the database is 1101, 160 voyages have been defined and 12 barges are used. All these, and other KPI's are shown in *table 6.7*;

KPI Real data set 3	**Value**	
Booking close dates horizon	2021-06-25	2021-07-02
Opening dates of containers between	2021-05-21	2021-09-26
Closing dates of container between	2021-06-22	2021-07-16
Percentage of close dates of containers between close date of horizon bookings	98,53%	
# of containers to be moved	1101	
# of voyages defined	160	
# of barges used	12	
# of containers moved on time	951	
# of containers on GHOST voyage	136	
# of containers delivered before opening time	146	
# of containers delivered after closing time	4	
# of defined voyages used	47	
Mean used capacity in weight for used voyages	7%	
Mean used capacity in **TEU** for used voyages	15%	
Used capacity in weight for most loaded voyage	98%	
Used capacity in **TEU** for most loaded voyage	100%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	0%	
Percentage of defined voyages used	29%	
Percentage of containers moved on time	86%	
Percentage of containers not moved (on time)	14%	

Table 6.7: KPIs for allocation of containers in real data set 3

T*able 6.7 shows* that the largest opening date is significantly far away from the plan horizon (2021-09-26) This is an error, since the opening date for the specific container lies before this date. As can be seen, most containers have their closing date in the same horizon as the booking closing dates. The percentage of moved containers is rather good, but in all real data sets, the used voyages are rather low. Therefore, one more real data set is introduced.

It is the same as real data set 3, but this time the filtering of the plan horizon will be done directly on the closing dates in *dfUNIT* rather than on the closing date of the bookings to see if any improvements may occur.

6.2.4 Real data set 4

To test whether it may be better to filter directly on the closing dates of the individual containers, rather than on the closing date as stated in the corresponding booking, the same query as real data set 3 is used to generate real data set 4. Only difference is that *dfBOOKING* is skipped and the horizon is directly filtered on *dfUNIT*.

This way, the window of closing dates get smaller, i.e. the same as the given horizon, and therefore less insignificant voyages should be retrieved. When only a couple of containers have a closing date that signifyingly differ from the plan horizon, all voyages with an **ETA** on the tail section of the horizon will be retrieved. Therefore, many voyages will be retrieved to potentially ship only a few containers. By filtering directly on the closing date of the containers, this may be countered. The results are shown in *table 6.8;*

KPI	**Value**	
Booking close dates horizon	-	-
Opening dates of containers between	2021-05-21	2021-09-26
Closing dates of container between	2021-06-25	2021-07-02
Percentage of close dates of containers between close dates of bookings	-	
# of containers to be moved	1093	
# of voyages defined	164	
# of barges used	13	
# of containers moved on time	986	
# of containers on GHOST voyage	59	
# of containers delivered before opening time	99	
# of containers delivered after closing time	7	
# of defined voyages used	69	
Mean used capacity in weight for used voyages	6%	
Mean used capacity in **TEU** for used voyages	11%	
Used capacity in weight for most loaded voyage	100%	
Used capacity in **TEU** for most loaded voyage	100%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	0%	
Percentage of defined voyages used	42%	
Percentage of containers moved on time	90%	
Percentage of containers not moved (on time)	10%	

Table 6.8: KPIs for allocation of containers in real data set 4

Chapter 7: Conclusions and recommendations

Chapter 7 will focus on conclusions based on the results from chapter 6. Furthermore, the quotative solutions to the problem will be formulated as a qualitative contribution to the research questions and the problem in general. This chapter will also include a discussion and will address the possible extensions for this model. At the end of this chapter an advice will be formulated for the company

7.1 Review and comparison of the results

This paragraph will review and analyze the results from chapter 6.2. A comparison of the KPIs for all the data set is shown in *table 7.1*;, the complete table with all the point indicators can be found in the attachments of this report.

KPI	Data set 1		Data set 2		Data set 3		Data set 4	
Booking close dates horizon	23-6-2021	24-6-2021	24-6-2021	28-6-2021	25-6-2021	2-7-2021	-	-
Opening dates of containers between	9-6-2021	1-7-2021	24-5-2021	2-7-2021	21-5-2021	26-9-2021	21-5-2021	26-9-2021
Closing dates of container between	21-6-2021	7-7-2021	22-6-2021	8-7-2021	22-6-2021	16-7-2021	25-6-2021	2-7-2021
# of containers to be moved	152		668		1101		1093	
# of voyages defined	79		144		160		164	
Percentage of defined voyages used	29%		23%		29%		42%	
Percentage of containers moved on time	95%		86,68%		86%		90%	
Percentage of containers not moved (on time)	5%		13,32%		14%		10%	

Table 7.1 (only KPIs): Comparison of KPI's for different datasets

In *table 7.1*, some important KPIs have been marked to indicate which set performs best and worst. Data set 1 moves the most containers on time, this is somehow predictable because this is the only dataset from the past. Overall, the percentage of containers moved on time do not have a worse score. With an average of 89% the conclusion can be drawn that most containers can be moved on time with the defined voyages. Foremost, all **mathematical program**s are solved optimally, thus for each set there is no better allocation with the given data.

The mean used capacity for used voyages, both in weight and **TEU** is rather low for all data sets. The following figures display the distribution of used capacity of weight of all used voyages for all data sets;

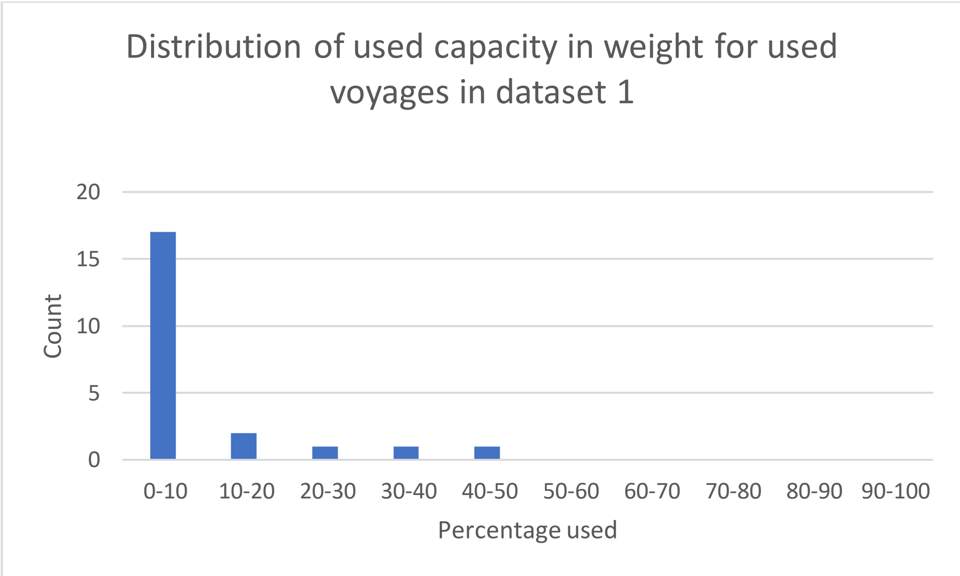


Figure 7.1: Distribution of used capacity in weight for used voyages in dataset 1

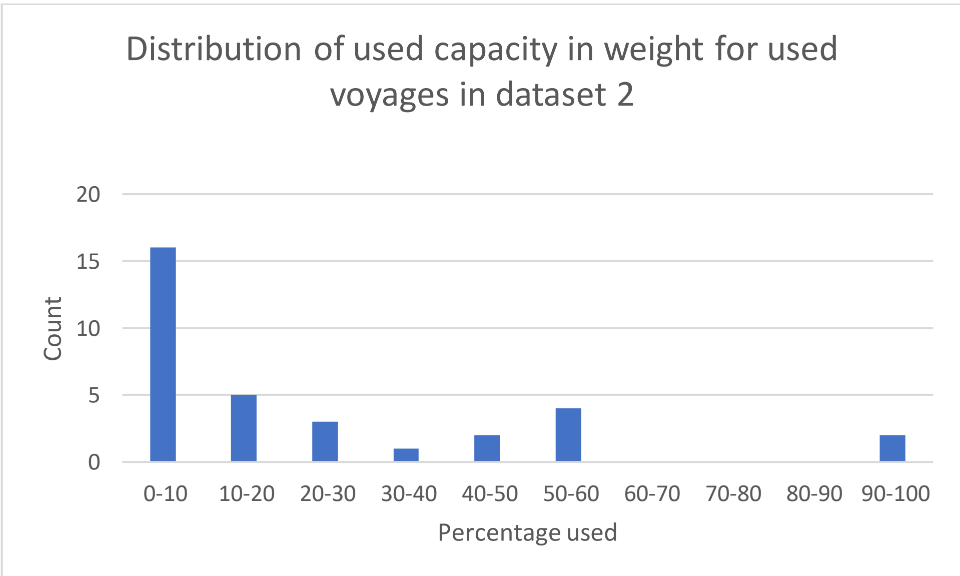


Figure 7.2: Distribution of used capacity in weight for used voyages in dataset 2

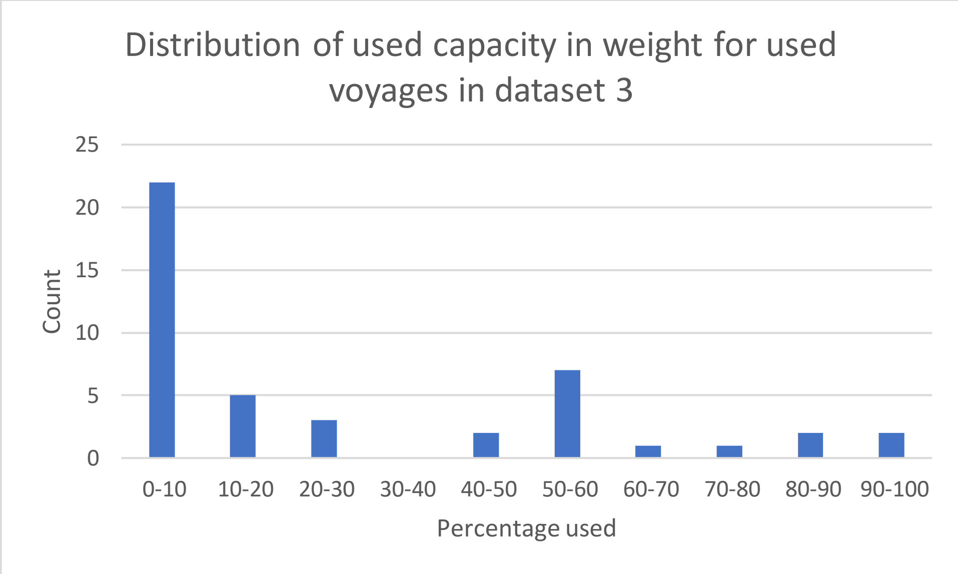


Figure 7.3: Distribution of used capacity in weight for used voyages in dataset 3

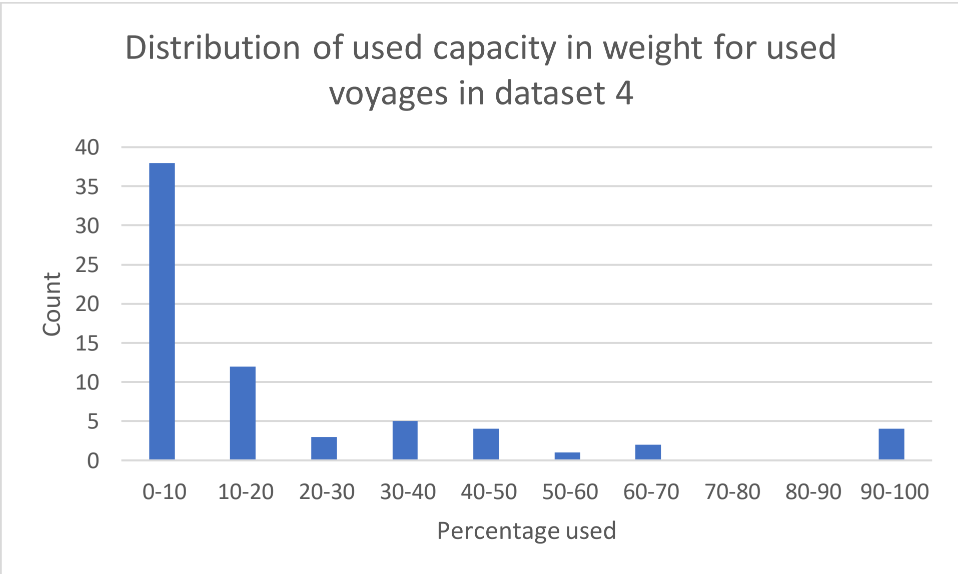


Figure 7.4: Distribution of used capacity in weight for used voyages in dataset 4

It can clearly be seen, that in all datasets most of the used voyages use less then 10% of the capacity.

Because this research only focusses on export from Veghel to Rotterdam, this result is not perse bad. Voyages are defined to also pick up containers at Cuijck and Oss. Also, there must be room for empty containers. The objective function could be extended in such a way that it maximizes the used capacity of each barge. But the main goal for the model is to allocate the containers. And because export from Veghel to Rotterdam does not include everything that should be moved on the voyages, maybe it is good to leave room on the voyages.

Table 7.2 compares the outcomes for real data set 3 and 4, because this is the same data with another filter it is good to compare the two separately;

KPI Real data set 3 vs 4	**Value set 3**		**Value set 4**	
Booking close dates horizon	2021-06-25	2021-07-02	-	-
Opening dates of containers between	2021-05-21	2021-09-26	2021-05-21	2021-09-26
Closing dates of container between	2021-06-22	2021-07-16	2021-06-25	2021-07-02
Percentage of close dates of containers between close dates of bookings	98,53%		-	
# of containers to be moved	1101		1093	
# of voyages defined	160		164	
# of barges used	12		13	
# of containers moved on time	951		986	
# of containers on GHOST voyage	136		59	
# of containers delivered before opening time	146		99	
# of containers delivered after closing time	4		7	
# of defined voyages used	47		69	
Mean used capacity in weight for used voyages	7%		6%	
Mean used capacity in **TEU** for used voyages	15%		11%	
Used capacity in weight for most loaded voyage	98%		100%	
Used capacity in **TEU** for most loaded voyage	100%		100%	
Used capacity in weight for least loaded voyage	0%		0%	
Used capacity in **TEU** for least loaded voyage	0%		0%	
Percentage of defined voyages used	29%		42%	
Percentage of containers moved on time	86%		90%	
Percentage of containers not moved (on time)	14%		10%	

Table 7.2: Comparison of KPIs for allocation of containers in real data set 3 & 4

At first, it looks like the statistics in data set 4 are slightly better than the ones of data set 3. Note that more voyages have been retrieved, this is due the fact there was some time between the queries. While the model was processing real data set 3, the data probably already changed inside the oracle database. For this comparison it should not be a problem, the main reason was to check whether more voyages would be used, not whether less would be retrieved. And it seems that the alternative model indeed is more efficient.

Most KPIs score better for the dataset filtered on only the containers instead of the bookings. And as can be seen in both *table 7.1* & *table 7.2*, the percentage of used voyages is by far the best. Data set 2 has the least percentage (67,33%) of the closing dates of containers within the closing dates of the bookings. This is probably also why it uses the least defined voyages (23%).

7.2 Contribution to answering the problem

In this paragraph the research questions will be answered briefly based the results of this research

How can the current situation be made transparent?

Maps of the important processes have been made and employees of the company have been interviewed to determine the requirements of a dashboard. Now, the company has to monitor different parameters, such as changing open and closing datetimes, from different data sources like the web. Data from the most frequently visited terminals is centralized in the model presented by this research. The company is a large company and still growing. A lot of data is available but not always used to its full potential. Schedules are created and changed continuously to make everything work, but this could be done more efficient.

Which factors/parameters are of influence at the inland date?

Many factors that can affect the **inland date** are unpredictable, such as the weather or a delay of a **deep-sea vessel**. Other parameters that affect the inland date can be monitored, such as open and closing datetimes. If a booking is not complete or no suitable voyage is scheduled on time, this could result in the container cannot be shipped on time. The biggest reason why **inland date***s* are not met is due to continuous changing open and closing datetimes.

What conditions must be taken into account when making a barge planning?

Capacity, closing/opening datetimes and destinations are the most obvious. Also the number of agreed possible moves at a terminal are important (chapter 2.1.7). These moves have not yet been implemented as a constraint in this model, but could be added in the future (chapter 7.4).

How is the quality of a planning determined?

KPIs have been defined to measure the quality of a planning (chapter 6). The most obvious one is the number of containers that is actually moved on time. Also different occupancy parameters will be calculated to say something about the quality of a planning.

How can a quantitative model be made to support/improve barge planning?

A connection to the database was established and an **integer linear program (ILP)** has been designed, created and implemented, providing a base solution for the export voyages from Veghel to Rotterdam. The model could be further extended, it does not yet consider import, moves at terminals, or other **inland terminal**s/deep seaports, as agreed upon. However it will resolve the issue of 'manual' planning and therefor support a part of the proves *bargeplanning* is working on daily.

How can the results and consequences be presented transparent for *customer service*

By displaying the results of the **mathematical program** and by combining these with the database, containers that threaten to go out of control are flagged and can be addressed quickly. A dashboard to display all the important KPIs and real time open and close datetimes pulled from the web to compare them with the initial open and close datetimes when the booking was created.

How can the process of planning barges be supported, and customer service be helped with insights, given (real time) data, in such a way that the company is more in control of its containers and thus inland dates?

With the combination of the **ILP** and the dashboard, a model has been created that serves both *bargeplanning* and *customer service*. By automating a part of the job for *bargeplanning*, the division will gain more time and less stress related to problem-solving. They can allocate containers to voyages in a later stadium, resulting in a more efficient transport of the containers and less truck transport which means less costs and more happy customers. When containers are allocated automatically, the *bargeplanning* can focus on containers that are harder to transport or which endure any kind of trouble to transport. Therefore, they will be more in control of the average container and **inland date**. A base solution of the planning problem is given by the model in this research. This base solution can be altered, but the easy to move containers will be allocated automatically and therefore the company is more in control of these containers.

Customer service will have a constant insight of the actual data available. Manual searches are no longer required neither as checking websites if specific terminals changed their open and close times. When containers are planned in the model, they will be flagged when data is missing, so only the containers that need attention will appear. By establishing a link to the database, these containers can be linked to the customers quickly and information about who to contact and how can be displayed in one view. Therefore, also *customer service* has accurate data automatically generated, allowing them to focus on the problem cases instead of doing a daily mandatory import and export checks for every container.

7.3 Discussion

This paragraph will discuss some choices that have or have not been made in this research.

The objective function for the **ILP** is rather simple still, it only punishes containers not delivered on time (too early or too late). The initial idea for an alternative objective function was to maximize the used capacity of the barges. But because this research only includes one part of the containers to move (only export from Veghel to Rotterdam), in practice voyages probably need more room to move the containers that are not taken into account with this model. For instance, if a voyage has to pick something up in Cuicjk between Veghel and Rotterdam, when planning a full voyage this would not be possible. Whether or not to extend it in the future is something that should be discussed about. Right now, the model contributes to the problem presented in this research, that is being more in control of the container. Whether or not to change or extend the objective function is to be determined.

The way to filter containers is also a point of discussion. In chapter 6, the last two data sets (data set 3 and 4) where the same, except that data set 3 was filtered on the closing dates of the bookings, and data set 4 was directly filtered on the closing dates of the containers. It seems that the KPIs of data set 4 score better, but whether or not to filter this way remains uncertain. It is up to the company to decide whether it may be better to filter directly on the units. By filtering on the bookings, the bookings will remain complete. i.e. when filtering on the bookings, all containers belonging to the bookings will be allocated, this may be preferred so bookings do not split up. The upside of filtering directly on the containers, is that the eligible voyages will be more precise and therefore better occupied. A possible solution to get the best out of both alternatives, is to filter only on container close dates, but extend the dashboard in such a way that bookings which are only partly moved will be shown so that the containers that are not moved can be checked. This could be due an error in the data, or because the container in fact has other open and close dates compared to the other containers in the same booking.

To actually solve the **ILP**, two default solvers where used in the implementation (GLPK & COIN). Which one is best is not determined. The outcomes are the same for both solvers and process time is also similar. When the model will extend and the data sets get larger, it may be a good idea to delve more into the possibilities of available solvers. This research has not done that because the solvers worked fine, but it is brought up here because it may be a significant part of the model to review when extending the scope.

The moves that *bargeplanning* agrees upon with the deep sea terminal, have not been taken into account in this model. In reality, when only 40 moves are agreed upon, this is a hard constraint for a voyage at the specific terminals. It would not be hard to add to the model, but the agreed moves are not in the database. They should either be included in the database, or be given as input at the R script for example before the **ILP** is solved. *Bargeplanning* will know how many moves they have, they have to check the voyages produced by the **ILP** either way so they can manually add or subtract containers when there are no sufficient moves, or they can first solve the **ILP** and only then make agreements with terminals based on the outcomes of the **ILP**. Either way, it has to be taken in to account, depending on what the companies preference is, they should either include it in the model or make agreements based on the model if fully functional.

Python was chosen for the implementation of the **ILP**, although other software may work better. Programs such as AIMMS come with a variety of solvers, capable of solving large complex problems. Although this software and its solvers where at hand at the time of this research, a conscious choice has been made to not use this software, because the license for the application had to be bought if used commercially. Therefore, the choice for Python was made.

When looking back on the solution strategy, it can be concluded this was a good one. The use of other software like AIMMS could've been a good measure to first test the program in a well known environment, because implementation in Python was rather new and therefore more of a challenge compared to AIMMS which is provided by Fontys. Only later on when writing this report the script to solve the **ILP** was completely bug free, at first the suspicion was there that due to computing power the script was not working as it is after writing this report. If first an implementation in AIMMS would have been made, this could have been ruled out. The choice for an **ILP**, the chosen software and the work order described in 3.3 however was a good one to follow. Especially when looking at possible extension of the model. It was not hard to serve both *customer service* and *bargeplanning* at the same time, because they both had other means with the model which where easy to combine.

One last point of discussion, is another type of capacity of the barges. If a barge is loaded in three layers, they have to have a minimum amount of weight in order to pass under certain bridges between Veghel and Rotterdam. This has not yet been included, but when the model will be used and/or extended, it should be taken into account just like the number of agreed moves. Basically it would mean that a constraint must be added, stating that if the sum of containers on a barge exceeds a certain level (from two to three layers high), the sum of the weight of those containers must be of a minimum value.

7.4 Possible extensions

The model has been designed in a way that it can be extended in multiple ways. First of all, the dashboard is still very simple as it is, it has not yet be linked to the **Octoparse** data from the web, and not all information is displayed on it yet. The data is all at hand, so improving this part of the module would be the easiest and most quick to do.

Besides the dashboard, the scope of the model can be extended as well. This can be done by extending the **ILP** step by step to match reality more. The order and how to do this will be punctually addressed;

- Include other **inland terminal**s (Veghel and Cuijck)
- Include more categories of containers such as reefers
- Include import of containers, i.e. allocate import containers to voyages from deep sea terminals to **inland terminal**s
- Include possible moves at the **deep-sea terminal** as a constraint
- Include trucks so containers can be assigned to them if there is no other way for transport
- Include demurrage costs and compare to trucking cost to determine whether a container has to be picked up by truck, or can be left at the **deep-sea terminal** for a given time
- Let the program not only assign to voyages, but also define the voyages based on available barges and the demand for transportation

7.5 Recommendations

The last paragraph will once more, briefly state and summarize the outcomes of this research, and will form my personal advice on what to do with these outcomes. The concept has been proven; the data which was already at hand has been used in such a way that the company can improve its control of its containers.

By allowing the data and the math do the work instead of visualizations in a planner's mind, part of the day-to-day operations have been automated, and problematic cases become visible and can be addressed instantly.

Following recommendations should be considered.

Recommendation 1: Continue with shadow run

Schedules can be generated at any given time, for the model is an online model directly linked to the database. Any changes will be directly translated and processed by the model. It's recommended to run the model as a shadow, meaning the planners should still make schedules as they are used to, but compare the results with the outcomes of the model. This is because some constraints are not included yet (such as number of moves). The differences can then be studied and based on these differences or similarities; the model can be extended step by step as described in chapter 7.4.

Recommendation 2: Frequency of planning

At first every day a planning was created and/or modified. Eventually this can be done once per week and compare the outcomes.

Recommendation 3: Apply alternative filter

As described in the discussion (chapter 7.3), the recommendation would be to filter directly on the closing datetimes of containers instead of closing datetimes of bookings and display not completely moved bookings on the dashboard. This way the most efficient schedule is made, and another parameter of flagging can be added to the dashboard for *bargeplanning* and *customer* service to react upon.

Recommendation 4: Extending and supervise the model

Eventually, the entire transportation process could be described mathematically. This research proves the concept that part of the process can be automated, hence this could also be applied to the entire value chain. In theory, a fully extended model could one day replace the *bargeplanning* and can make schedules that are always 100% error free. In practice, I think it is good to extend the model, but always let *bargeplanning* supervise the model. Their role would ultimately be to manage by exception, the math program can provide a solid base solution, and the supervisor can act on those containers that cause trouble. The model can give solutions, but the eventual decision is to be made by the planner.

Ultimate end-goal is to extend and deploy the quantitative model to Increase control over individual containers in **synchromodal** transportation

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Attachments

Chapter 2

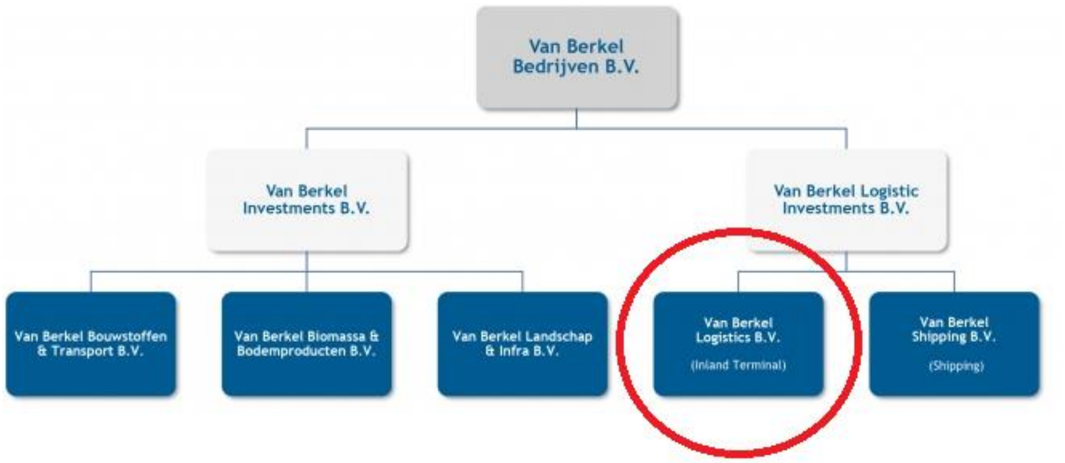


Figure 2.1: organization chart of Van Berkel Group

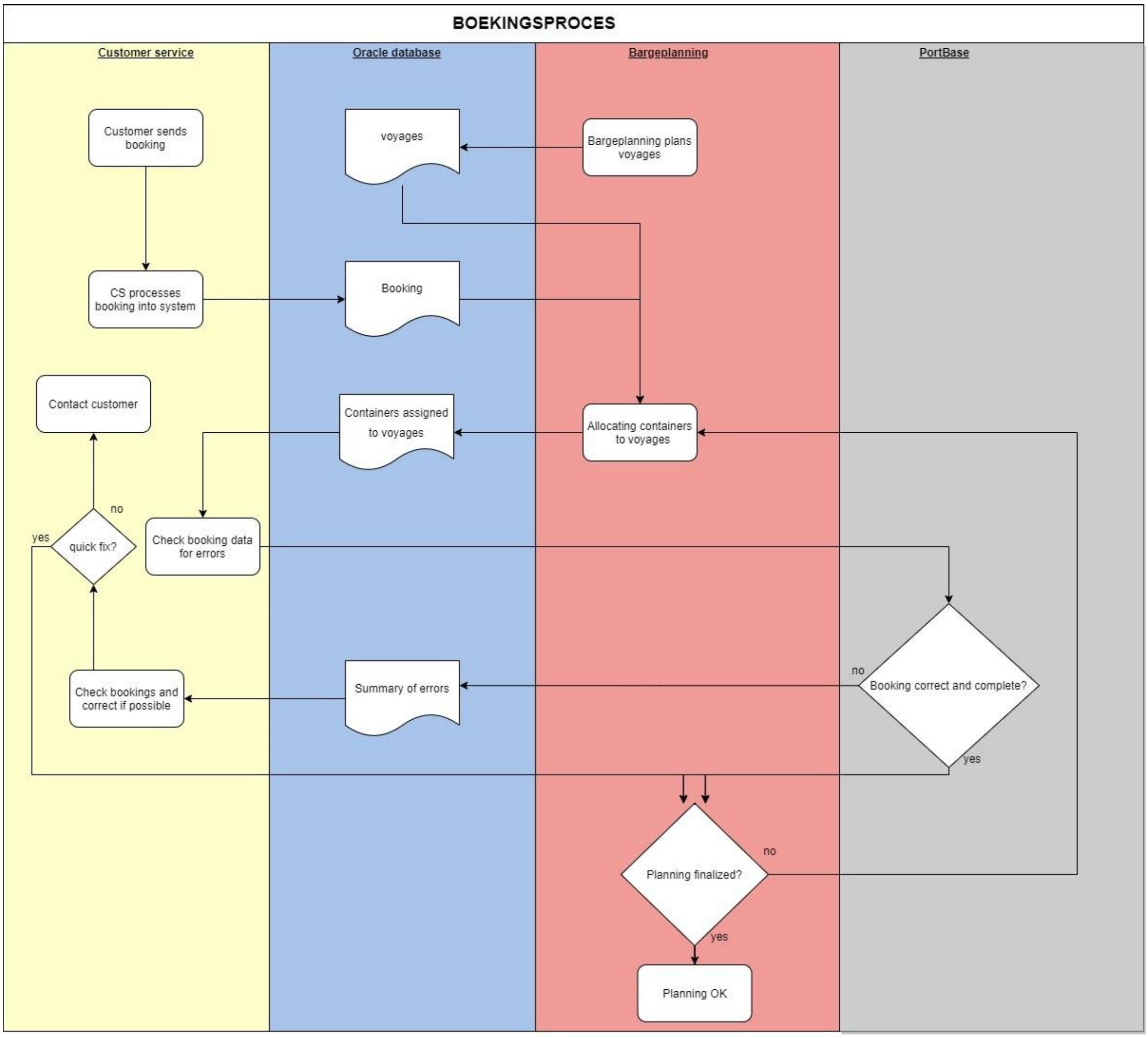


Figure 2.2: Schematic view of booking process

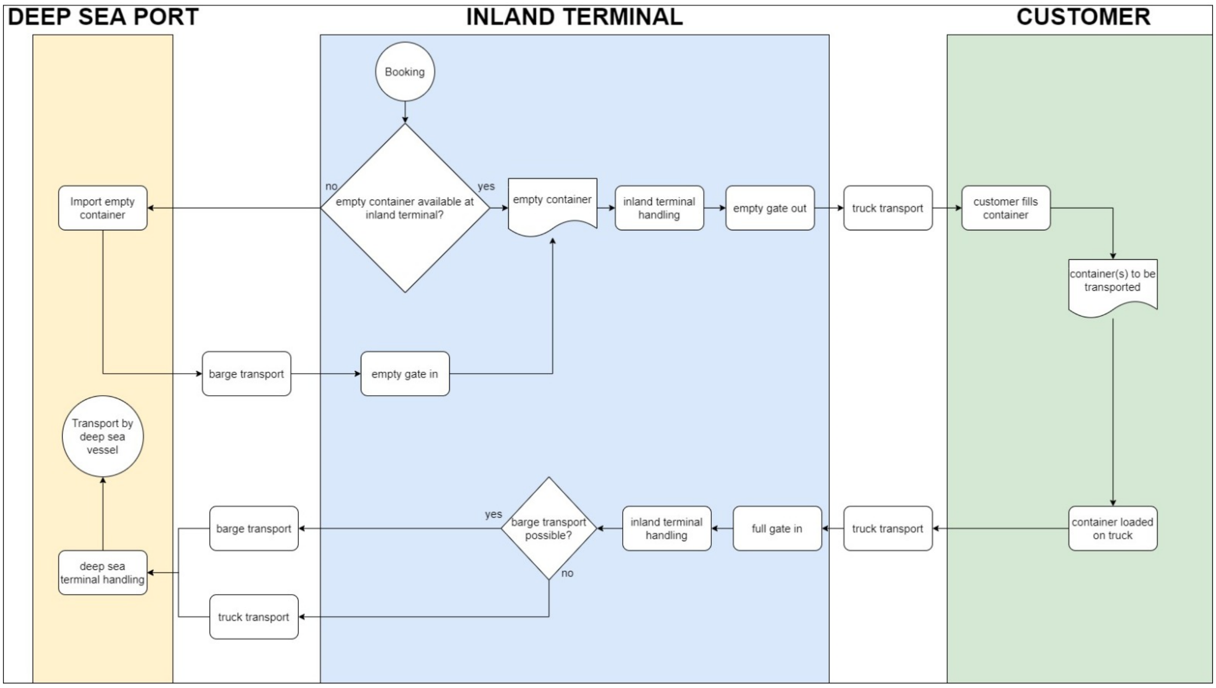


Figure 2.3: Schematic view of an export journey

Chapter 3

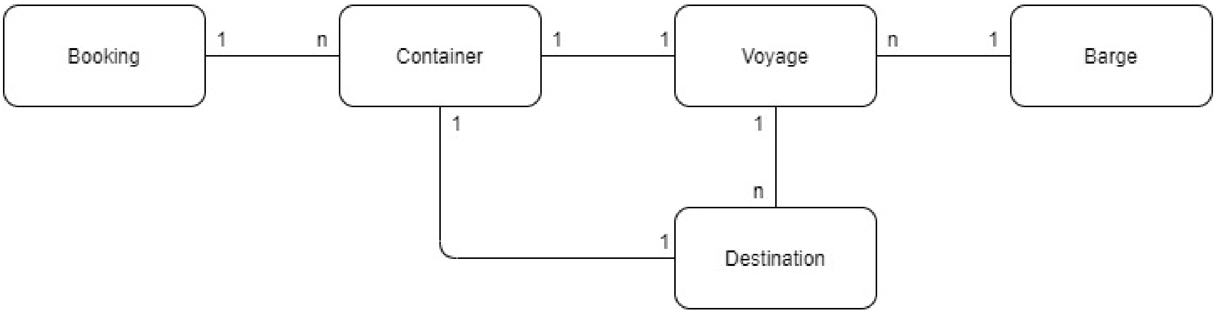


Figure 3.1: Significant objects for container transport and its relations

Chapter 4

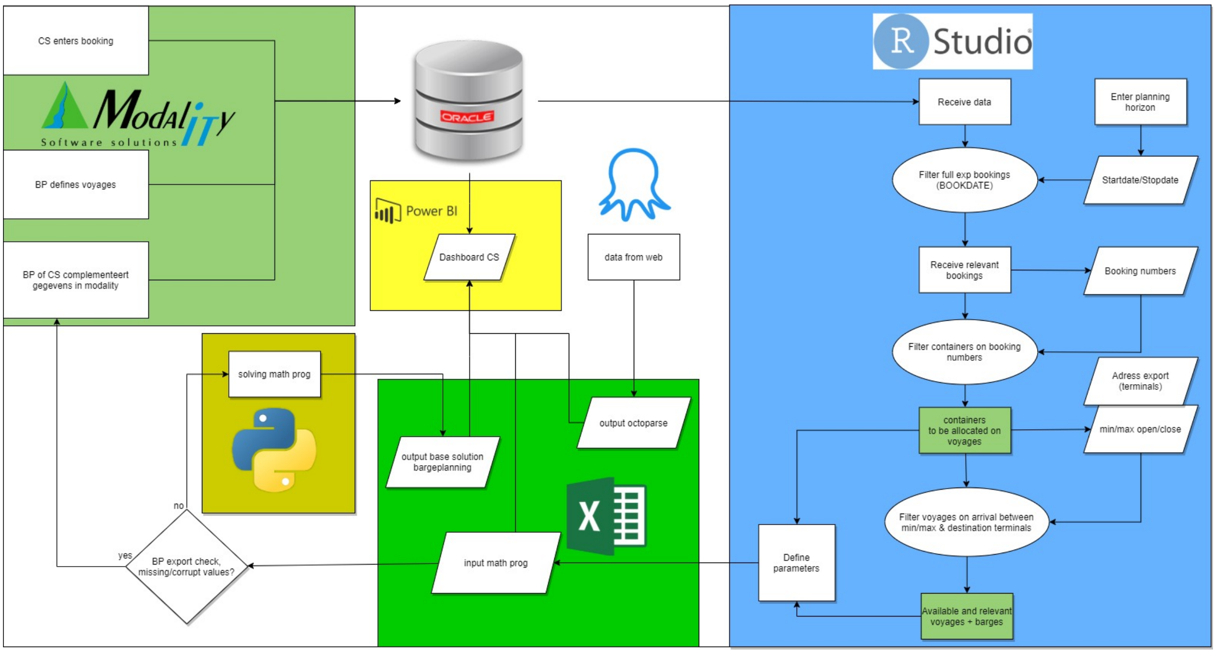


Figure 4.1: Schematic view of global model

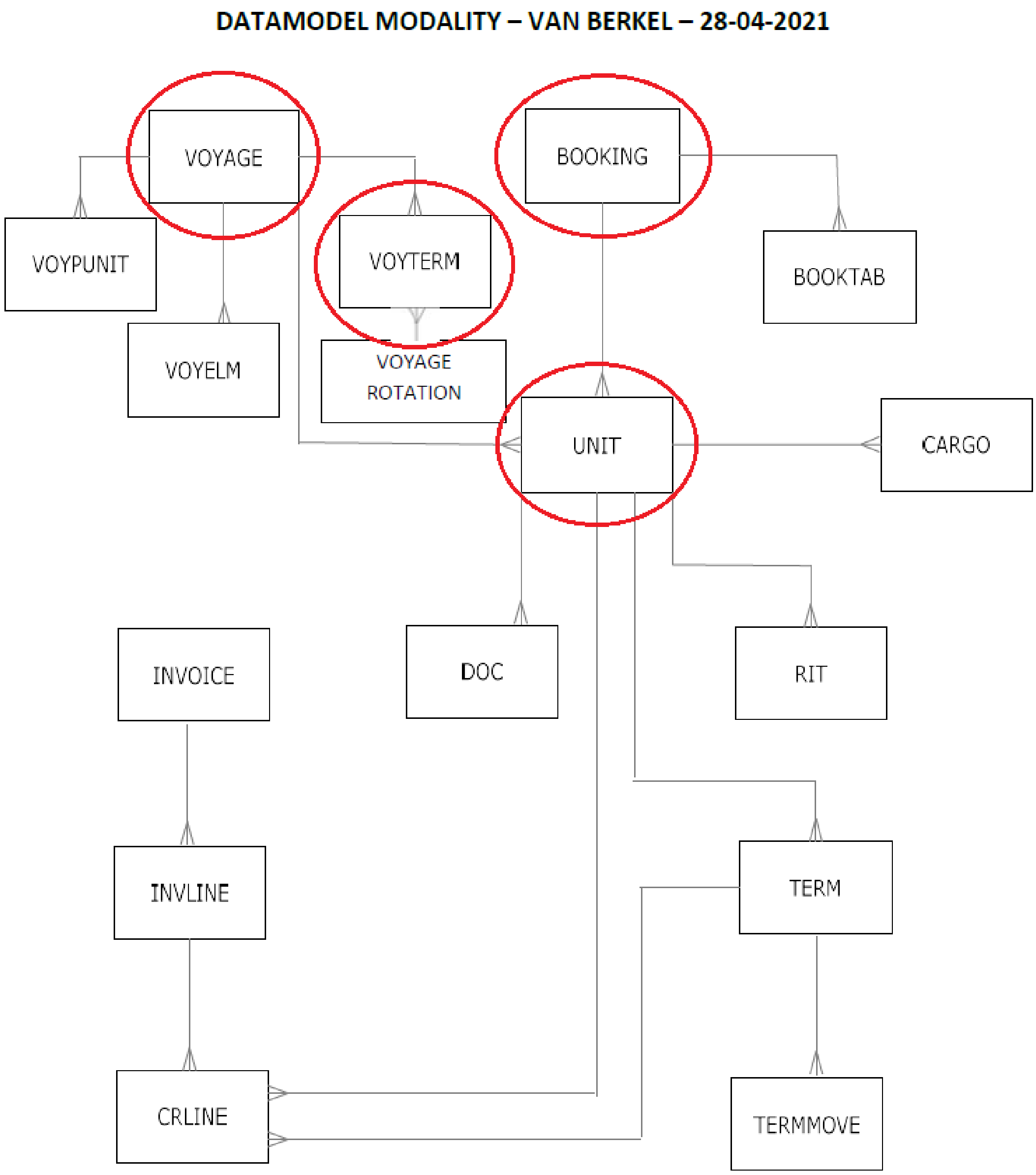


Figure 4.2: Complete database structure of the Oracle database

Parameter	*Description*	*Data type*
SEQ	Booking number	Integer
ADRESS_EXP	Destination for containers in the booking (deep sea terminal)	String
DELDATE	Delivery date, the closing date (last possible date to deliver)	Date
DELTIME	Delivery time, the closing time on the closing date	Integer
FIRSTDELDATE	First delivery date, the opening date (first possible date to deliver)	Date
FIRSTDELTIME	First delivery time, the opening time on the opening date	Integer

Table 4.1: Parameters received from BOOKING table

	SEQ	ADDRESS_EXP	DELDATE	DELTIME	FIRSTDELDATE	FIRSTDELTIME	FIRSTDELNUM	DELNUM
1	199263	ECTDDE	2021-06-09	0200	2021-05-26	0001	550.02	888.00
2	201077	ECTDDE	2021-06-09	0200	2021-05-30	0001	646.02	888.00
3	201585	RWG	2021-06-09	1200	2021-06-01	0001	694.02	898.00
4	202116	APMR2	2021-06-09	1400	2021-06-04	0001	766.02	900.00
5	202225	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
6	202229	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
7	202239	ECTDDE	2021-06-09	1400	2021-06-02	0001	718.02	900.00
8	202271	RWG	2021-06-09	1500	2021-05-25	0001	526.02	901.00
9	202323	ECTDDE	2021-06-09	0700	2021-05-27	0001	574.02	893.00
10	202324	ECTDDE	2021-06-09	0700	2021-05-27	0001	574.02	893.00

Table 4.2: Example of dfBOOKING

Parameter	*Description*	*Data type*
BOOKING	Booking number	Integer
CNTR	Container number	String
NETT	Net weight (Kg)	Double
ADRESS_EXP	Destination for container in the booking (deep sea terminal)	String
UNITTYPE	Container type	String
FIRSTDELDATE	First delivery date, the opening date (first possible date to deliver)	Date
FIRSTDELTIME	First delivery time, the opening time on the opening date	Integer
DELDATE	Delivery date, the closing date (last possible date to deliver)	Date
DELTIME	Delivery time, the closing time on the closing date	Integer

Table 4.3: Parameters received from UNIT table

	BOOKING	CNTR	NETT	ADDRESS_EXP	UNITTYPE	FIRSTDELDATE	FIRSTDELTIME	DELDATE	DELTIME	FIRSTDELNUM	DELNUM	TEU	CAT
1	202324	BEAU 546386 0B	20100.0	ECTDDE	40HC	2021-05-27	0001	2021-06-09	0700	574.02	8.9300e+02	40	40ft
2	201585	BMOU 521601 5A	19145.0	RWG	40HC	NA	0001	NA	0001	1.00	2.0000e+07	40	40ft
3	203983	CAAU 563723 8H	1338.0	APMR	40HC	NA	0001	2021-06-09	0300	1.00	8.8900e+02	40	40ft
4	202229	CAIU 790251 8A	6072.0	ECTDDE	40HC	2021-06-02	0001	2021-06-09	1400	718.02	9.0000e+02	40	40ft
5	202271	CAIU 924367 8A	20809.0	RWG	40HC	2021-05-25	0001	2021-06-10	1100	526.02	9.2100e+02	40	40ft
6	202602	CMAU 772248 4A	23305.0	ECTDDE	40HC	2021-06-05	0001	2021-06-09	2000	790.02	9.0600e+02	40	40ft
7	202354	FFAU 317749 8A	23488.0	RWG	40HC	2021-06-04	0001	2021-06-09	1500	766.02	9.0100e+02	40	40ft
8	203618	FSCU 827620 5A	20300.0	APMR2	40HC	2021-06-03	0001	2021-06-09	1400	742.02	9.0000e+02	40	40ft
9	199263	GESU 623815 6G	25500.0	NA	40HC	NA	0001	2021-06-09	0200	1.00	8.8800e+02	40	40ft
10	201077	HAMU 122222 3A	0.1	NA	40HC	NA	0001	NA	0001	1.00	2.0000e+07	40	40ft

table 4.4: Example of dfUNIT

Parameter	*Description*	*Data type*
VOYAGE	Voyage number	String
EDATE	**ETA** (date)	Date
ETIME	**ETA** (time)	Integer
ADRESS	Destination of voyage	String

Table 4.5: Parameters received from VOYTERM table

	VOYAGE	EDATE	ETIME	ADDRESS	BARGE	EDATENUM
1	ALF0635	2021-05-25	1250	RWG	ALF	538.83
2	ALF0639	2021-05-29	0425	RWG	ALF	626.42
3	ALF0641	2021-06-02	1700	APMR2	ALF	735.00
4	ALF0641	2021-06-02	1600	RWG	ALF	734.00
5	ALF0643	2021-06-05	0730	APMR	ALF	797.50
6	ALF0645	2021-06-06	1900	APMR2	ALF	833.00
7	ALF0647	2021-06-09	1330	ECTDDE	ALF	899.50
8	ALF0649	2021-06-11	1805	ECTDDE	ALF	952.08
9	ALF0651	2021-06-13	1430	APMR	ALF	996.50
10	ALF0653	2021-06-15	1445	EMX	ALF	1044.75

Table 4.6: Example of dfVOYAGE

Parameter	*Description*	*Data type*
BARGE	Code of the barge	String
DESCR	Full name of the barge	String
MAXTEU	Maximum **TEU** to be loaded on barge	Integer
MAXWEIGHT	Maximum weight to be loaded on barge	Integer
MAX20FT	Maximum amount of 20ft containers	Integer
MAX40FT	Maximum amount of 40ft containers	Integer
MAX45FT	Maximum amount of 45ft containers	Integer

Table 4.7: Parameters received from BARGE

	BARGE	DESCR	MAXTEU	MAXWEIGHT	MAX20FT	MAX40FT	MAX45FT
1	MOR	MORDOMIA	1080	1702000	54	24	0
2	VIR	VIRAGE	3120	3000000	156	72	52
3	VIC	VICTOR	2080	1200000	104	52	9
4	FAR	FAR YSKAR	1800	1507000	90	45	36
5	ALF	ALFA NERO	1800	1500000	90	45	0
6	MRB	MER-BLUE	1800	1500000	90	45	0
7	SIN	SINCERO	2160	1200000	108	54	0
8	ERC	ERCULANO	1980	1750000	99	45	0
9	ESP	ESPERO	2160	2000000	108	48	0
10	SPE	SPECTER	3120	3000000	156	72	54

Table 4.8: Example of dfBARGE

Chapter 5

BOOKING	CNTR	NETT	ADRESS_EXP	UNITTYPE	FIRSTDELDATE	FIRSTDELTIME	FIRSTDELNUM	DELDATE	DELTIME	DELNUM	TEU	CAT
1	i1	100	t1	20hc	1-5-2021	0	44317,00	15-5-2021	0	44331,00	20	20ft
1	i2	200	t2	20hc	3-5-2021	0	44319,00	15-5-2021	0	44331,00	20	20ft
1	i3	300	t3	40hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	40	40ft
2	i4	400	t4	40hc	7-5-2021	0	44323,00	16-5-2021	0	44332,00	40	40ft
2	i5	500	t5	40hc	9-5-2021	0	44325,00	17-5-2021	0	44333,00	40	40ft
2	i6	600	t6	20hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	20	20ft
3	i7	700	t1	20hc	1-5-2021	0	44317,00	15-5-2021	0	44331,00	20	20ft
3	i8	800	t2	40hc	3-5-2021	0	44319,00	15-5-2021	0	44331,00	40	40ft
3	i9	900	t3	40hc	5-5-2021	0	44321,00	15-5-2021	0	44331,00	40	40ft
4	i10	1000	t4	45hc	7-5-2021	0	44323,00	16-5-2021	0	44332,00	45	45ft

Table 5.1: dfUNIT test set

VOYAGE	EDATE	ETIME	ADRESS	BARGE	EDATENUM
v1	10-5-2021	7	t1	b1	44326,29
v1	10-5-2021	9	t2	b1	44326,38
v2	11-5-2021	7	t3	b2	44327,29
v2	11-5-2021	9	t4	b2	44327,38
v3	12-5-2021	7	t5	b3	44328,29
v3	12-5-2021	9	t6	b3	44328,38

Table 5.2: dfVOYAGE test set

BARGE	DESCR	MAXTEU	MAXWEIGH	MAX20FT	MAX40FT	MAX45FT
b1	Barge1	2200	31000	110	55	48
b2	Barge2	1400	30000	70	35	0
b3	Barge3	1800	32000	90	45	40

Table 5.3: dfBARGE test set

	v1	v2	v3	GHOST
b1	1			
b2		1		
b3			1	
OTHER				1

Table 5.4: assignment of barges to voyages in matrix form of test set

	v1	v2	v3	GHOST
t1	1			1
t2	1			1
t3		1		1
t4		1		1
t5			1	1
t6			1	1

Table 5.5: Destination of the voyages in matrix form of test set

	t1	t2	t3	t4	t5	t6
i1	1					
i2		1				
i3			1			
i4				1		
i5					1	
i6						1
i7	1					
i8		1				
i9			1			
i10				1		

Table 5.6: Destination of the containers i at terminal t in matrix form of test set

	t1	t2	t3	t4	t5	t6
i1	44317					
i2		44319				
i3			44321			
i4				44323		
i5					44325	
i6						44321
i7	44317					
i8		44319				
i9			44321			
i10				44323		

Table 5.7: Open time for container I at terminal t of test set

	t1	t2	t3	t4	t5	t6
i1	44331					
i2		44331				
i3			44331			
i4				44332		
i5					44333	
i6						44331
i7	44331					
i8		44331				
i9			44331			
i10				44332		

Table 5.8: Close time for container I at terminal t of test set

	v1	v2	v3	GHOST
t1	44326,29			-10
t2	44326,38			-10
t3		44327,29		-10
t4		44327,38		-10
t5			44328,29	-10
t6			44328,38	-10

Table 5.9: Arrival time at terminal t of voyage v of test set

cap	b1	b2	b3	OTHER
20hc	110	70	90	1E+99
40hc	55	35	45	1E+99
45hc	48	0	40	1E+99

Table 5.10: Capacity of category c of barge b of test set

VOYAGE	MAXWEIGHT	MAXTEU
v1	31000	2200
v2	30000	1400
v3	32000	1800
GHOST	1E+99	1E+99

Table 5.11: Capacities of voyages in weight and total **TEU** *of test set*

Chapter 6

```
x_var_('i1',_'v1') = 1
x_var_('i10',_'GHOST') = 1
x_var_('i2',_'v1') = 1
x_var_('i3',_'v2') = 1
x_var_('i4',_'v2') = 1
x_var_('i5',_'v3') = 1
x_var_('i6',_'v3') = 1
x_var_('i7',_'v1') = 1
x_var_('i8',_'v1') = 1
x_var_('i9',_'v2') = 1
yo_var_i10 = 1
1
```

Figure 6.1: Raw output from python console for **mathematical program** *of test set*

CNTR	VOYAGE	yo	yc
i1	v1	0	0
i10	GHOST	1	0
i2	v1	0	0
i3	v2	0	0
i4	v2	0	0
i5	v3	0	0
i6	v3	0	0
i7	v1	0	0
i8	v1	0	0
i9	v2	0	0

Table 6.1: Allocation and variable values for each container of test set

VOYAGE	MAXWEIGHT	MAXTEU	totWeight	totTEU	BARGE	max20ft	max40ft	max45ft	tot20ft	tot40ft	tot45ft
v1	31000	2200	1800	100	b1	110	55	48	3	1	0
v2	30000	1400	1600	120	b2	70	35	0	0	3	0
v3	32000	1800	1100	60	b3	90	45	40	1	1	0
GHOST	1E+15	1E+15	1000	45	OTHER	0	0	0	0	0	1

Table 6.2: Capacity and occupation of the voyages of test set

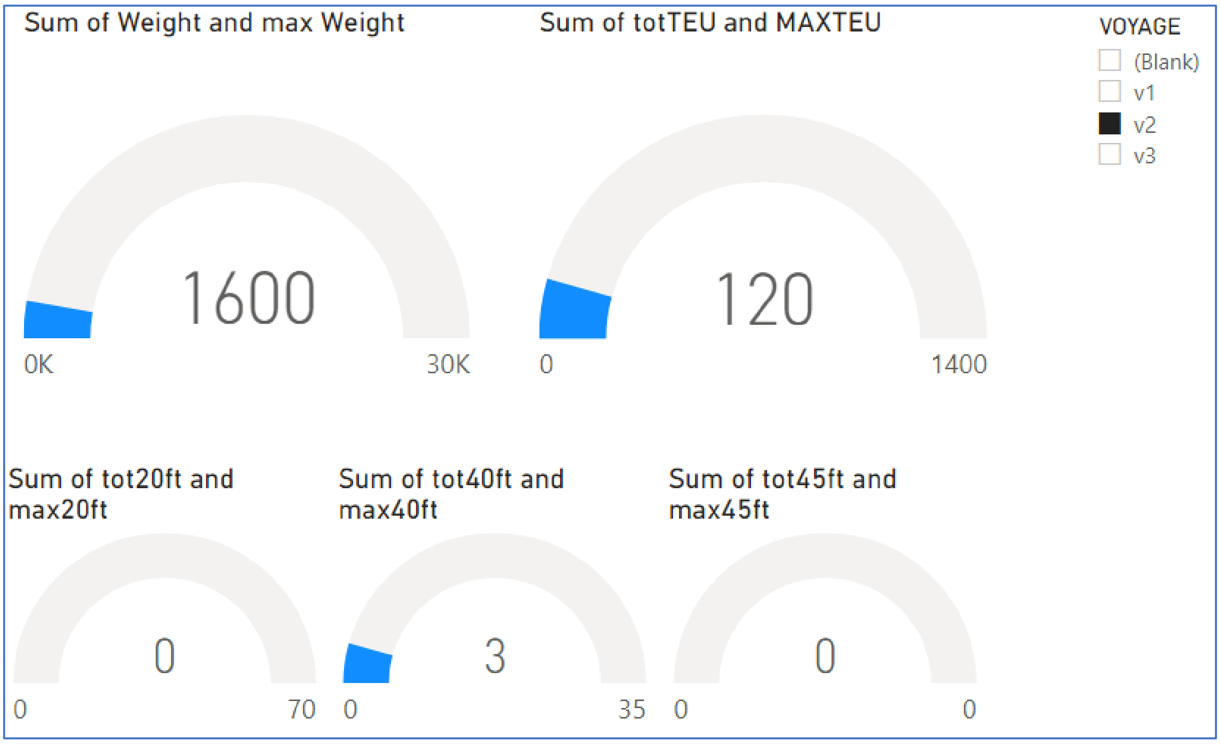


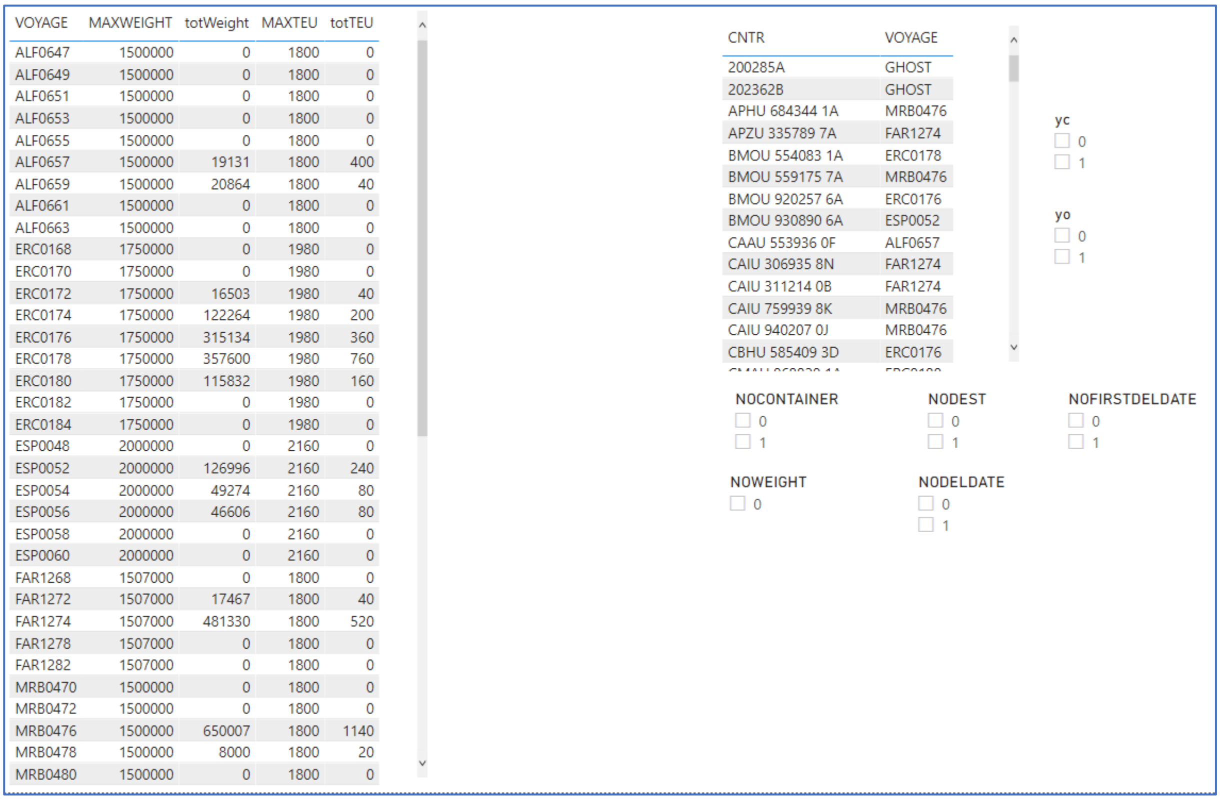
*Figure 6.2: Sketch of interactive dashboard in **PowerBI**, occupancy tab of test set*

CNTR	VOYAGE
i1	v1
i10	GHOST
i2	v1
i3	v2
i4	v2
i5	v3
i6	v3
i7	v1
i8	v1
i9	v2

VOYAGE	MAXTEU	totTEU	MAXWEIGHT	totWeight
v1	2200	100	31000	1800
v2	1400	120	30000	1600
v3	1800	60	32000	1100

VOYAGE	max20ft	tot20ft	max40ft	tot40ft	max45ft	tot45ft
v1	110	3	55	1	48	0
v2	70	0	35	3	0	0
v3	90	1	45	1	40	0

*Figure 6.3: Sketch of interactive dashboard for test data set in **PowerBI**, allocation tab of test set*



VOYAGE	MAXWEIGHT	totWeight	MAXTEU	totTEU
ALF0647	1500000	0	1800	0
ALF0649	1500000	0	1800	0
ALF0651	1500000	0	1800	0
ALF0653	1500000	0	1800	0
ALF0655	1500000	0	1800	0
ALF0657	1500000	19131	1800	400
ALF0659	1500000	20864	1800	40
ALF0661	1500000	0	1800	0
ALF0663	1500000	0	1800	0
ERC0168	1750000	0	1980	0
ERC0170	1750000	0	1980	0
ERC0172	1750000	16503	1980	40
ERC0174	1750000	122264	1980	200
ERC0176	1750000	315134	1980	360
ERC0178	1750000	357600	1980	760
ERC0180	1750000	115832	1980	160
ERC0182	1750000	0	1980	0
ERC0184	1750000	0	1980	0
ESP0048	2000000	0	2160	0
ESP0052	2000000	126996	2160	240
ESP0054	2000000	49274	2160	80
ESP0056	2000000	46606	2160	80
ESP0058	2000000	0	2160	0
ESP0060	2000000	0	2160	0
FAR1268	1507000	0	1800	0
FAR1272	1507000	17467	1800	40
FAR1274	1507000	481330	1800	520
FAR1278	1507000	0	1800	0
FAR1282	1507000	0	1800	0
MRB0470	1500000	0	1800	0
MRB0472	1500000	0	1800	0
MRB0476	1500000	650007	1800	1140
MRB0478	1500000	8000	1800	20
MRB0480	1500000	0	1800	0

CNTR	VOYAGE
200285A	GHOST
202362B	GHOST
APHU 684344 1A	MRB0476
APZU 335789 7A	FAR1274
BMOU 554083 1A	ERC0178
BMOU 559175 7A	MRB0476
BMOU 920257 6A	ERC0176
BMOU 930890 6A	ESP0052
CAAU 553936 0F	ALF0657
CAIU 306935 8N	FAR1274
CAIU 311214 0B	FAR1274
CAIU 759939 8K	MRB0476
CAIU 940207 0J	MRB0476
CBHU 585409 3D	ERC0176

*Figure 6.4: sketch of interactive dashboard for real data set 1 in **PowerBI**, allocation tab*

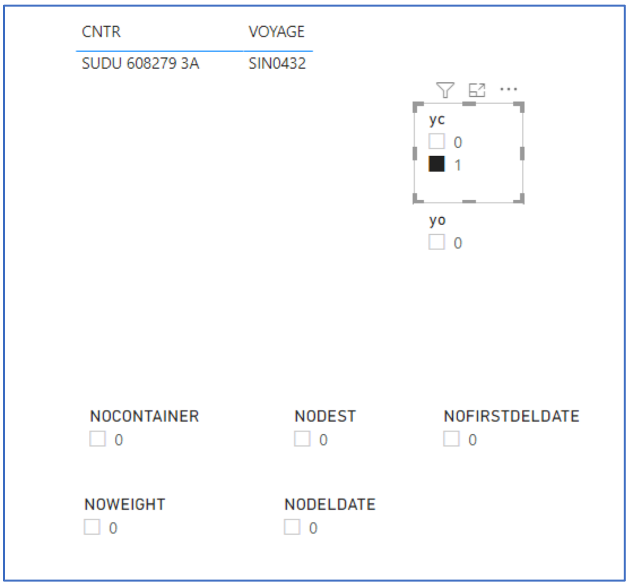


Figure 6.5: subset of containers with yc == 1, arrival after closing datetime of real data set 1

VOYAGE	EDATE	ETIME	ADDRESS	BARGE	EDATENUM		>1198 & < 1233?
ALF0651	44360	1430	APMR	ALF	996,5		FALSE
ALF0657	44367	0830	APMR	ALF	1158,5		FALSE
ESP0052	44364	2330	APMR	ESP	1101,5		FALSE
ESP0056	44371	1715	APMR	ESP	1263,25		FALSE
FAR1278	44372	1600	APMR	FAR	1286		FALSE
SIN0418	44356	0830	APMR	SIN	894,5		FALSE
SIN0432	44376	1100	APMR	SIN	1377		FALSE
SPE0002	44356	2000	APMR	SPE	906		FALSE
SPE0012	44370	1945	APMR	SPE	1241,75		FALSE
VIC1413	44362	1945	APMR	VIC	1049,75		FALSE
VIC1415	44364	2000	APMR	VIC	1098		FALSE
VIC1417	44366	2200	APMR	VIC	1148		FALSE
VIC1423	44374	1200	APMR	VIC	1330		FALSE

Table 6.4: All voyages going to APMR of real data set 1

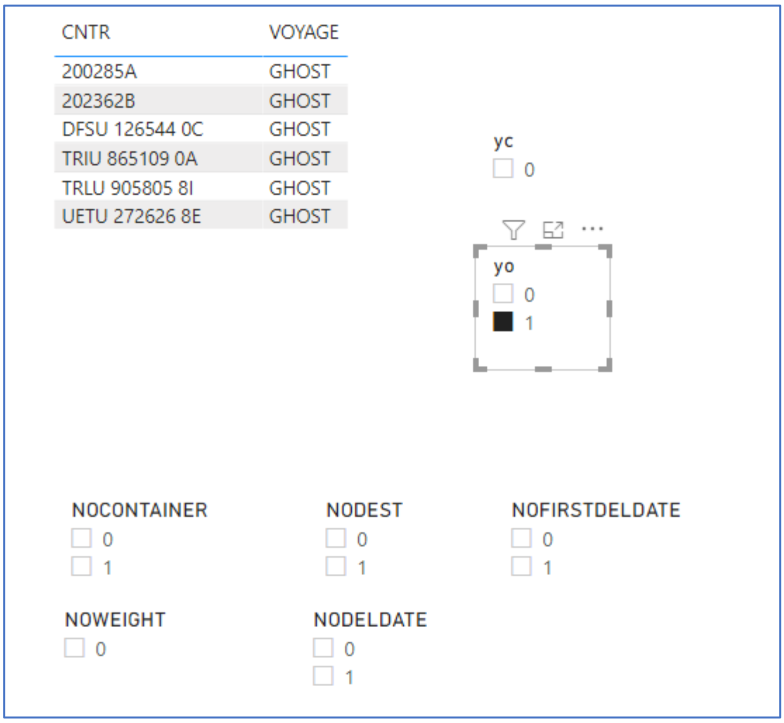


Figure 6.6: subset of containers with yo == 1, arrival before open datetime of real data set 1

Figure 6.7: subset of containers with NOCONTAINER == 1, no container yet assigned of real data set 1

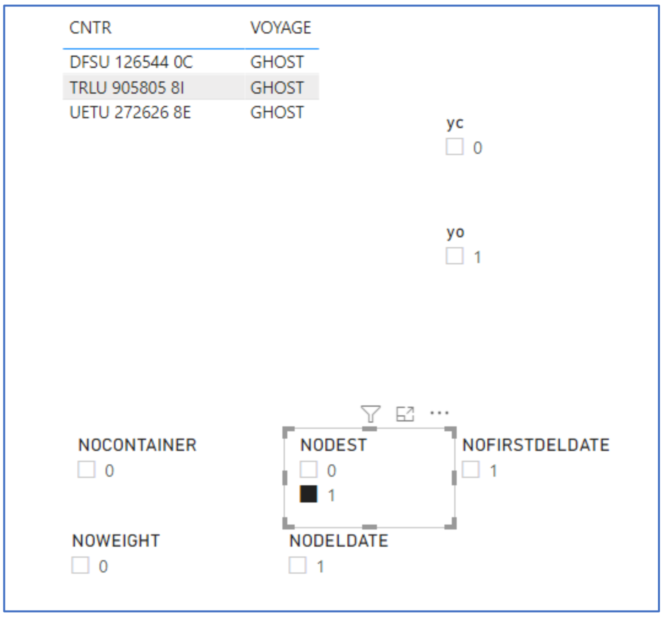


Figure 6.8: subset of containers with NODEST == 1, destination unknown of real data set 1

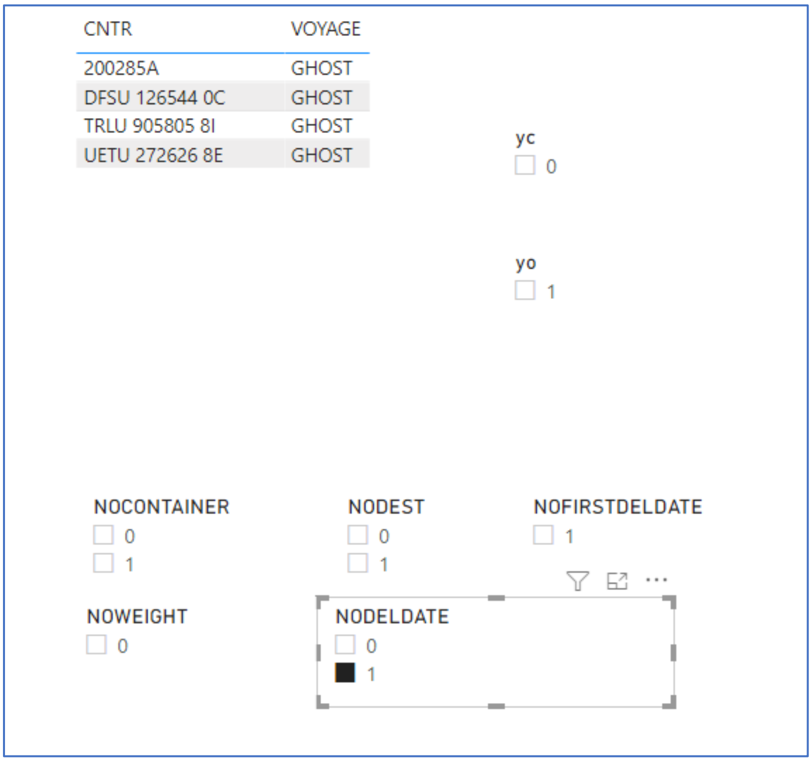


Figure 6.9: subset of containers with NODELDATE == 1, closing datetime unknown of real data set 1

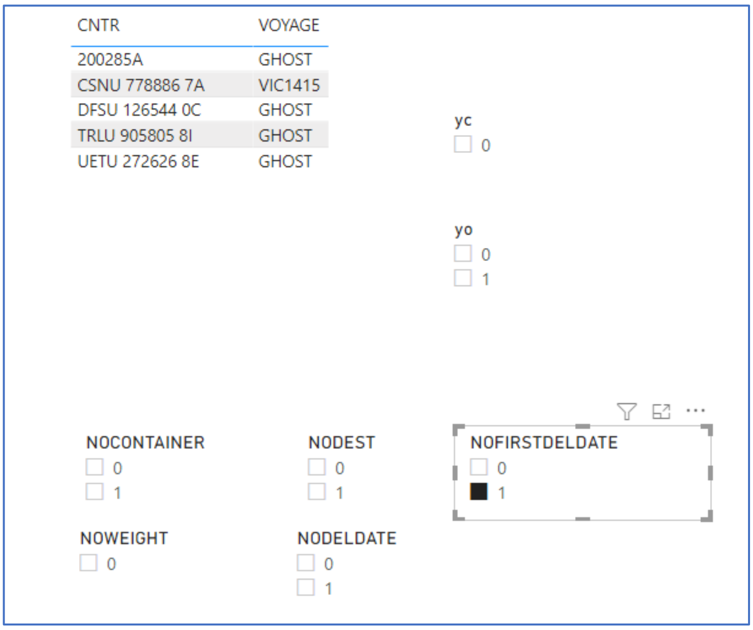
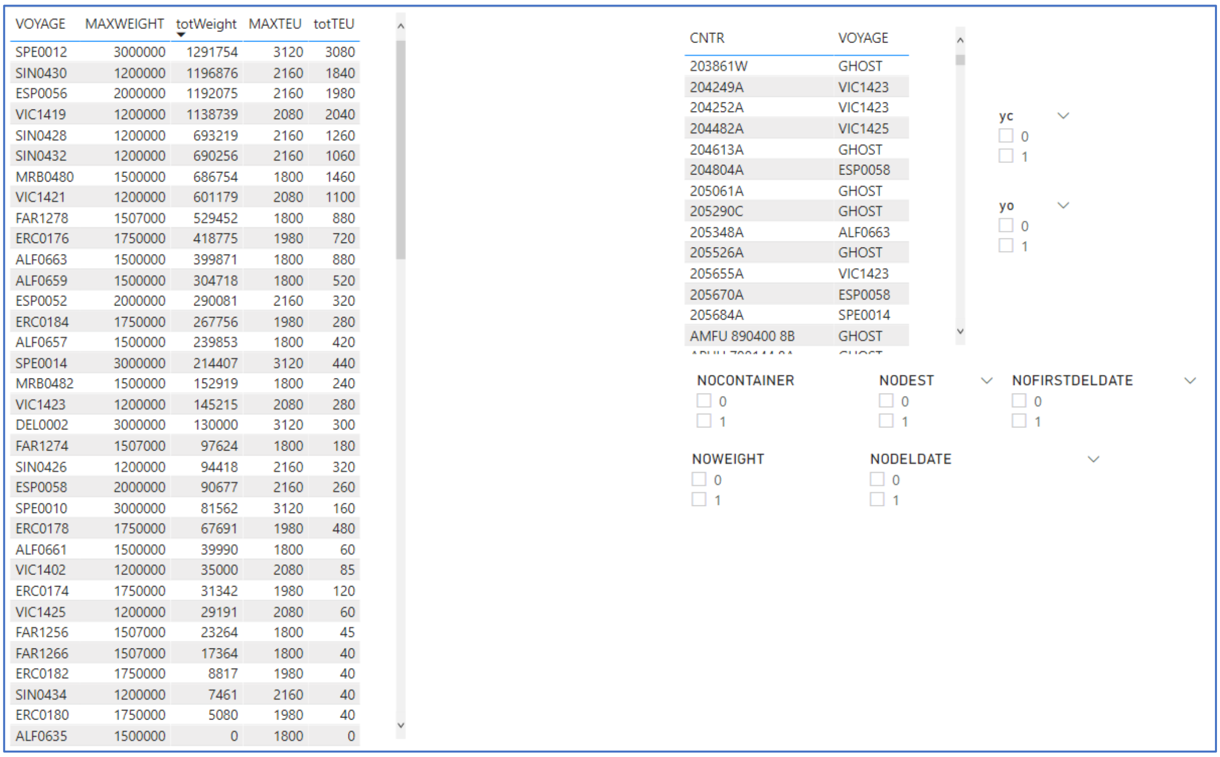


Figure 6.10: subset of containers with NOFIRSTDELDATE == 1, open datetime unknown of real data set 1

Figure 6.11: Occupation of the voyage and their maximum capacity of real data set 1

KPI Real data set 1	**Value**	
Booking close dates horizon	2021-06-23	2021-06-24
Opening dates of containers between	2021-06-09	2021-07-01
Closing dates of container between	2021-06-21	2021-07-07
Percentage of close dates of containers between close date horizon of bookings	89,86%	
# of containers to be moved	152	
# of voyages defined	79	
# of barges used	8	
# of containers moved on time	145	
# of containers on GHOST voyage	6	
# of containers delivered before opening time	6	
# of containers delivered after closing time	1	
# of defined voyages used	23	
Mean used capacity in weight for used voyages	3%	
Mean used capacity in **TEU** for used voyages	6%	
Used capacity in weight for most loaded voyage	43%	
Used capacity in **TEU** for most loaded voyage	63%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	2%	
Percentage of defined voyages used	29%	
Percentage of containers moved on time	95%	
Percentage of containers not moved (on time)	5%	

Table 6.5: KPIs for allocation of containers in real data set 1



VOYAGE	MAXWEIGHT	totWeight	MAXTEU	totTEU
SPE0012	3000000	1291754	3120	3080
SIN0430	1200000	1196876	2160	1840
ESP0056	2000000	1192075	2160	1980
VIC1419	1200000	1138739	2080	2040
SIN0428	1200000	693219	2160	1260
SIN0432	1200000	690256	2160	1060
MRB0480	1500000	686754	1800	1460
VIC1421	1200000	601179	2080	1100
FAR1278	1507000	529452	1800	880
ERC0176	1750000	418775	1980	720
ALF0663	1500000	399871	1800	880
ALF0659	1500000	304718	1800	520
ESP0052	2000000	290081	2160	320
ERC0184	1750000	267756	1980	280
ALF0657	1500000	239853	1800	420
SPE0014	3000000	214407	3120	440
MRB0482	1500000	152919	1800	240
VIC1423	1200000	145215	2080	280
DEL0002	3000000	130000	3120	300
FAR1274	1507000	97624	1800	180
SIN0426	1200000	94418	2160	320
ESP0058	2000000	90677	2160	260
SPE0010	3000000	81562	3120	160
ERC0178	1750000	67691	1980	480
ALF0661	1500000	39990	1800	60
VIC1402	1200000	35000	2080	85
ERC0174	1750000	31342	1980	120
VIC1425	1200000	29191	2080	60
FAR1256	1507000	23264	1800	45
FAR1266	1507000	17364	1800	40
ERC0182	1750000	8817	1980	40
SIN0434	1200000	7461	2160	40
ERC0180	1750000	5080	1980	40
ALF0635	1500000	0	1800	0

CNTR	VOYAGE
203861W	GHOST
204249A	VIC1423
204252A	VIC1423
204482A	VIC1425
204613A	GHOST
204804A	ESP0058
205061A	GHOST
205290C	GHOST
205348A	ALF0663
205526A	GHOST
205655A	VIC1423
205670A	ESP0058
205684A	SPE0014
AMFU 890400 8B	GHOST

Figure 6.12: Allocation tab of the interactive dashboard for real data set 2

Figure 6.13: Occupation tab of the interactive dashboard for real data set 2

KPI Real data set 2	**Value**	
Booking close dates horizon	2021-06-24	2021-06-28
Opening dates of containers between	2021-05-24	2021-07-02
Closing dates of container between	2021-06-22	2021-07-08
Percentage of close dates of containers between close date horizon of bookings	67,33%	
# of containers to be moved	668	
# of voyages defined	144	
# of barges used	11	
# of containers moved on time	579	
# of containers on GHOST voyage	82	
# of containers delivered before opening time	85	
# of containers delivered after closing time	3	
# of defined voyages used	33	
Mean used capacity in weight for used voyages	9%	
Mean used capacity in **TEU** for used voyages	15%	
Used capacity in weight for most loaded voyage	100%	
Used capacity in **TEU** for most loaded voyage	85%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	2%	
Percentage of defined voyages used	23%	
Percentage of containers moved on time	86,68%	
Percentage of containers not moved (on time)	13,32%	

Table 6.6: KPIs for allocation of containers in real data set 2

KPI Real data set 3	**Value**	
Booking close dates horizon	2021-06-25	2021-07-02
Opening dates of containers between	2021-05-21	2021-09-26
Closing dates of container between	2021-06-22	2021-07-16
Percentage of close dates of containers between close date of horizon bookings	98,53%	
# of containers to be moved	1101	
# of voyages defined	160	
# of barges used	12	
# of containers moved on time	951	
# of containers on GHOST voyage	136	
# of containers delivered before opening time	146	
# of containers delivered after closing time	4	
# of defined voyages used	47	
Mean used capacity in weight for used voyages	7%	
Mean used capacity in **TEU** for used voyages	15%	
Used capacity in weight for most loaded voyage	98%	
Used capacity in **TEU** for most loaded voyage	100%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	0%	
Percentage of defined voyages used	29%	
Percentage of containers moved on time	86%	
Percentage of containers not moved (on time)	14%	

Table 6.7: KPIs for allocation of containers in real data set 3

KPI	**Value**	
Booking close dates horizon	-	-
Opening dates of containers between	2021-05-21	2021-09-26
Closing dates of container between	2021-06-25	2021-07-02
Percentage of close dates of containers between close dates of bookings	-	
# of containers to be moved	1093	
# of voyages defined	164	
# of barges used	13	
# of containers moved on time	986	
# of containers on GHOST voyage	59	
# of containers delivered before opening time	99	
# of containers delivered after closing time	7	
# of defined voyages used	69	
Mean used capacity in weight for used voyages	6%	
Mean used capacity in **TEU** for used voyages	11%	
Used capacity in weight for most loaded voyage	100%	
Used capacity in **TEU** for most loaded voyage	100%	
Used capacity in weight for least loaded voyage	0%	
Used capacity in **TEU** for least loaded voyage	0%	
Percentage of defined voyages used	42%	
Percentage of containers moved on time	90%	
Percentage of containers not moved (on time)	10%	

Table 6.8: KPIs for allocation of containers in real data set 4

Chapter 7

KPI	Data set 1		Data set 2		Data set 3		Data set 4	
Booking close dates horizon	23-6-2021	24-6-2021	24-6-2021	28-6-2021	25-6-2021	2-7-2021	-	-
Opening dates of containers between	9-6-2021	1-7-2021	24-5-2021	2-7-2021	21-5-2021	26-9-2021	21-5-2021	26-9-2021
Closing dates of container between	21-6-2021	7-7-2021	22-6-2021	8-7-2021	22-6-2021	16-7-2021	25-6-2021	2-7-2021
Percentage of close dates of containers between close dates of bookings	89,86%		67,33%		98,53%		-	
# of containers to be moved	152		668		1101		1093	
# of voyages defined	79		144		160		164	
# of barges used	8		11		12		13	
# of containers moved on time	145		579		951		986	
# of containers on GHOST voyage	6		82		136		59	
# of containers delivered before opening time	6		85		146		99	

# of containers delivered after closing time	1	3	4	7
# of defined voyages used	23	33	47	69
Mean used capacity in weight for used voyages	3%	9%	7%	6%
Mean used capacity in **TEU** for used voyages	6%	15%	15%	11%
Used capacity in weight for most loaded voyage	43%	100%	98%	100%
Used capacity in **TEU** for most loaded voyage	63%	85%	100%	100%
Used capacity in weight for least loaded voyage	0%	0%	0%	0%
Used capacity in **TEU** for least loaded voyage	2%	2%	0%	0%

Percentage of defined voyages used	29%	23%	29%	42%
Percentage of containers moved on time	95%	86,68%	86%	90%
Percentage of containers not moved (on time)	5%	13,32%	14%	10%

Table 7.1: (K)PIs for allocation for all data sets compared

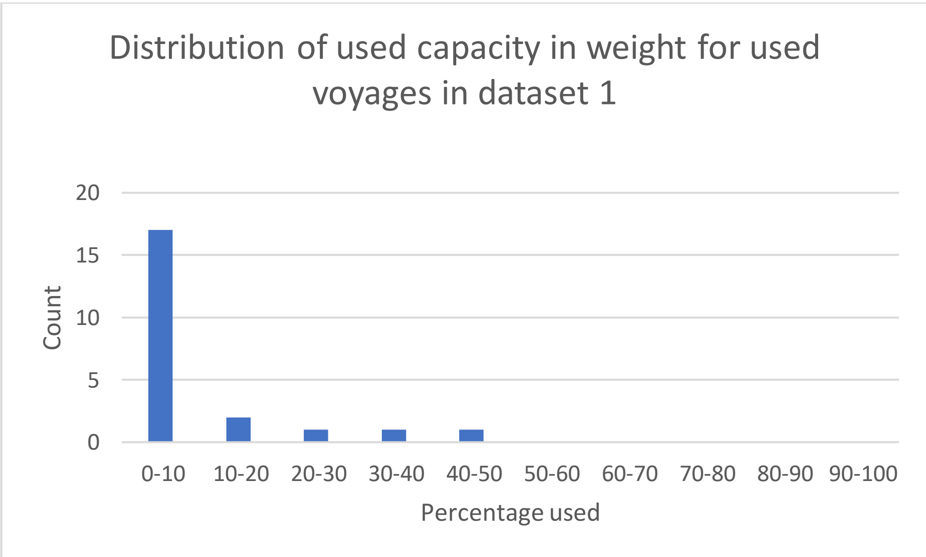


Figure 7.1: Distribution of used capacity in weight for used voyages in dataset 1

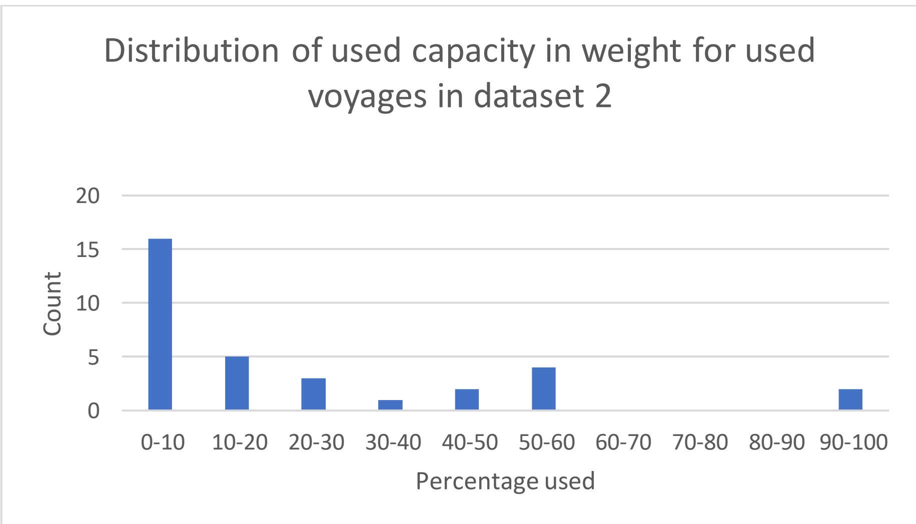


Figure 7.2: Distribution of used capacity in weight for used voyages in dataset 2

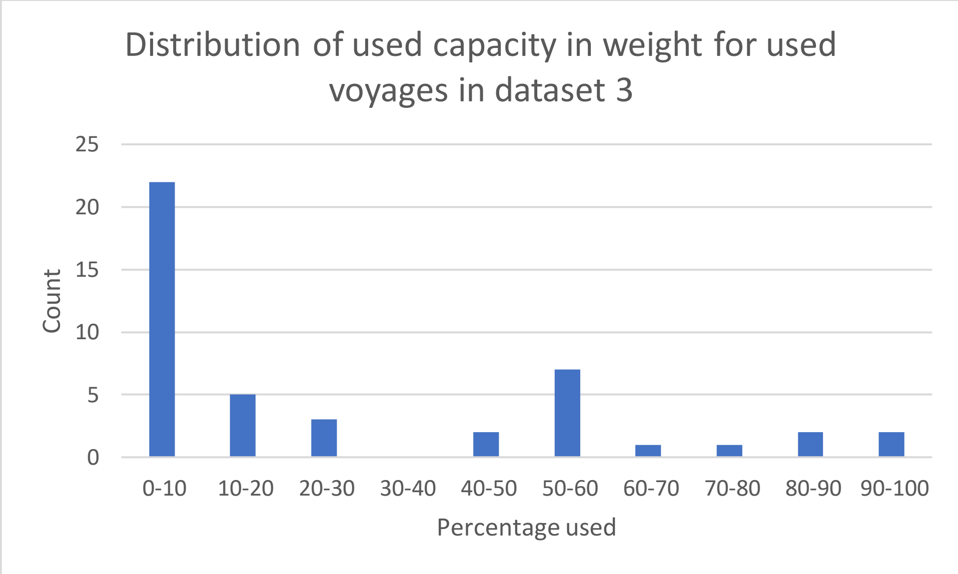


Figure 7.3: Distribution of used capacity in weight for used voyages in dataset 3

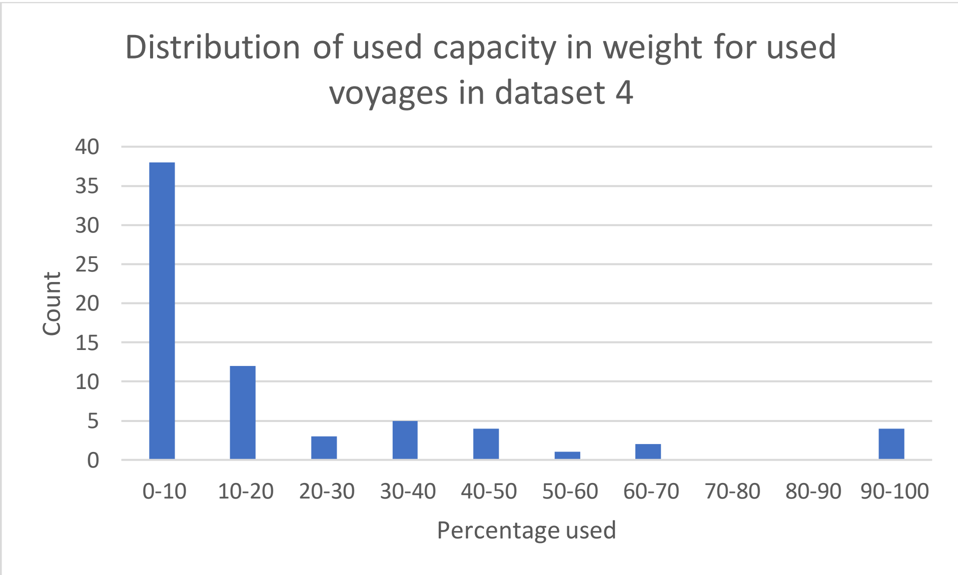


Figure 7.4: Distribution of used capacity in weight for used voyages in dataset 4

KPI Real data set 3 vs 4	**Value set 3**		**Value set 4**	
Booking close dates horizon	2021-06-25	2021-07-02	-	-
Opening dates of containers between	2021-05-21	2021-09-26	2021-05-21	2021-09-26
Closing dates of container between	2021-06-22	2021-07-16	2021-06-25	2021-07-02
Percentage of close dates of containers between close dates of bookings	98,53%		-	
# of containers to be moved	1101		1093	
# of voyages defined	160		164	
# of barges used	12		13	
# of containers moved on time	951		986	
# of containers on GHOST voyage	136		59	
# of containers delivered before opening time	146		99	
# of containers delivered after closing time	4		7	
# of defined voyages used	47		69	
Mean used capacity in weight for used voyages	7%		6%	
Mean used capacity in **TEU** for used voyages	15%		11%	
Used capacity in weight for most loaded voyage	98%		100%	
Used capacity in **TEU** for most loaded voyage	100%		100%	
Used capacity in weight for least loaded voyage	0%		0%	
Used capacity in **TEU** for least loaded voyage	0%		0%	
Percentage of defined voyages used	29%		42%	
Percentage of containers moved on time	86%		90%	
Percentage of containers not moved (on time)	14%		10%	

Table 7.2: Comparison of KPIs for allocation of containers in real data set 3 & 4

Interview customer service

Hoe vaak is CS (naar eigen inzicht) niet in controle over een container/inland date/uithaal datum

Als je kijkt naar de uithaaldatum, dus wanneer we hem op moeten halen in Rotterdam, gebeurd dat honderden keren per week. Dat wil zeggen, dat we de door ons op gegeven kaders niet halen. Dat geld voor alle containers, dus zowel lege containers die moeten worden opgehaald, of containers die van de **inland terminal** naar Rotterdam moeten. Als je kijkt naar de inland date, dus het laden of lossen bij de klant, gebeurd dat ook nog te vaak, tien tot twintig keer per week moeten we actief gaan schakelen met de klant. Maar dat heeft als voornaamste reden, dat heel veel klanten containers gaan afroepen op het moment dat ze al hier zijn. Dus ze kijken in onze webportal wanneer er een vrij stabiele **ETA** in staat en dan pas gaan ze plannen wanneer ze kunnen lossen. In de ideale situatie gebeurd dit al bij het invoeren van de boeking. Helaas doen steeds minder klanten dit. Vooral de grote klanten die bepaalde afspraken hebben met warehouses of met hun voorraad zitten, maken wel een planning. De kleinere klanten waar minder druk achter zit kijken het liever eerst af voordat ze iets betrouwbaars hebben en plannen het dan in.

Is er prioriteit in bepaalde klanten?

Deels is dat wie het hardst roept, maar niet altijd. Dat is verder ook afhankelijk van de grootte van de klant. Een klant die 1 container per jaar doet heeft dan minder te zeggen. Wij zijn dagelijks in gesprek met klanten, dus we weten precies welke containers het meest urgent zijn. Als je weet dat een klant zit te wachten op een container omdat er veel contact is, zet je je wel meer in voor deze klant.

Als een inland date niet gehaald wordt, hoe en wanneer komt CS daar dan achter?

Het liefst weten we dat natuurlijk zo vroeg mogelijk, maar dat is standaard eigenlijk te laat. Dat heeft er mee te maken dat het altijd uit de import check volgt. Als bargeplanning deze doorstuurt komen daar containers uit die te laat gaan komen, dat is 99/100 keer het gevolg van het feit dat de terminal afspraken uitlopen. Terminal afspraken lopen niet een week van tevoren uit, maar slechts 2 of 3 dagen van te voren, daarom kom je er altijd te laat achter. Deze check komt dagelijks binnen, op basis hiervan nemen wij dus contact op met de klant.

Hoe vaak zijn jullie naar eigen inzicht niet in controle over een container?

Dat is erg afhankelijk van sentiment. Op het moment dat het slecht gaat lijkt alles fout te gaan, maar als het goed loopt dan heb je daar minder last van. Nu zitten we in een slechte periode, weinig terugkoppeling en vertraging van een week in Rotterdam, dat voelen wij (Suez kanaal). Wij moeten emails gaan sturen naar klanten, zo hebben we 1 klant die 50 containers in Rotterdam heeft staan die we niet kunnen uithalen. Als dat vaak gebeurd heb je vrij snel een vrij negatief gevoel er over. Er is dus grote afhankelijkheid van de terminals, als zij tijden veranderen moet je het daar maar mee doen.

Wat voor informatie zou CS graag inzichtelijk gemaakt hebben op een dashboard?

Cargo open, cargo close en de **ETA** van de zeeboten zijn het belangrijkste.

Wat is de meest voorkomende reden dat een inland date niet gehaald wordt?

Het verschuiven van afspraken in Rotterdam is de grootste reden dat planningen mislukken. Op het laatste moment kan een afgesproken tijd een dag naar voren schuiven en dan moet je opnieuw gaan plannen. Import heeft hier meer last van dan export, omdat export het grote voordeel heeft dat als een zeeboot vertraging heeft, dat de closing date ook verder naar achter gaat en er dus meer tijd is om de container in te leveren. Bij import werkt dat juist averechts, de spullen moeten steeds dringender gelost worden. Als je er bijvoorbeeld van uitgaat bij het boeken dat je container 40 dagen onderweg is, en dat worden er dan 45, dan raakt je magazijn een keer leeg. Vertraging voor export is dus prettig, voor import niet, hoe meer het daar vertraagd hoe meer containers via de vrachtwagen moeten worden getransporteerd.

Glossary

Inland terminal

A distribution node in the hinterland where barges dock and trucks pick up and deposit containers

Deep sea terminal

A distribution node in the port where barges and deep sea vessels dock and trucks pick up and deposit containers

Deep sea vessel

A large container ship that sails the oceans around the world

Intermodal transport

A form of transportation where goods are shipped in only one transportation unit, like a container

Multimodal transport

A transport chain where ate least two kinds of modes are used to transport goods

Synchromodal transport

Transportation where there is synchronization between multiple modes of transportation based on certain conditions the best way to transport is chosen

Inland date

The date a customer has to receive goods (import), in this report it also refers to the date the goods should be at a deep sea terminal (export)

Twenty foot equivalent unit (TEU)

An exact unit of measurement used to determine the cargo capacity of a container, 1 **TEU** is a 20 foot container.

Barges

A long flat bottomed boat for carrying freight on canals and rivers

ETA

Estimated time of arrival

ETD

Estimated time of departure

(Integer) linear program

A method to achieve the best outcome in a mathematical model whose requirements are represented by linear relationships. Linear programming is a special case of **mathematical programm**ing. Integer **linear programs** (**ILP**) are a special case of **linear program**ing, where the decision variables are forced to be integers rather then arbitrary integers.

Mathematical programing

The selection of a best element, with regard to some criterion, from some set of available alternatives

single objective function

The goal of a **single**-**objective** optimization problem is to find the best solution for a specific criterion or metric, such as execution time (or performance) and/or a combination of this metric with energy consumption or power dissipation metrics.

Multi-objective function

Multi-objective programming is a part of **mathematical program**ming dealing with decision problems characterized by multiple and conflicting objective functions that are to be optimized over a feasible set of decisions.